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UNDERGRADUATE THESIS

On the Origin of Quantum Discreteness in Plasma Electron Scattering: A Non-Archimedean Exploration of the Amplituhedron and Its p-adic Structure

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A thesis submitted in fulfillment of the requirements for the degree of Bachelor of Engineering

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Abstract

Division of Electronics and Electrical Engineering,
Dongguk University

Bachelor of Engineering

**On the Origin of Quantum Discreteness
in Plasma Electron Scattering:
A Non-Archimedean Exploration of the Amplituhedron and Its p-adic Structure**

by SUNG-HYUK YOON

As semiconductor manufacturing approach to nanometer scale, the stochastic fluctuation of particles shows a limit on process control that does not vanish even the process getting precisely. This thesis takes that limit as its starting point and asks a question one step behind it: Does the probabilistic character of the microscopic world is the real aspect of the nature? Is there any possibility that probablic came out from other reasons—such like an artifact of the real-number language in which nature is customarily described.

From this the study proposes a hypothesis—that probability may be the noise of the mapping between physical reality and the mathematical language chosen to represent it, a language that cannot hold the infinitesimal discrepancies the phenomenon may carry. Rather than leaving the hypothesis a matter of philosophical discourse, the study seeks a setting in which it can be examined concretely, and finds one in the amplituhedron—a geometric object defined by the positive geometry of scattering. And to validate it, the study turns to the amplituhedron, a geometric object defined by the positive geometry of scattering, and computes its canonical form in parallel over the rational numbers (QQ) and a p-adically perturbed track (QP). The comparison, however, does not yield the difference in computational precision the hypothesis first anticipated: because both tracks are evaluated in exact rational arithmetic, there is no error to improve upon.

When examined through its p-adic valuation, instead of magnitude, however, the difference between the two tracks reveals a definite and reproducible arithmetic structure—one that no part of the construction was designed to produce in the first place. This structure is not an accident of the computation, but traceable to the way the amplitude is built and carried through its recursive decomposition. The study therefore does not establish that the p-adic framework computes the amplitude more accurately, nor that probabilistic uncertainty has been reduced. What it establishes is narrower but not negligible: that the choice of number system leaves a definite, reproducible, and measurable imprint on an exactly computed amplitude.

In doing so, the study moves the conjecture from a philosophical supposition toward a well-posed and computationally tractable question. The forward computation examined here is the necessary precondition for the direction in which the inquiry ultimately points: the inverse problem of recovering, from a given amplitude, the input that produced it. Here the two number systems are expected to part most sharply—over the reals the candidate inputs form a continuum, while over the p-adic numbers they resolve into a discrete set. This difference in the shape of the solution is what the forward computation, by its nature, cannot reveal, and it is the direction in which the inquiry is to be carried forward.

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Chapter 1

Introduction

This study originates from an investigation into the stochastic limitations inherent in plasma etching. As the critical dimension (CD) scales down to the 2 nm node in contemporary semiconductor manufacturing, the probabilistic fluctuation of particles throughout fabrication has emerged as a critical bottleneck restricting overall yield. These stochastic limits were originally defined in terms of photon shot noise during extreme ultraviolet (EUV) lithography [1]. And the roughness formed at that stage is then transferred and amplified during etching, where identical statistical fluctuations arise independently within the etching step itself [2, 3]. The present work focuses specifically on particle behavior during this plasma etching stage.

During etching, the behavior and positioning of individual particles are inherently stochastic, and this microscopic uncertainty poses a fundamental challenge for precise process control. Mean-value-based statistical descriptions remain valid when the particle count is sufficiently large; in regimes where only tens to hundreds of particles interact within a single etch feature, however, statistical fluctuations become dominant. Most conventional plasma models therefore rely on probabilistic computations, yet as these computations accumulate, the stochastic nature of particle behavior propagates and amplifies uncertainty—sharply increasing the variance of predictive outcomes and posing severe challenges to precise process control.

These difficulties are compounded by computational cost. The widely used *Monte Carlo method* suffers from a statistical error that decreases only as $1/\sqrt{N}$ with sample size N, requiring an unrealistically large number of samples to capture rare physical events [4]. *Particle-in-Cell/Monte Carlo Collision (PIC/MCC)* frameworks, which self-consistently track particle trajectories, are in turn constrained in their time steps and grid sizes by the plasma frequency and the Debye length, demanding prohibitive resources for practically scaled problems [4]. Ultimately, these constraints restrict plasma modeling to mere predictions of containment within a specific range at a certain probability.

Recently, a paradigm shift toward data-driven methodologies—training neural network surrogate models or large language models (LLMs) on extensive process datasets—has shown significant potential for enhancing the efficiency of traditional Monte Carlo simulations [5, 6]. Such models exhibit a "scaling law" trend, in which predictive accuracy improves as data volume and model scale expand[1]. Yet they excel at capturing macro-scale statistical correlations while lacking the causal rigor of deterministic physical laws. To capture the microscopic interactions of individual particles, first-principles computation must supply the physical consistency that statistical inference alone cannot.

[1]The "scaling law" describes an empirical trend in machine learning. It states that model performance improves following a power-law as model size, data volume, and compute increase. This relationship was first quantified in the domain of large language models [7].

It is undeniable that current data-driven and statistical methods represent the most advanced computational approaches available. These methods, however sophisticated, describe the microscopic world in the language of probability. This study steps back to a question beneath them—whether a part of what we treat as probability belongs to the phenomenon, or to the language we describe it with. The possibility it pursues is that "probability" is not an inherent essence of nature, but an artifact of the mathematical language we have chosen for the microscopic world: *the real number system*. It seeks the root of the problem not in the phenomenon, but in the language used to formulate it. Guided by this awareness, the thesis develops its argument as follows.

1.1 Stochastic Limitations in Plasma

Below the 2 nm node, the number of particles interacting within a single etch feature becomes extremely small. In this regime, conventional statistical "averages" lose their physical meaning [3]. This leads to uneven etching on the sidewalls, known as Line Edge Roughness (LER), and inconsistent line widths, known as Line Width Roughness (LWR). This roughness does not scale down proportionally with feature size. Therefore, as devices miniaturize, its relative impact grows [2].

This roughness arises from particle scarcity, which manifests in three ways.

First, the scarcity intrinsic to 2 nm features. A single 2 nm trench holds only tens to hundreds of ions. In legacy nodes at 10 nm and above, statistical fluctuations were absorbed into mean values; at 2 nm, the same fluctuations damage the circuitry [3].

Second, the trajectory scattering of positive ions. During physical etching, ions collide with neutral particles in the plasma sheath and deviate from their intended paths. In larger nodes, abundant ions absorbed these deviations into an average; at 2 nm, stray ions degrade feature integrity [8].

Third, the uncertainty of reaction placement. During chemical etching, radicals diffuse and adsorb randomly onto the surface, and etching occurs only where they meet incident ions. At 2 nm, particle scarcity raises the chance of spatial mismatch between radical and ion, which directly produces inconsistent line widths [9].

Together, these limitations degrade yield and stall further scaling.

1.2 Calculations in Modern Physics

Before shifting our perspective, we must examine our current standpoint. It is remarkable that modern physics still begins with Newton's mathematical structure [10, 11]. This implies that classical mechanics, constructed centuries ago, still profoundly shape our scientific reasoning [12].

Newtonian mechanics gave humanity the grand vision of a "deterministic universe." The continuous real number system he constructed through calculus —with breathtaking elegance—has reigned as the absolute language of science for centuries. The first crack appeared in 1900, when Max Planck introduced his constant (h) to resolve the blackbody radiation problem [13, 14]. This revealed that the universe might not be an infinitely divisible continuum, but a discrete realm governed by minimum units of energy. This marked the dawn of quantum mechanics.

Quantum Field Theory (QFT), the relativistic extension of quantum mechanics, can be called one of humanity's greatest intellectual achievements. QFT serves as the

foundational pillar of the *Standard Model* —the pinnacle of modern physics— and has achieved unparalleled success in describing the fundamental interactions of nature. In particular, calculations of the electron's anomalous magnetic moment $(g-2)$ agree with experiment to ten significant figures —an accuracy surpassing any other tool humanity has possessed [15, 16]. Yet this success rests on probability amplitudes. However precise its predictions, quantum field theory describes the world in fundamentally statistical terms, and *Heisenberg's uncertainty principle* ($\Delta x \Delta p \geq \hbar/2$) stands as the emblem of that irreducible probability [2][17, 18].

The point to note is that all these calculations still rest on a deterministic mathematical language:*the real number system*. This is why we began this section with Newton. We still compute on the calculus he invented, and every such computation inherits the *Archimedean property* —the system in which $3+4=7$ holds true [3].

Here a fundamental question arises. Planck's constant already told us the world may not be entirely continuous. *Why, then, does the mathematical language we use to describe that world still remain strictly continuous?*

1.2.1 Discrete World in a Continuous Language

If the world's true nature differs from our assumed continuum, why do we still cling to it? Our *macroscopic* world runs smoothly on Newton's continuum, which presumes infinite precision. The *microscopic* world no longer seems fully expressible in real numbers alone.

In his later years, Newton famously compared himself to a boy on the sea-shore, finding a smooth pebble while the great ocean of truth lay undiscovered before him [4]. The remark is usually read as intellectual humility. However, this study wishes to read it differently. The pebble he found was a shining stone, polished smooth by centuries of weathering. Yet, unlike Newton, who remained on the shore, we have since developed the vision to look further and deeper. Examined closely with that vision, his pebble reveals tiny scratches hidden beneath its *seemingly* smooth surface. Newton explored the truth by picking up "smooth pebbles" on the sand. We now stand in a position where we must explain the rough textures on the surface of those very pebbles. It is perhaps a natural consequence that errors arise when we attempt to calculate these rough surfaces using the perfectly smooth real number system.

This is only a metaphor, but it outlines the mismatch between our tool and our subject. The discrepancy cannot be resolved by more sophisticated techniques or greater computing power; it demands that we doubt the *very language* we have chosen to describe the world.

1.3 Philosophical Approach

Introducing philosophy into a study of plasma and the amplituhedron may seem unconventional. Readers in physics, a field prioritizing quantitative rigor, might dismiss philosophical speculation as unverifiable rhetoric; this approach admittedly diverges from the mainstream. For the following three reasons, however, this study

[2]Whether this bound can itself be reduced is examined in Section 1.4.3.

[3]This point will be discussed in greater detail in Section 1.4.2.

[4]This anecdote originates from a later biography (Brewster, 1855), and the exact phrasing varies slightly across different editions.

argues that approaching the current problem philosophically is not merely permissible but actively necessary.

First, philosophy is not a sub-discipline of science but its origin [5]. Today we study science, mathematics, and philosophy as separate disciplines, yet this separation is relatively recent. In ancient Greece, where organized academia truly emerged, these three remained undivided [12], coexisting within a single framework that sought the ultimate truth of the world; philosophy was the study of the world itself, and science and mathematics were cultivated within it. This is no mere historical footnote, for the very names of our disciplines testify to their origins: the word "physics" originates from Aristotle's work *Physica*(Φυσική) [19], and the most fundamental unit of modern physics, the "atom," derives from the Greek *atomon*(ἄτομον), meaning "indivisible" [20]. Even more remarkably, the concept emerged not from experiment but from the speculation of a philosopher. Democritus asked whether matter could be divided without end, and concluded it could not—reaching this through logical deduction alone, in an era devoid of empirical tools. The aim here is direction, not superiority. Science has evolved toward measuring *how* the world operates, while philosophy has deepened by questioning the *why* and *what* of our premises; the two do not compete but divide the labor, both aiming at an understanding of the world. Neither discipline can therefore claim superior academic depth—they rightfully stand as equals. Today, when measurement precision has hit physical limits such as the 2 nm node and the quantum scale, what is required is not finer measurement but the philosophical direction of doubting our foundational premises.

Second, whenever physics faces a limit, it seeks breakthroughs by descending into its mathematical and logical foundations—this is not an external observation but a precedent from within physics itself. At the dawn of quantum mechanics, physicists realized that microscopic phenomena defy classical logic itself: the noncommutativity of observables, which forbids determining position and momentum simultaneously, breaks the classical framework of "A or not-A." John von Neumann refused to remain confined to existing physical perspectives. With Garrett Birkhoff, he created a new logical system for quantum phenomena —"*quantum logic*"—proposing that even the distributive law of classical propositional logic might not hold universally [21]. The crucial point is that a physicist created a new logic—not just new physics—to explain the world. The logic of physics is never immutable; our taken-for-granted premises can change. When physics meets a limit, it dives into its mathematical and logical roots to uncover flawed premises, and this study's doubt of the describing *language* stands firmly in that tradition.

Third, information and logic have already entered the foundational ontology of physics. If introducing philosophical thought into physics seems unfamiliar, one must examine the changes already occurring within the physics community, where the ontological foundations are being rewritten in the name of information and logic. Through his "*It from Bit*" doctrine, John A. Wheeler argued that all physical reality fundamentally arises from information—the binary unit of "yes or no"— and foresaw that physics would soon be expressed in the *language of information* [22]. Today this view, placing information and logic beneath matter itself, is the subject of serious discourse. The logic now entering physics originally branched off from philosophy, just as science did. Since logic already lives in the foundations of physics, admitting philosophical reasoning—the mother of logic—is no stranger a step. If physics accepts information as a legitimate way to describe the world, it is natural

[5] For ease, in this section, Mathematical Physics will be called Science, and the two terms will carry the same meaning

to ask philosophical questions about the language carrying that information, and ultimately about the *language system itself*. This is not an awkward intrusion but the natural next step.

For these three reasons, this study considers it entirely justified to view the current problem from a philosophical perspective. There is a philosopher who honed the proposition "*the limits of my language mean the limits of my world*" more sharply than anyone else; in the next section, we will lean heavily on his thought to re-examine the mismatch between our tool and our subject.

1.3.1 Ludwig Wittgenstein: Philosophy of Language

This study borrows the viewpoint of a philosopher who rigorously organized the relationship between language and the world: *Ludwig Wittgenstein*. The question of this study begins from his thought, specifically his proposition that "*language defines the limits of the world*."

The reason for adopting Wittgenstein's perspective is that his philosophy of language does not linger in ambiguous speculation; it is closer to an analytical process that rigorously examines propositions and logic. His early masterpiece, the *Tractatus Logico-Philosophicus* [6], breaks from conventional philosophical texts: instead of continuous prose, each proposition is numbered hierarchically (e.g., 1, 1.1, 1.11), orming an axiomatic structure where each statement supports a higher one [23, 24, 25]. Reading it feels less like a philosophical treatise than an analytical manual of interlocked propositions.For this rigor the Vienna Circle, the logical positivists of his time, embraced the *Tractatus* as a model for treating philosophy with scientific precision [26, 27]. This analytical nature—scrutinizing the logic of propositions rather than indulging in vague speculation—makes his insight well suited to the current problems in physics.

A clear distinction is needed. Wittgenstein's philosophy divides into an early period, represented by the *Tractatus*, and a later one, the *Philosophical Investigations*, in which he critiqued and largely retracted his early stances. [7]. The framework adopted here is strictly that of the early *Tractatus*; his later views fall outside this discussion. What follows is an intentional borrowing of specific propositions, not an interpretation of his entire philosophy.
The central proposition adopted by this study is Proposition 5.6 of the *Tractatus*:

> ***"Die Grenzen meiner Sprache bedeuten die Grenzen meiner Welt."***
> (The limits of my language mean the limits of my world.)
> *- Ludwig Wittgenstein*[23]

This proposition states that the scope within we perceive and describe the world is defined by the scope of our language. We can treat as *the world* only what our language can hold; whatever language cannot reach lies outside our describable world.

Wittgenstein's original intent must be established first. He drew a boundary around cognitive limits: we cannot speak of what lies beyond language, and so must remain silent. This aligns with the famous closing proposition of the *Tractatus* — *Whereof one cannot speak, thereof one must be silent.* For Wittgenstein, the limits of language and the world were given, insurmountable boundaries.

[6] for ease, in this section, *Tractatus Logico-Philosophicus* will be written as *Tractatus*

[7] While this shift of perspective is closely tied to Wittgenstein's biography, a detailed explanation is omitted. This study is not a biographical study, and further elaboration falls outside its scope.

This study, however, intends to read the proposition from a different angle. The "*language*" Wittgenstein analyzed was no singular natural tongue, but the logical form permeating all language—the correspondence between propositions and states of affairs. His insight was in this sense universal. Yet the *Tractatus* focused on defining *how* language depicts the world and drawing the limits of *what* can be said within it; the possibility that the language system *itself* could be replaced was not his concern. Language, however, is not fixed: it is constructed and refined by humans, and it can be expanded or replaced by other systems. If the limits of the world are defined by the limits of language, and if that language can be replaced, then the boundaries of the world we can describe are *not* permanently fixed to the world itself, but shift according to the language we select.

Interestingly, this idea is not entirely disconnected from Wittgenstein's thought. In Proposition 6.43 [8], he suggested that the limits of the world—though not the facts themselves—can alter. He attributed the driver of that change to the "will"; this study diverges by attributing it to the replacement of the *describing* language. This study therefore reads Proposition 5.6 not as a fixed *declaration* of limits, but as a proposition suggesting their *variability*. The reading diverges from Wittgenstein's intent, yet it denies none of his insight —it takes one step further upon it.

This interpretation directly intersects with our current situation. The wall we have met below the 2 nm scale may not be a limit of the world itself, but of the *language* we use to describe it. If so, changing the language might relocate that wall, or reveal a side of it we could not see before.

Proposition 5.6 must now be brought into the context of this study. The "*language*"Wittgenstein referred to was everyday language and logical propositions, but he did not restrict language to one kind. This study extends it to mathematical language—the number system we use to describe the world quantitatively. If the language physics uses to describe the world is mathematics, Proposition 5.6 can be translated as follows: "*The limits of the mathematical language we use mean the limits of the physical world we can quantitatively describe.*" The next question follows: *what are the limits of the mathematical language—the real number system—that we currently use to describe the world?*

1.4 Limits of Current Mathematical Language

We found that the limits of our mathematical language mean the limits of the physical world we can quantitatively describe. What, then, are the limits of the mathematical language —*the real number system*— that we currently use to describe the world? In this section, the study proposes a main sentence:

"The real number system reveals its limits when describing the microscopic world, and a portion of what we have called *probability* may originate from this very limitation."

The "probability" mentioned here is not the everyday probability based on the number of possible cases in the macroscopic world; it strictly refers to the probability analyzed as observational outcomes in the *microscopic realm*. This is not an *attempt* to deny or discard the real number system, but to pinpoint its limits and explore room for refinement.

[8] Original text of Proposition 6.43 (Ogden trans.[24]) trans.): "*If the good or bad exercise of the will alters the world, it can only alter the limits of the world, not the facts.*" This study adopts solely the insight that "the limits of the world can alter." The context of ethics and the 'will,' which Wittgenstein posited as the driver of this change, falls outside the scope of this discussion [24].

At the extreme limits of this precision, physics encountered the boundary of "probability." The behavior of the microscopic world can no longer be predicted *deterministically*; we can only describe it as existing within a specific range with a certain probability, and this probability has widely been accepted as the intrinsic nature of quantum mechanics.

Here, we must pay attention to a fundamental *mismatch*. Since the discovery of Planck's constant, it has been revealed that physical quantities—including energy—are quantized and take *discrete* values in the microscopic realm. This does not imply that our macroscopic world is discrete; it means, rather, that at the microscopic scale physical quantities are no longer infinitely and smoothly continuous. And yet these discrete physical quantities are still described using the language of the continuous real number system—a system that *presumes* infinite precision.

This is exactly where the mismatch occurs: the structure of the subject being described and the structure of the language describing it are misaligned. Where, then, does this discrepancy go during the calculation process? Does it vanish without a trace, or does it manifest in the final results in some form? And is the premise that the continuous real number system presumes infinite precision truly an unquestionable axiom? [9]

1.4.1 Probability as Mapping Noise

This study proposes the following hypothesis: at least a portion of what has thus far been accepted as "*stochasticity*" may actually be information loss—a form of noise—arising during the mapping of a discrete reality onto the continuous language of real numbers.

This proposition is not unfamiliar. Translating an object into a representational system of different structure incurs inevitable information loss—a well-established principle in signal processing. Just as distortion emerges when continuous values are mapped into finite representations, a reverse mismatch can occur when a discrete reality is mapped onto a continuum of infinite precision [10]. In the microscopic world, this discrepancy may amplify to a non-negligible degree, unlike at macroscopic scales; considering the ultra-scaled dimension of 2 nm, this possibility is acceptable.

However, it must be clarified what this hypothesis asserts and what it does not. This study does *not* claim that the entirety of quantum-mechanical probability is merely linguistic noise. This hypothesis also differs fundamentally from classical *local hidden variable theories*, which assume that particle properties are strictly determined prior to measurement. To coin a term, the study focuses not on a "hidden variable" but on a "hidden *language*"—namely, our chosen number system.

[9] This final question—is it unquestionable axiom?— will be discussed in greater detail in Section Section 1.4.2.

[10] The inevitability of information loss when transferring signals to a representational system with a finite or differing structure is an established principle in information theory. Quantization noise, which occurs when continuous-amplitude signals are represented by finite bits, is a prime example. The fundamental limits of such loss are formalized in Shannon's *rate-distortion theory* (definition of the rate-distortion function $R(d)$)[28]. However, while these theorems deal with the conversion from "continuous to discrete," the study analogously proposes that a similar loss can occur in the reverse direction—mapping a discrete reality onto a continuous real-number language [29, Chapter 3.1.3].

Upon encountering this hypothesis, some readers might immediately recall Bell's inequality[11][30]. This hypothesis does not contradict Bell's theorem. The violation of Bell's inequality rules out the conjunction of locality and realism; it is not a theorem concerning the language we select to describe the world. That "choice of *language*" is precisely what this study brings into focus, so the hypothesis remains outside the jurisdiction of Bell's inequality. To reiterate, this is not presented as a proven fact, but strictly as a *hypothetical* starting point for this study's examination.

1.4.2 Reason of Incomplete Description

To prevent the preceding hypothesis from remaining mere intuition, the study must demonstrate exactly which property of the real number system induces this mismatch. The crucial clue can be found in the Archimedean property of the real number system[12]:

$$\forall x, y \in \mathbb{R}, \quad x > 0 \implies \exists n \in \mathbb{N} \quad \text{such that} \quad nx > y \tag{1.1}$$

The Archimedean property dictates that for any positive number x, adding it to itself a sufficient number of times will eventually exceed any large number y. As a direct consequence, the real number system cannot contain any number greater than zero yet smaller than every arbitrary positive number—a non-zero infinitesimal. The real numbers simply lack the linguistic resources to contain infinitesimals.

This is directly tied to the concept of limits in the real numbers. Continuity and limits are rigorously defined by the ϵ-δ definition, which requires a specific condition to hold for an arbitrarily small positive number ϵ[13]:

$$\lim f(x) = L \iff \forall \varepsilon > 0, \quad \exists \delta > 0 \quad \text{such that} \quad 0 < |x - c| < \delta \implies |f(x) - L| < \varepsilon \tag{1.2}$$

The core issue here is the premise that ϵ can be driven infinitely close to zero. This premise contains no mathematical flaw: the ϵ-δ definition defines limits without ambiguity, and completeness is the axiom that makes the real numbers real. This study points to no contradiction within the real number system itself.

The problem arises when mapping this mathematical system onto physical reality. As discussed previously, the discrete microscopic world, invoked through Planck's constant, implies the existence of a strict physical lower bound. In a reality where such a lower bound exists, there is no physical counterpart to the mathematical process of driving ϵ infinitely toward zero. An infinitesimal limit is thus perfectly defined within the real number system, yet lacks a corresponding physical entity in the microscopic reality it attempts to describe.

[11] Bell's inequality establishes a strict upper bound on the measurement correlations of two entangled particles, provided a theory simultaneously assumes locality (the absence of faster-than-light influence) and realism (properties are determined prior to measurement). *Quantum mechanics* predicts a violation of this bound, and a fact firmly corroborated by numerous experiments. Thus, what Bell's theorem actually rules out is any local hidden-variable theory; it does not dictate the mathematical language—*the number system*— we select to describe reality. Because the study's hypothesis does not posit a hidden "variable" determined prior to measurement but rather questions the very layer of the descriptive "language" itself, it falls entirely outside the category excluded by Bell's theorem.

[12] The completeness (least upper bound property) of the real number system strictly precludes the existence of non-zero infinitesimals. This Archimedean property can be found in [31, Theorem 1.20]. Also can be found in [32, Theorem 2.13].

[13] For the ϵ-δ definition of limits and continuity, see [31, Definition 4.5]. Also can be found in [32, Definition 5.5].

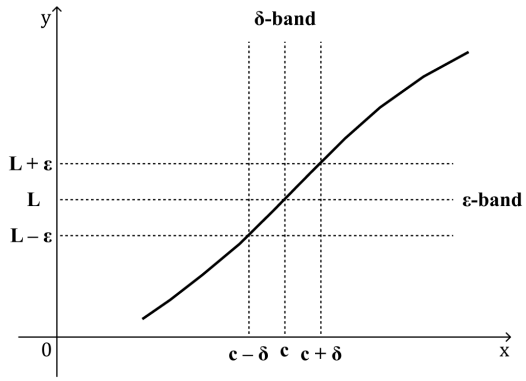


FIGURE 1.1: graph of the epsilon-delta definition

The limitation, then, lies not within the real number system but in the correspondence *mapping*— the attempt to describe a structure with a physical lower bound using an Archimedean language that cannot contain infinitesimals. This is the true mathematical identity of the mismatch proposed in the hypothesis of Section 1.4.1. The study will now trace how this mismatch is observationally manifested to us in the form of "probability," guided by a perfectly simple arithmetic proposition.

1.4.2.1 Limitation in Representation, not Measurement

Before proceeding with the main argument, a crucial distinction must be established: the mismatch this study speaks of is not an error arising from the physical act of measurement.

Heisenberg, who birthed quantum mechanics, dismantled the very premise of classical determinism in his 1927 paper. In the rigorous formulation of causality—*if we know the present exactly, we can calculate the future*—it is not the conclusion that is false, but the premise [14][17]. We inherently cannot know the present in precise detail; thus the future emerges only as a selection among possibilities. This is the limit of measurement—a fundamental constraint intrinsic to the physical act of observation.

This study, however, focuses on a different layer. Even if the "exact measurement" Heisenberg deemed impossible were somehow achievable —even if we could perfectly observe a value—another mismatch intervenes the moment we *map* that value onto the real number system, which presumes infinite precision. While Heisenberg questioned the premise of *measurement*, this study temporarily grants that premise only to question the subsequent step: transcribing the measured value into the language of real numbers. This is not a limit of *observation* but a limit of *representation*; not a problem of *measurement* but a problem of *language*. This is why this study previously invoked Wittgenstein to discuss the "*limits of language*."

1.4.2.2 A Thought Experiment: 3 + 4 = 7

Let us trace this limit of representation through the simplest arithmetic proposition. The proposition $3 + 4 = 7$ holds evidently true; at least within the macroscopic reality we can observe, there is nothing flawed in it. The situation changes, however, when we magnify these numbers down to microscopic precision—here referring to the microscopic realm expressed in nanometer units, such as 2 nm.

[14]This statement appears in the concluding section of the paper. An English translation is available in [18].

First, a distinction must be drawn between a mathematical idealized point and a physical resolution limit. Mathematically, we can pinpoint a location on a number line exactly, without dimension; the idealized point "3" possesses no dimensions. This is the most fundamental idealization mathematics has maintained since Euclid defined a point in his *Elements* as "that which has no part" [15][33]. It implies that we can, *ideally*, plot this point with whatever precision we desire. Yet such idealization harbors an inherent gap from physical reality: no matter how precisely one attempts to plot a point—even with the most precise instruments available—a physically marked point will *always* occupy some finite size. The mark merely grows smaller, representing the idealized point ever more closely, but it can never become the "*partless*" point itself. In other words, a dimensionless point is a physically unattainable object, belonging strictly to the realm of *mathematical idealization*.

For a long time, this gap posed no problem. Just as the idealization in classical mechanics—which treats an object's mass as a concentrated single center of mass—has always been valid, the finite size of a point in the macroscopic world does not affect any calculation. The focus here, however, is precisely on the realm where this gap can no longer be ignored: the microscopic world. Physically, no representation can resolve a region smaller than the physical lower bound—a presumed minimum unit denoted here as m for convenience [16]. Therefore, when we place the mathematical idealized point designated "3" onto the real number system, it smears into a region of indeterminacy on the scale of m in physical reality. Thus, in the subsequent discussions of this chapter, the expression "to plot a point" refers to this *mathematical idealization* and does not imply physical attainability.

Building upon this distinction, a question arises: is the number we write down and put into our calculations as "3" truly a flawless "3," devoid of even a microscopic discrepancy on the scale of m? In the macroscopic world, this discrepancy causes no issues. We have, in effect, been calculating without perceiving the difference among $3+m$, $3+2m$, and so forth, with no anomalies; m exerts no influence on macroscopic outcomes. Perhaps for this reason, as seen earlier, there is no corresponding place for this m in the real number system: due to the Archimedean property, an infinitesimal-scale unit like m is absorbed into the limit and vanishes. What is being scrutinized here is precisely that absorbed, ignored portion—the part erased for being too minuscule.

The problem surfaces in the microscopic realm, where this discrepancy exerts actual influence. Let us express this in the form of error propagation. If "3" and "4" are represented as real numbers, each is accompanied by an indeterminacy on the scale of m, which can be written as:

$$\{3 \pm am\} + \{4 \pm bm\} = \{7 \pm cm\} \tag{1.3}$$

(where a, b, and c are coefficients representing the magnitude of the discrepancy).

From a macroscopic perspective, no matter how large a and b become, the result $3+4=7$ remains unshaken. From a microscopic perspective where m actively operates, however, fixing the resulting discrepancy c to a single value becomes exceedingly difficult: the range of values a and b can take during representation is broad, and consequently their sum c scatters across a multitude of possible cases.

[15] The tension between this idealization—viewing a point as a dimensionless object—and the fact that a physically marked point always possesses a finite size has been recognized since antiquity.

[16] Denoting the unit of discrepancy as m is inspired by the concept of a monad—the infinitesimal neighborhood of a point—discussed in non-standard analysis. The concept of the monad itself will be addressed in detail in Section 2.3, when discussing alternative number systems.

What form, then, does this scattering take? Here an assumption is introduced: the discrepancies a and b are not distributed in a completely disordered manner, but are concentrated around the *true value*. The rationale is as follows. The laws and constants we have reached through experiments are the values closest to reality, regardless of the mathematical language we use [17]. Furthermore, the procedure of theoretical calculation—performed by substituting measured values into real numbers—has historically been refined to align ever more closely with these realistic values. As mentioned earlier in Section 1.2.1, the extreme precision humanity has achieved in calculating the anomalous magnetic moment of the electron is evidence of this. This implies that our representational procedure of transcribing values into real numbers, even while unaware of the existence of m, is already on a trajectory that minimizes the discrepancy around the true value. It is therefore natural to view the discrepancy as having a distribution that grows increasingly sparse the further it moves from the true value.

The problem is that we do not know the exact shape of that distribution. Given only minimal information—a center (the true value) and a finite scattering (variance)—what distribution expresses this with the fewest additional assumptions? In information theory, the answer is provided by the *maximum entropy principle* [34]: under a given mean and variance, the distribution that maximizes entropy—the most unbiased distribution—is the normal distribution [18]. In accordance with this principle, this study adopts the normal distribution as the form of the discrepancy caused by m. This is not a claim that it is the true distribution, but a provisional toy model established with the least bias at our current level of precision.

If precision improves to the point where this discrepancy becomes measurable, its distribution would sharpen at the center into a narrow, nearly straight-line form—precision and the sharpening of this distribution being one and the same. Observed at present, it would presumably manifest in a *nearly* linear form.

With each input's discrepancy now modeled as a normal distribution, we can return to the thought experiment. If the discrepancies in representing "3" and "4" each follow a normal distribution, and these two discrepancies are independent of each other, what distribution characterizes the discrepancy of their sum, "7"? The sum of two independent normally distributed variables remains normally distributed [19].

In other words, the output of a calculation comprising addition is likewise scattered in the form of a normal distribution.

One point requires precise clarification here. This "preservation of normality" is a property strictly limited to addition and subtraction; for operations such as multiplication or division, the resulting distribution no longer remains normal but assumes a far more complex morphology [36, Chapter 6.3, 6.7]. Yet what is essential is not that the result retains the specific form of a normal distribution. The crux is that the representational discrepancy at the input —whatever its specific distributional

[17] These laws and constants are conclusions reached through literal "real-world experiments." Therefore, denying them is equivalent to denying experimental reality itself, which is a problem on a fundamentally different layer from the uncertainty originating from measurement limits, and thus does not affect the thesis of this study.

[18] It is a standard result in information theory that among continuous distributions with a given mean and variance, the one that maximizes differential entropy is the normal (Gaussian) distribution.[35, Chapter 8, 12]

[19] The sum of two independent normal random variables follows a normal distribution (where the mean and variance are the sums of the respective means and variances), and this property is predicated strictly on independence. Conversely, the distribution of the product or ratio of normal random variables is generally not normal and takes on a more complex form. For discussions on the distributions of sums, products, and ratios of random variables, see [36, Chapter 6.3, 6.7].

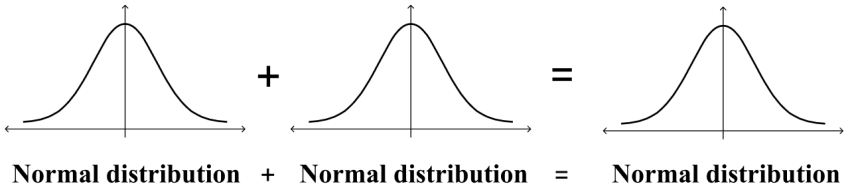


FIGURE 1.2: Simplified graph model of the sum of two independent normal distributions.

form—propagates through the calculation to manifest as a distribution in the final result. The normal form here merely happens to be preserved. For other operations the morphology may alter, but the discrepancy still disperses the outcome into a *distribution* rather than a single value. Consequently, as values carrying such discrepancies are recursively subjected to calculation, the ultimate outcome is materialized not as a deterministic scalar, but as a *distribution*.

This is precisely the focal point of this study. The "stochasticity" encountered in microscopic calculations might well be the manifestation of these representational m-discrepancies, cumulatively structured as a distribution in the final result. In other words, it is entirely possible that a fraction of the probability we have long accepted as the *inherent nature* of quantum mechanics is, in truth, merely the *statistical footprint of the discrepancy* generated during the representational mapping of reality onto the language of real numbers.

At this juncture, an anticipated counterargument must be addressed:

"If a probabilistic model—the normal distribution—is assumed to explain probability, does this not constitute *circular reasoning*, explaining probability via probability?" This is a misconception. What enters at the input is the *representational* discrepancy between language and reality; wemerges at the output is the *observable* stochasticity. The latter follows logically from the former; it does not recycle the same premise. The outcome's emergence as a probability distribution does not refute the hypothesis but aligns exactly with its prediction.

All of the above is not presented as proven fact, but remains a provisional *toy model*. Nevertheless, this hypothesis leads to at least one testable inquiry:

"If a fraction of this probability stems from the language itself, could this uncertainty be minimized if we were to change the descriptive language?"

1.4.3 Reducing Uncertainty

The preceding section reached a hypothesis: a *fraction* of the probability we have accepted as the essence of quantum mechanics might actually be a discrepancy originating in the transcription of reality into the language of real numbers. If this discrepancy stems from language, could it be reduced by changing the language? One prior point must be addressed first: is what we call "*uncertainty*" truly an untouchable, singular, and immutable law? Recent revelation suggests it is *not*—one formulation of the uncertainty principle has already been superseded by a more precise form.

The original *error-disturbance relation* proposed by Heisenberg in 1927 is expressed as an inequality between the error of a measurement $\epsilon(q)$ and the resulting disturbance $\eta(p)$ [17]:

$$\epsilon(q)\eta(p) \geq \frac{\hbar}{2} \tag{1.4}$$

Long accepted as the standard expression of the principle, this relation was shown by Masanao Ozawa in 2003 to hold only in specific cases, not universally. The universally valid inequality he proposed is [37]:

$$\epsilon(q)\eta(p) + \epsilon(q)\sigma(p) + \sigma(q)\eta(p) \geq \frac{\hbar}{2} \tag{1.5}$$

Equation (1.5) adds two terms—$\epsilon(q)\sigma(p)$ and $\sigma(q)\eta(p)$—to the left side of (1.4), leaving the right side unchanged. Because of these terms, the product $\epsilon(q)\eta(p)$ can itself fall below $\frac{\hbar}{2}$: Heisenberg's original inequality is thus subsumed as a special case of the more general (1.5) and can be violated under certain conditions. This result was subsequently confirmed through experiments such as neutron spin measurements [38].

A precise distinction is needed between what has been revised and what remains intact. What Ozawa corrected is strictly the error-disturbance relation of the *measurement process*. The *Kennard-Robertson inequality* [39, 40] [20]:

$$\sigma(q)\sigma(p) \geq \frac{\hbar}{2} \tag{1.6}$$

which describes the intrinsic variance the quantum state possesses prior to any measurement, remains valid and independent of that process. It is therefore inaccurate to say the uncertainty principle has been discarded: only the error-disturbance formulation (1.4) was dismantled, while the intrinsic variance of the state (1.6) stands unaltered.

Yet it is precisely within this distinction that the *possibility* explored in this study opens up. The uncertainty principle has proven to be not a singular, immutable law but a structure with room for refinement depending on its formulation. Just as Heisenberg's formulation could be superseded by a more precise one, the possibility that the description of uncertainty might likewise shift when we change the language of description is *not*, in principle, closed.

1.4.4 Renormalization Problem

Thus far, this study has centered on the limits of the real number language and the possibility of its supplementation. There exists, however, one highly orthodox and formidable counterargument against this reasoning, emerging from *renormalization*—one of the most robust achievements of modern physics [21][41, 42, Chapter 12]. This study does not evade this issue, but intends to confront it head-on.

The crux of the counterargument is as follows. According to the renormalization group (RG) framework established by Kenneth Wilson, the detailed physics of the short-distance (high-energy) regime influences low-energy (macroscopic) observations only through a few effective parameters, while the minute details are absorbed and washed away [41, Chapter 12]. Furthermore, the *Appelquist–Carazzone decoupling theorem* shows that the effects of heavy (high-energy) degrees of freedom, of

[20] While Heisenberg proved the relation for the special case of Gaussian states, Kennard rigorously proved the modern inequality for the standard deviations of position and momentum in 1927, which Robertson later generalized to arbitrary observables in 1929. Thus, it is most accurate to refer to (1.6) as the Kennard-Robertson inequality.

[21] The historical flow regarding the renormalization problem will be addressed in detail in Section 2.1.2 of the following chapter.

mass M, are suppressed by powers of $1/M$. They effectively decouple from low-energy physics [43][41, Chapter 12]. From this, the counterargument follows: the m-scale discrepancy spoken of here would likewise be *washed out* in macroscopic calculations by this *decoupling*, which is precisely why macroscopic predictions remain accurate even when the microscopic realm is treated with the real number system. If so, is a new mathematical language not unnecessary in the first place?

This counterargument is powerful, and this study acknowledges its weight. Nevertheless, from *two perspectives*, it fails to dismantle the hypothesis proposed here.

First, the very fact that renormalization is necessary suggests that a discrepancy arises when describing the microscopic realm with a real continuum. In quantum field theory, infinities—ultraviolet (UV) divergences—occur, and only by treating them through a renormalization procedure can one obtain finite predictions [41, Chapter 10]. This divergence originates in the continuum assumption itself, which integrates down to arbitrarily short distances [41, Chapter 10]. Introducing a cutoff to control the divergence can thus be interpreted as endowing the continuum with a kind of minimum interval.

Interpretations of the status of this divergence vary. The dominant view today holds that UV divergence is not an actual physical problem but a sign marking the limits of the perturbative framework. Quantum field theory, on this view, is an approximate effective theory, not valid up to arbitrarily high energies [44, Chapter 5]. Regardless of interpretation, however, one thing remains common: descriptions based on the real continuum do not apply perfectly in the microscopic realm of arbitrarily short distances, demanding a cutoff—a minimum interval—in some form. This is exactly the point this study focuses on.

This prescription is not unique to this study. Lattice quantum field theory regularizes divergences by approximating spacetime as a discrete lattice with spacing ℓ, an established computational technique [45]. Its approach, however, differs decisively. The spacing ℓ is a regularization tool, removed in the continuum limit ($\ell \to 0$). It discretizes spacetime but leaves the underlying real number system intact. The m in this study, by contrast, is postulated not as a *temporary scaffold* to be removed but as a *structure of reality*. It points toward replacing the number system itself, not discretizing spacetime. Where lattice QFT addresses microscopic problems within the language of real numbers, this study operates on a different layer by questioning that language.

It is also worth noting that renormalization was *never* smoothly accepted from the start. P. A. M. Dirac, who laid the foundations of quantum electrodynamics (QED), rejected it to the end, dismissing it as "not sensible mathematics"—one should disregard a quantity because it is small, he insisted, not neglect it merely because it is infinitely large and unwanted [22]. Even R. P. Feynman, who helped establish the theory, later called renormalization a "dippy process." Its necessity, he remarked, had kept physicists from proving quantum electrodynamics to be mathematically self-consistent. [23].

Their criticism does not deny the empirical success of renormalization, which is the most precise theoretical tool humanity possesses and whose predictive power is beyond question [15, 16]. Yet the fact that even the greatest physicists who built it felt the procedure to be *a kind of stopgap* demonstrates that describing the microscopic realm in the language of the real continuum is not entirely natural. Renormalization

[22] P. A. M. Dirac, 1975 lecture; the same criticism is reiterated in his final paper. [46, Page 194-198]
[23] R. P. Feynman's criticism is in his book.[47, Page 128]

is a tool that sophisticatedly manages that discomfort, and its very necessity serves as circumstantial evidence for the discrepancy this study raises.

Second, the *decoupling theorem* operates on a different layer from this study—a more fundamental response. What the theorem addresses is strictly the relationship between physical degrees of freedom: whether high-energy physics contributes to low-energy observations [41, Chapter 12]. The m-discrepancy discussed here, however, is not a physical degree of freedom but a linguistic error, one that arises when reality is represented as real numbers. The renormalization group, moreover, operates on the premise of the real continuum to begin with. Decoupling thus describes the separation between scales *within* the language of real numbers, while the representational limits of that language *itself* lie outside the theorem's scope. This is the same vein as the earlier point on Bell's inequality: a theorem about hidden variables is not a theorem about the choice of descriptive language.

When it absorbs microscopic details into a few effective parameters, the discrepancy has not vanished but *folded behind* measured constants such as the electron mass or charge[24]; what this study questions is precisely that folded portion, latent in the microscopic description though unrevealed in macroscopic predictions. Seen this way, decoupling does not dismantle the present hypothesis but upholds it in the *form of a theorem*: its guarantee that microscopic details do not affect macroscopic observations is exactly the claim that "no matter how large a and b become, the macroscopic result does not change" in the $3 + 4 = 7$ thought experiment of Section 1.4.2.2.

Finally, it is worth noting the core attitude of the effective field theory (EFT) upon which decoupling is founded. EFT treats physics beyond the cutoff—at shorter distances—as unknown, taking an agnostic stance by pushing it into a few effective constants [42, Chapter 12]. In other words, the framework of renormalization bypasses the sub-cutoff structure by constructing the theory so that there is no need to know it. Yet "having no need to know" is fundamentally different from "not existing." What this study intends to scrutinize is precisely the realm renormalization has set aside as unnecessary to know—the fine structure submerged below the cutoff when representing reality as real numbers. What this study seeks to describe is not the general macroscopic world, but the very layer that renormalization chose to bypass and cease guaranteeing.

In synthesis, renormalization is not a refutation of this study but illuminates it from two directions: through its very necessity, it evidences that a discrepancy arises when the microscopic realm is described with a real continuum; through decoupling, it explains why that discrepancy is isolated from the macroscopic realm. Rather than removing the discrepancy *a posteriori*, as renormalization does, this study attempts to find a language for the microscopic realm in which less discrepancy occurs *a priori*. If renormalization has handled divergences by introducing a cutoff to the continuum, this study can be seen as an attempt to resolve a similar problem through a different language capable of harboring infinitesimals.

We have now examined the limits of the real number language, the possibility of its supplementation, and the most formidable counterargument against it. Renormalization does not close the question but sharpens it. The discrepancy that renormalization manages so successfully is the same as this study seeks to describe. What

[24]The process by which high-energy information enters measured effective constants (renormalized mass and charge) is typically termed "absorption" in renormalization. The phrase "folded behind" is used here to emphasize that this absorption conceals information rather than erasing it. [41, 42, Chapter 12]

that description requires is a language that can hold the infinitesimals the real numbers cannot.

1.4.5 Direction for Alternative Mathematical Languages

This study has argued that the real number system, owing to its Archimedean property, cannot contain infinitesimals, and that this limitation can manifest as a discrepancy in the representation of the microscopic realm. A final question naturally follows: what language can hold the infinitesimals that the real numbers cannot?

This study locates that possibility in number systems that reject the Archimedean property —*the non-Archimedean number systems*. Among these, the focus falls on two distinct yet complementary languages: *non-standard analysis* and the p-adic numbers.

The first is non-standard analysis [25]. This is where the problem-awareness of this study originated, and the foundational language for reasoning about the discrepancy. Non-standard analysis restores infinitesimals not as vague intuitions but as rigorously defined objects. Through the concept of a *monad*—the infinitesimal neighborhood of a point—it directly describes the minute structure that the real numbers absorb into limits. The earlier denotation of the discrepancy unit as m was, in fact, derived from this very monad. In this sense, non-standard analysis is the mathematical language that most directly confronts the problem raised here.

The second is the p-adic number system [26]. Non-standard analysis is the starting point, but its hyperreal system is non-constructive, and so unsuitable for direct calculation. The simulation to come requires a computable number system, and here the p-adic numbers enters. While sharing the non-Archimedean property with non-standard analysis, p-adic numbers constitute a distinct, explicitly computable system. They are introduced not to replace or approximate non-standard analysis, but as an operational language for implementing and verifying the same non-Archimedean problem-consciousness.

1.5 Aim and Scope of the Study

The preceding sections converged on a single possibility: that a portion of what we call probability may originate not in nature, but in the real-number mathematical language we use to describe it. Left as a philosophical claim, this can be argued indefinitely but settled by nothing. The aim of this study is to move it onto ground where it can be examined by computation.

To do so, the study fixes a definite object and a definite pair of languages. The object is *the canonical form of the amplituhedron*, a quantity determined purely by the positive geometry of scattering and exact by construction. The languages are the *rational numbers* and the *p-adic numbers*, one *Archimedean* and one *non-Archimedean*. The study computes the canonical form over each and compares the two descriptions, asking whether the choice of number system leaves any trace on the object it describes.

[25] Non-standard analysis, founded by A. Robinson, is a theory that rigorously restores infinitesimals as well-defined mathematical objects [48]. For a pedagogical exposition on the concepts of infinitesimals and monads [49, Chapter 1]. Regarding the applications of non-standard analysis [50].

[26] The p-adic number system is a non-Archimedean framework constructed via the completion of the rational numbers with respect to the p-adic absolute value for a given prime p. For a comprehensive overview of the p-adic number system [51, Chap 1-4][52, Chapter 1]

The scope of this comparison must be stated plainly, since the claim it supports is narrow. This study does not attempt to locate the origin of quantum discreteness, nor to prove that probability is an artifact of the real numbers; a single idealized computation could not carry such a conclusion. It does not model plasma scattering directly, but treats an idealized amplitude as a *controlled surrogate* in which the arithmetic question can be posed exactly. And it does not claim that the p-adic description is more precise; it asks only whether that description differs in structure. What the study offers is not an answer to the guiding conjecture, but a *first test* of whether the conjecture can be posed as a matter of calculation at all.

Two things are needed before that test can begin: the object must be built, and the alternative language must be made precise. The next chapter supplies both, tracing the amplituhedron from its origins in quantum field theory and the p-adic numbers from their non-Archimedean foundations.

Chapter 2

Background of The Study

This chapter organizes the theoretical foundations upon which this study is grounded into two distinct branches. The first is the physical background (Section 2.1-2.2), which elucidates the physical targets of this study and the domain in which they manifest. The second is the mathematical language background (Section 2.3), which articulates the mathematical tools adopted to describe those targets. The rationale for this bifurcation lies in the different criteria by which each branch is narrowed down: the physical background converges along the axes of *scale* and *abstraction*, whereas the mathematical background converges strictly along the axis of structure.

Specifically, the discourse on the physical background (Section 2.1) is guided by two distinct trajectories. First, guided by the degree of abstraction, the discussion traces the descriptive evolution of scattering amplitudes, narrowing the focus from early quantum field theory to the state-of-the-art amplituhedron (Section 2.1.1-2.1.3). Second, guided by physical scale, it traces the domain where these amplitudes are physically realized, descending from the macroscopic stage of plasma to the microscopic process of e^--e^- Møller scattering (Section 2.2.1-2.2.4).

These two trajectories converge upon e^--e^- Møller scattering, which subsequently serves as the foundation for the next chapter, where the hypothesis of this study is computationally verified and critically examined through simulation.

The mathematical background (Section 2.3) occupies an equally paramount position in this dissertation. This study is not a historical treatise, however; the historical emergence of these mathematical theories, intertwined with the trajectory of physics, is addressed only briefly, in Section 2.1.2. Section 2.3 instead delineates *what these theories are* in concrete terms and *how they operate* as mathematical languages—the tools themselves, constructed to describe the world.

2.1 Physical Background of the Study

2.1.1 From Quantum Mechanics(QM) to Quantum Field Theory(QFT)

In the early 20th century, a series of phenomena that could not be explained by classical physics were observed in succession. The spectrum of black-body radiation, the photoelectric effect, and the discreteness of atomic spectra clashed with the classical premise that energy is exchanged continuously. Attempts to elucidate these phenomena converged into a coherent theory by the mid-1920s. In 1925, Werner Heisenberg established its foundational principles in the form of *matrix mechanics*; the following year, Erwin Schrödinger proposed a more tractable formulation known as *wave mechanics*. These two formulations were soon proven mathematically equivalent. The interpretive framework that came to accompany them, formulated by Bohr and Heisenberg in 1927, is known as the*Copenhagen Interpretation* [53].

In the framework of quantum mechanics (QM) thus established, the state of a particle is described by a wavefunction. According to Max Born's interpretation (1926), the absolute square of this wavefunction yields the probability of finding the particle at a specific position. Observables such as position, momentum, and energy are represented by operators acting upon the wavefunction, and the outcomes of measurements are predicted not as deterministic values but as probability distributions [54, Chapter 1.2]. Quantum mechanics achieved unprecedented success in describing the microscopic atomic world, opening a new chapter in physics. It could not, however, explain *everything*. Upon reaching the scattering processes addressed in this study, the framework of non-relativistic quantum mechanics confronts three fundamental limitations.

First, quantum mechanics postulates a system composed of a *fixed number of particles*. The wavefunction describes the probability amplitude for a definite number of particles, and that number is *conserved* during time evolution. In actual high-energy scattering, however, particles are created and annihilated. Photons are emitted or absorbed in collisions between electrons and atoms, or among particles themselves; if the collision energy is sufficient, particle-antiparticle pairs are generated [8, Chapter 3]. Photons themselves are not explicitly described within non-relativistic quantum mechanics; only the formalism of *field* theory captures their emission and absorption [55, Chapter 2, 3]. A framework predicated on particle-number conservation cannot describe such processes [56, Chapter 8].

Second, early attempts to introduce relativistic effects into quantum mechanics engendered the problem of *negative-energy solutions* [57][56, Chapter 8.2]. When P. A. M. Dirac proposed a relativistic wave equation for the electron (the Dirac equation) in 1928, it permitted not only positive-energy solutions but also arbitrarily low negative-energy ones. Under a single-particle interpretation, this implies inherent instability: an electron in a positive-energy state could cascade endlessly into lower negative-energy states, precluding any stable ground state. To resolve this, Dirac proposed in 1930 that the vacuum is in fact a *Dirac sea*, completely filled with electrons in negative-energy states, and that the Pauli exclusion principle prevents positive-energy electrons from falling into it [58][55, Chapter 3].

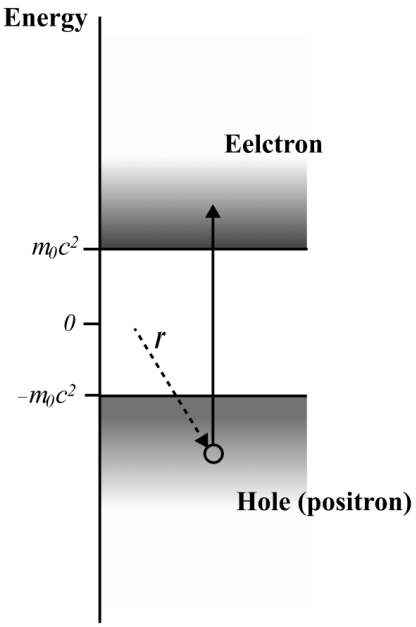


FIGURE 2.1: Simplified graph model of Dirac sea. Adaped from [56, Fig.8.1.].

In this conceptual diagram, a vacancy (hole) created in the negative-energy sea manifests as a particle possessing positive energy and opposite charge—that is, an *antiparticle* [56, Chapter 8.2.3.].

This reinterpretation was experimentally corroborated in 1932 when Carl Anderson discovered the *positron* in cosmic rays [59][56, Chapter 8.2.3.].

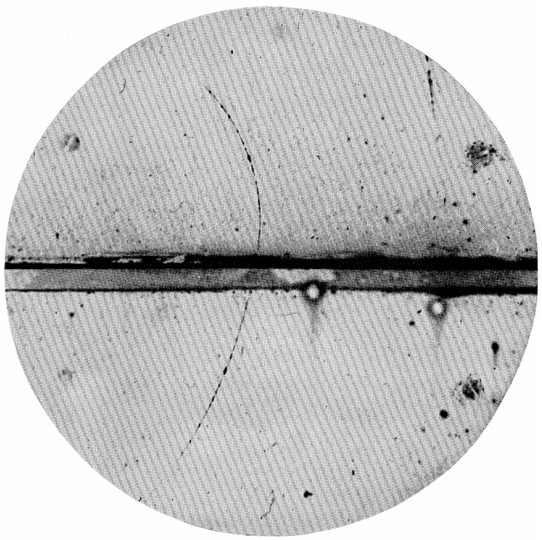


FIGURE 2.2: Carl Anderson's cloud chamber photograph of a positron track. Reproduced from [59, FIG. 1].

The reinterpretation carried a clear imperative: even the vacuum must be viewed as a many-body system filled with an infinite number of particles. What this demanded was the language of a *field*, in which particle number is not fixed.

Third, relativistic quantum mechanics raises profound issues regarding *causality* and *locality*. *Special relativity* demands that no signal propagate faster than light. Yet a direct calculation of the relativistic propagation amplitude of a single particle

reveals that even when two events are separated by a space-like interval—lying outside each other's light cones, unconnectable even by light—the amplitude remains exponentially small yet non-zero [55, Chapter 2.4]. For propagation from $\mathbf{x}$ to $\mathbf{y}$ this amplitude reads,

$$U(x \to y) \;=\; \langle \mathbf{y}|e^{-iHt}|\mathbf{x}\rangle \;\sim\; e^{-m\sqrt{r^2-t^2}}, \tag{2.1}$$

where $t = y^0 - x^0$ and $r = |\mathbf{x} - \mathbf{y}|$. A space-like interval means $r > t$, so the exponent stays real and the amplitude, however suppressed, never reaches zero.

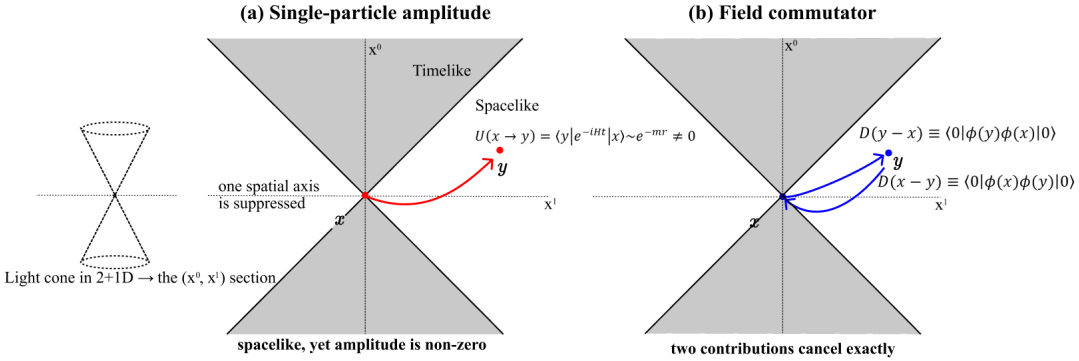


FIGURE 2.3: Space-like separation and the vanishing commutator. Two events at space-like separation, with one spatial dimension suppressed. Natural units ($\hbar = c = 1$) are used, and r denotes the spatial separation of the two events. (a) The single-particle propagation amplitude U between $\mathbf{x}$ and $\mathbf{y}$. (b) The two field-theoretic contributions $D(x-y)$ and $D(y-x)$. Based on the discussion in [55, Chapter 2.4].

This signifies that a single-particle description conflicts with relativistic causality. Quantum field theory resolves the problem by restructuring particle propagation in terms of field operators. What governs causality is then not the amplitude itself but the commutator of two field operators, built from the two-point function $D(x-y) \equiv \langle 0|\phi(x)\phi(y)|0\rangle$:

$$[\phi(x), \phi(y)] \;=\; D(x-y) - D(y-x) \;=\; 0, \qquad (x-y)^2 < 0. \tag{2.2}$$

The two terms—propagation in either direction, which for a complex field are the particle and the antiparticle contributions—cancel precisely over space-like intervals [1]. Consequently, local operators at different spacetime points commute outside the light cone, guaranteeing that no measurable signal can travel faster than light [55, Chapter 2.4].

It is intriguing, moreover, that the second and third limitations converge at the same root: the *antiparticle*, born of the negative-energy problem, becomes the protagonist of the cancellation that restores causality.

Quantum field theory (QFT) emerged to overcome these limitations. In QFT, the fundamental object is not the particle but the field defined over spacetime; the particle is described merely as an *excitation* of that *field*. Through this paradigm, the creation and annihilation of particles are naturally expressed as changes in the state of the field, negative-energy solutions are coherently repositioned as antiparticles, and causality is safeguarded by the local interactions of the fields.

[1] The two terms coincide because for a space-like interval a Lorentz transformation can carry $(x-y)$ into $-(x-y)$, so that $D(x-y) = D(y-x)$. The cancellation is therefore a consequence of Lorentz invariance rather than an accident of the free-field calculation [55, Chapter 2.4].

In quantum field theory, scattering amplitudes are standardly calculated through a *perturbative expansion* from a *Lagrangian* defined over spacetime. The S-matrix is decomposed into a non-interacting part and an interacting part T, such that $S = 1 + iT$, and the invariant matrix element $\mathcal{M}$ is extracted from the T-matrix by isolating 4-momentum conservation, conventionally structured as follows [55, Eq 4.72-3]:

$$\langle p_1 p_2 \cdots | iT | k_A k_B \rangle = (2\pi)^4 \delta^{(4)} \left(k_A + k_B - \sum p_f \right) \cdot i\mathcal{M}(k_A, k_B \to p_f) \tag{2.3}$$

Each term of this perturbative expansion is represented by a *Feynman diagram* and subsequently summed [55, Chapter 4.5]. The stage for this computation is a *continuous real spacetime*, and the amplitude is given by integrals defined over it. The core principle governing the form of interactions in QFT is *gauge symmetry*; the choice of gauge theory determines the structure of the amplitude [2]. The amplitude addressed in this study is likewise defined within a specific gauge theory, discussed in further detail in Section 2.1.3.

This quantum field theory exhibits phenomenal precision. For the anomalous magnetic moment of the electron ($g - 2$), perturbative calculation and experimental measurement agree to more than ten significant figures (Section 1.2). Yet within the very interior of this theory a striking discovery has been made. When the perturbative amplitudes of gauge theories are expanded via Feynman diagrams, the number of terms explodes into extreme complexity; the *final result*, paradoxically, reveals an astonishing simplicity and hidden symmetries completely obscured in the original formulation [61, 62].

This unnatural gap—the disconnect between the complexity of the computational process and the ultimate simplicity of the result—suggests that the true starting point of scattering amplitudes *might* not be spacetime, locality, and unitarity, but something else entirely. Following this clue, a current in modern physics since the late 20th century has begun to view spacetime not as a fundamental reality but as an emergent phenomenon. Among the various attempts to substantiate this paradigm shift, this study directs its focus toward the amplituhedron, proposed by Nima Arkani-Hamed and his colleague Jaroslav Trnka [61]. The ensuing section explores this sequence of theoretical currents.

2.1.2 Trajectory of Modern Physics

The preceding section reached an interesting point: the scattering amplitudes of quantum field theory, summed laboriously over Feynman diagrams drawn upon a continuous real spacetime, collapse in the end into an astonishing simplicity that the original formulation gave no reason to expect. This gap is the clue this study follows, for it implies that *spacetime*, *locality*, and *unitarity* may not be where scattering truly begins. To see why such a suspicion is even permissible, This section steps back to watch how the language of physics has shifted across eras; the amplituhedron, the object toward which this chapter converges, is the offspring of the most recent such shift.

By *mathematical language* is meant here not the real and complex number systems—which remain the intact scaffold beneath everything—but the structures through which physical theories organize and express those numbers: differential equations, operator algebras, topological invariants, geometric objects. These languages have

[2] The general theory of gauge theories and their historical development are addressed in Section 2.1.2. For standard expositions, see [55, Chapter 16],[60, Chapter 8].

not replaced one another so much as accumulated, each built upon the sediment of the last.

2.1.2.1 Evolution of Modern Mathematical Physics

The evolution may be divided into four phases [3], each named for the mathematical language that stood at its center: the analytical, the algebraic and operator, the topological and geometric, and finally the informational and positive-geometric.

The earliest phase, **from the late nineteenth century into the 1920s**, wrote physics in the language of differential geometry and analysis, from Newtonian mechanics through general relativity. It rested upon a single premise—that the world sits upon a smooth, continuous structure, spacetime as a differentiable manifold and forces as continuous fields defined without gaps at every point [4]. What most repays attention here is not the manifold but the *variational* principle that matured within it. The principle of least action fixes a particle's path by requiring the action to be stationary [5].:

$$S = \int L\,dt, \quad \delta S = 0 \tag{2.4}$$

Among all conceivable paths, the one realized is that for which S takes a stationary value. This conception—many paths collapsed into one value—stands in quiet foreshadowing of the moment, much later, when the amplituhedron replaces the summation over intricate paths with the computation of a single geometric *volume*.

In the decades **from the 1920s to the 1960s**, quantum mechanics shifted the center of the language from continuous functions over spacetime to *operators* acting upon abstract vector spaces. Two features of that shift bear on what follows. The first is *non-commutativity*. Unlike classical scalars, where $AB = BA$, operators generally obey $AB \neq BA$, and this failure of order-independence furnishes the mathematical ground of the uncertainty principle. A number had become an algebraic object bearing an inherent sequence —a transformation that would later germinate into a geometry of its own. The second is the predicament that shadowed the era's great computational engine. Interactions were organized through the S-matrix and rendered visible as Feynman diagrams (2.3), yet as interactions grew intricate the diagrams proliferated combinatorially, and *infinities* surfaced within the individual integrals [6]. It was this intractability—the explosion of local, diagram-by-diagram labor—that pressed physics to lift its gaze from local calculation toward the global structure of spacetime.

[3] This classification is a reconstruction that takes the logical layers presented by Roger Penrose in *The Road to Reality* (2004), together with the paradigm shifts of modern physics, as its point of departure. The intervals mark when each language rose to prominence, not rigid boundaries: the seminal ideas of each current often preceded its interval and continued to resonate long after.

[4] This smoothness was never a proven fact but a prerequisite for mathematical description—differential operations demand it—and the status of that prerequisite is precisely what this study will later interrogate. For the analytical framework, see [63, Chapter 3-4].

[5] The principle of least action is described by defining the action S from the Lagrangian L and setting its variation to zero ($\delta S = 0$). In the context of general relativity, the exposition of this action principle was referenced from [63, Chapter 3.3]. This Lagrangian formalism later serves as the foundation for the Feynman path integral—the formulation of integrating $e^{iS/\hbar}$ over all paths—which connects to the Feynman diagram calculations mentioned in Section 2.1.1.

[6] The explicit diagrammatic expansions, which display this combinatorial proliferation concretely, are collected in Appendix A. Renormalization—the procedure that tames these infinities by cancellation—was examined in Chapter 1 (Section 1.4.4) as an anticipated critique of this study's hypothesis, and is not reprised here; for a standard exposition see [55, Part II].

That turn defined the next phase, **from the 1970s into the 1990s**, whose language was topological and geometric, and three of its currents bear directly upon what was to come. The first is topology itself, owhere more tellingly displayed than in the Chern–Simons theory, whose action carries no dependence upon the spacetime *metric*: however spacetime is stretched or bent, so long as its topological structure is preserved, the identical result follows [64]. This is what *background independence* establishes: a physical result can be something narrow but not negligible—that a physical result can be severed from spacetime and fixed instead by an underlying geometric structure. It is exactly this severance that the amplituhedron inherits, drawing its amplitudes from geometry rather than from any metric. The second current carries the non-commutativity of the previous phase to its conclusion: Alain Connes recast it from a property of position and momentum into a property of the spatial coordinates themselves, yielding a geometry in which the coordinates no longer commute [65]. The third, and for this study the most consequential, is *twistor theory*: in 1967 Roger Penrose proposed taking twistors, corresponding to light rays, rather than the points of spacetime as the fundamental entities, so that a point of spacetime becomes a *derived* notion [66]. Twistor theory did not prevail as a computational tool in its own day; but its premise—that spacetime is not foundational but derived—was carried, in the era that followed, directly into the calculation of scattering amplitudes, and twistor theory became the immediate ancestor of the amplituhedron.

Across all three of these phases one premise had gone unchallenged: that *spacetime* is the fundamental stage upon which physics is enacted, a postulate held fast since Galileo and Newton and undisturbed even by the two revolutions of relativity and quantum mechanics. It is this premise that the most recent phase, **from the 2010s to the present**, has begun to interrogate, and it did so almost at once from two unrelated directions—spacetime *emerging from information*, and scattering amplitudes *emerging from positive geometry*. The first traces to Wheeler's *It from Bit*(Section 2.2.1), which matured into a concrete program when the Ryu–Takayanagi formula (2006) identified the entanglement entropy of a boundary region with the area of a minimal surface in the bulk [67]:

$$S_A = \frac{\text{Area}(\gamma_A)}{4G_N} \tag{2.5}$$

where S_A is the entanglement entropy of the boundary region A, γ_A the corresponding minimal surface in the bulk, and G_N Newton's constant. That a purely informational quantity should resolve into a geometric one—an *area*—was sharpened by Van Raamsdonk (2010) into the claim that spacetime is held together by entanglement itself [68]. The second direction returns to Penrose's twistors, brought back into amplitude calculations by Witten (2004) [69]. The BCFW recursion relation (2005) then permitted intricate amplitudes to be reconstructed from a minimal set of terms[7], and this recursive structure was organized into the geometry of the *positive Grassmannian*. In 2013 Arkani-Hamed and Trnka condensed the whole into a single geometric object, the amplituhedron, whose *volume* is the amplitude [61].

Here two conclusions, reached along orthogonal paths, come to intersect. Section 1.4 concluded that the real number system is an inadequate instrument for describing the world completely; the trajectory of this chapter concludes that spacetime is not itself fundamental. These are not rival claims but convergent ones. Both the informational and the geometric axis describe the world, in the end, through *computation*, and both terminate in a geometric quantity—an *area* in the one (2.5), a *volume*

[7]The BCFW recursion relation is treated in Section 3.1.3.4, on the mechanics of the amplituhedron [70].

in the other[8]. And since Galileo and Newton, that spacetime has invariably been computed over the continuous real numbers; if the spacetime so computed is not fundamental, then this is one more piece of evidence—arriving now from the side of spacetime, as Section 1.4's arrived from the side of the quantum—that the real number system is not the language in which the bedrock of the world is written. What this convergence establishes is not a proof; it is narrower, and harder to dismiss: that a single question now presses from two directions at once—*can this area and this volume be represented completely over the continuous real number system?*

It is in extension of this doubt that the following section (Section 2.3) introduces the non-Archimedean number systems—non-standard analysis and p-adic numbers—as candidates to stand where the real numbers fall short. Before that, however, the two pillars of the present era—emergent spacetime and positive geometry—warrant a closer conceptual examination, for it is upon them that the amplituhedron is built.

2.1.2.2 Emergent Spacetime and Positive Geometry

The preceding section arrived, as the terminus of a century-long trajectory, at a single conclusion: that both axes—information and positive geometry—regard spacetime not as fundamental but as emergent. That conclusion was reached historically; here it must be understood conceptually. What, precisely, does emergent spacetime assert, and by what route does it lead to positive geometry?

Conventional physics posited spacetime as a fixed stage—a *background*—upon which particles interact and from whose coordinates every calculation departs. The emergent view inverts that order: spacetime and its properties are not the *starting point* of computation but its *destination*. One manipulates not temporal or spatial variables directly, but quantities such as energy and momentum, and the properties of spacetime declare themselves only afterward, as outcomes [67]. This is what *background independence* means, and the word *emergence* carries no metaphor here; it is a concrete claim about the structure of computation itself. Were spacetime fundamental, it would have to be presupposed in every calculation; if it is emergent, the calculation must be able to begin without it. The contrast with the Feynman-diagram predicament of the previous era is exact: to compute without presupposing spacetime is precisely the possibility of replacing that combinatorial explosion with a computation upon a *single geometric object*.

If spacetime is not fundamental, what stands in its place? Emergent spacetime is only the negative assertion—*spacetime is not fundamental*. *Positive geometry* is the affirmative reply to the question that denial leaves open. The reply is that the scattering amplitude is given not as a sum of diagrams upon spacetime, but as the *volume* of a geometric object: certain amplitudes admit interpretation as the volumes of high-dimensional polytopes, so that the quantity encoding scattering probabilities in energy and momentum acquires a second reading as a volume within an abstract space [62, Chapter 10].

[8] Intriguingly, the computational targets to which both axes culminate are geometric quantities. In the information axis, entanglement entropy corresponds to the area of a minimal surface (2.5), and in the positive geometry axis, the scattering amplitude is given by the volume of a geometric object called the amplituhedron. The former's area is the Arera (γ_A) of the Ryu-Takayanagi formula presented as 2.5, while the latter's volume corresponds to the canonical form of the amplituhedron, which will be detailed in Section 2.1.3. The question this study raises is precisely whether this area and volume can be completely represented over the continuous real number system

The term "positivity" signifies in positive geometry. It denotes a *region* of a geometric space singled out by a sign condition: in the *positive Grassmannian*, the region where certain coordinates all carry positive sign constitutes the domain from which the geometry is built. Positivity therefore excludes neither negative nor complex numbers from the computation. It is the condition that separates inside from outside by sign, constituting the interior as a well-defined domain. It is from the volume of that interior that the amplitude is drawn [61, Section 4]. Which coordinates carry the condition, and how they fix the amplitude, is the business of the section to follow.

The decisive element is the object's *boundary*. If the volume of the polytope is the amplitude entire, the facets that bound it answer to the individual physical structures composing that amplitude [62, Chapter 10]. It is exactly at this juncture that *emergence* takes on its concrete form. *Locality* and *unitarity*—the two principles quantum field theory had held as its founding premises—are read off from this boundary structure. In the expression of Arkani-Hamed and Trnka, who proposed the amplituhedron, they are not the central assumptions of the computation but *consequences* that co-emerge from positive geometry [61]: the two pillars that once held up quantum field theory appear only at the boundaries of an object defined, from the first, without any spacetime at all.

What emergent spacetime and positive geometry jointly establish, then, is not merely that spacetime recedes, but that its very properties—locality and unitarity—surface from the boundaries of a more fundamental geometric structure. The concrete embodiment of this view, for scattering amplitudes, is the *amplituhedron*: what that object is, how its volume yields the amplitude, and what bearing it holds for this study are the matters the ensuing section takes up.

2.1.3 The Amplituhedron

The objective of this section is not a general overview of the amplituhedron. Because the target of this study's simulation (Chapter 3) is the volume of the amplituhedron —its canonical form—the exposition is confined to establishing that computational target at tree level: the positive Grassmannian, the definition of the *amplituhedron*, $\mathcal{N} = 4$ *Super Yang-Mills Theory*, *cell decomposition*, and the *canonical form*. Extensions beyond this—loop-level integrands, the minutiae of superamplitudes —lie outside the scope of this study and are relegated to footnotes[9]. For a fuller treatment the reader is referred to the standard reviews [62, Chapter 10][71].

The amplituhedron asserts that scattering amplitudes can be computed as geometric volumes entirely *independent* of spacetime—a perspective that declines to posit spacetime as the fundamental stage of physics, yet furnishes a concrete framework for computing amplitudes all the same. What makes this more than a formal reformulation is the theory beneath it. The $\mathcal{N} = 4$ supersymmetric Yang-Mills (SYM) theory on which it rests is UV finite: unlike conventional quantum field theories, its beta function $\beta(g)$, which governs the running of the coupling, vanishes exactly at

[9]The amplituhedron is extended beyond the tree-level to incorporate loop integrands; the entity encompassing this extension is defined in the foundational paper as $\mathcal{A}_{n,k,L}(Z)$. Because this study solely addresses the tree-level ($L = 0$), this extension is not covered. For the complete structure [61].

every order of perturbation [10], and the theory is conformally invariant [11] in consequence.

$$\beta(g) = \mu \frac{\partial g}{\partial \mu} = 0 \tag{2.6}$$

The coupling g thus does not vary with the energy scale μ, and the theory is *finite*, free of ultraviolet (UV) divergences. Within it, amplitudes can be defined geometrically without any renormalization, liberated from both spacetime and renormalization. The object is no familiar low-dimensional figure, however, and computing its volume is vastly more involved than for ordinary geometries: visual representation reaches *only* a single facet of the four-dimensional amplituhedron, and nothing beyond that renders in three-dimensional space —the more so once $k \geq 2$, where the object forsakes flat facets for a curved structure reflecting the nonlinearity of the positive Grassmannian[12].

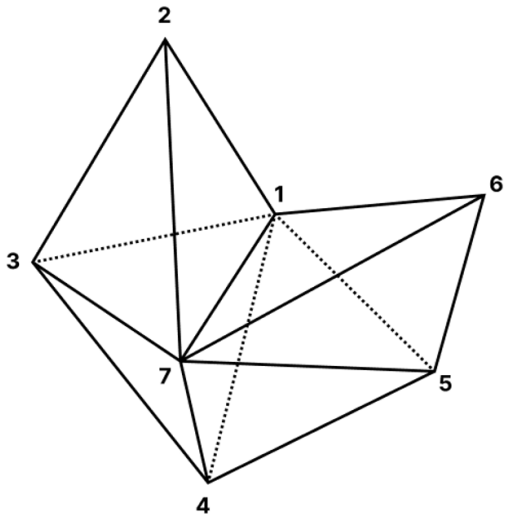


FIGURE 2.4: Depicts a 3-dimensional facet of a 4-dimensional amplituhedron projected into the projective space. Adapted from [61, Section 3].

What this study fixes upon is that the amplitude is thereby reduced to a *deterministically* fixed value—the volume of a polytope. This is the stepping stone for the thesis developed throughout: the hypothesis that a portion of the stochasticity surfacing at the quantum scale may originate in the computational limits of the real number system. The reasoning is simple. If the logic of the volume computation is itself deterministic, then for a given input it must always return an identical true value; should any deviation appear in the outcome, its origin can be attributed not to the logic but to the *number system* used to implement that logic. It is this separation—deterministic logic on one side, the number system that carries it on the other—that constitutes the core of how this study treats the amplituhedron.

[10] That the beta function vanishes to all orders in $\mathcal{N} = 4$, rendering the theory UV finite, is a standard result; its basis is addressed alongside the symmetries of the theory in Section 2.1.3.1.

[11] Conformal invariance means the theory is invariant under angle-preserving transformations—rotations, translations, and scale transformations. In $\mathcal{N} = 4$ this is extended to dual superconformal symmetry, and the two together form an infinite-dimensional *Yangian symmetry* [62, Chapter 5].

[12] Figure 2.4 depicts a three-dimensional facet of a four-dimensional amplituhedron projected into projective space $\mathbb{P}^3$; this is the $k = 1$ case, drawn as a polytope with flat facets. In the toy model of this study, where $k = 2$, such visualization becomes impossible.

Accordingly, the simulations of this study treat the amplituhedron as a *black box*. The term does not imply that its interior is unknown—on the contrary, that interior is set out in this section to the extent this study requires; it signifies that in the simulation the internal logic is left untouched, held as a single fixed deterministic function that maps an input (momentum data) to a unique output (the volume, that is, the canonical form). With the logic so fixed, the *number system* chosen to implement it remains the sole variable governing the precision of the final output. Whether the deviation in that precision is bound up with quantum stochasticity is not the question of this section but of the simulations, which examine it through the comparative passage across different number systems[13].

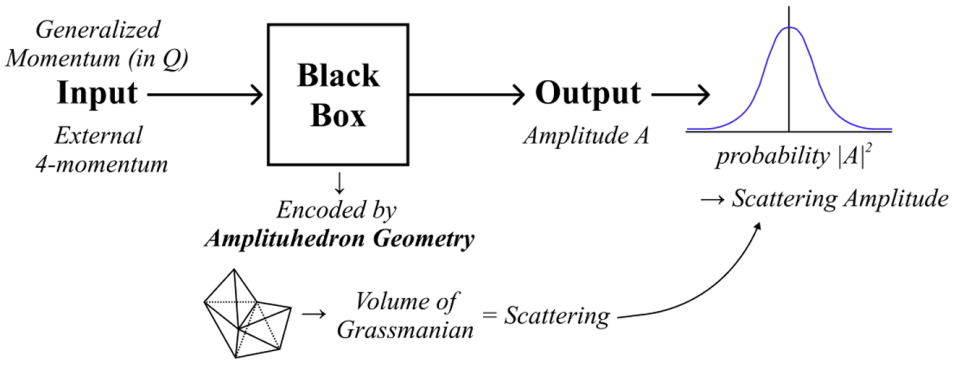


FIGURE 2.5: Depicts a black box model in this study.

2.1.3.1 $\mathcal{N} = 4$ Super Yang-Mills Theory

The amplitudes the amplituhedron reproduces as volumes belong not to an arbitrary quantum field theory but to a specific one: $\mathcal{N} = 4$ Super Yang-Mills theory (hereafter $\mathcal{N} = 4$ SYM), and its nature must be examined first. It has served as the standard laboratory of scattering-amplitude research and is often called the simplest quantum field theory [62, Chapter 4.4]—where *simplest* implies no physical triviality but rather that its amplitudes carry the most systematically ordered, symmetrically rich structure. It is precisely this order that allows amplitudes once given as sums of intricate Feynman diagrams to be condensed into the volume of a single geometric object.

The theory's distinguished status rests on its symmetry. In $\mathcal{N} = 4$ SYM, the $\mathcal{N} = 4$ designates the amount of supersymmetry in four-dimensional spacetime—the maximal value permissible for a theory built exclusively from particles of spin no greater than one. This constraint of *maximal supersymmetry* fixes the theory uniquely, up to the choice of gauge group and the value of the coupling [62, Chapter 4.2–4]; $\mathcal{N} = 4$ SYM is thus not one arbitrary selection among many possible theories, but the single theory whose form is all but dictated by the mandate of symmetry alone.

This maximal supersymmetry is also the ground of the UV finiteness noted in the preceding section. Supersymmetry induces an exact cancellation between bosonic and fermionic contributions; under maximal supersymmetry that cancellation is absolute, eradicating UV divergences entirely, so that the beta function vanishes at

[13]Figure 2.5: when external momenta given over the rational numbers are input, the internal logic encoded by the amplituhedron geometry outputs the volume of the Grassmannian, corresponding to the amplitude A and subsequently its probability $|A|^2$. The simulation fixes this internal logic as a constant function and attends to the character of the number system computing the volume.

every perturbative order ($\beta(g) = 0$) and the theory is finite and scale-independent and *conformally invariant*. The property invoked earlier—that amplitudes may be defined geometrically without renormalization—is anchored precisely here [14].

This *uniqueness* and *finiteness* carry a specific weight for the present study. The simulations to follow are executed not in the actual physical world but upon the mathematical space of $\mathcal{N} = 4$ SYM—not an arbitrarily contrived stage, but a rigorously refined entity extant within theoretical physics, a point revisited when the significance of the simulations is discussed in Chapter 5. The specific *field content* and the minutiae of the supersymmetry algebra, not being directly linked to the simulation mechanics, are left to a footnote [15].

Only one classification of amplitudes from this theory is required to handle the amplituhedron. The tree-level amplitudes of $\mathcal{N} = 4$ SYM are classified by the number of gluons of negative helicity [16]. From the simplest upward, the hierarchy runs Maximally Helicity Violating (MHV), Next-to-MHV (NMHV), N^{2}MHV, and so on, and it corresponds directly to k, one of the defining integers of the amplituhedron. Under the momentum-twistor convention of the foundational paper [61] adopted here, the correspondence is $k = 0$ for MHV, $k = 1$ for NMHV, and $k = 2$ for N^{2}MHV; it is stated in advance that the toy model of Chapter 3, assigned $k = 2$, therefore corresponds precisely to N^{2}MHV.

With this, the target of the simulations—the tree-level amplitudes of $\mathcal{N} = 4$ SYM, designated by the helicity hierarchy k—is definitively established. What remains is to find the geometric space that reduces these amplitudes to volumes; the ensuing section examines the *positive Grassmannian*, the material from which that space is built.

2.1.3.2 The Positive Grassmannian

What geometric space is to harbor these amplitudes? Its foundation is the Grassmannian—more precisely, the *positive Grassmannian*.

Consider first the Grassmannian itself. The Grassmannian $G(k, n)$ is the set of k-dimensional planes through the origin of an n-dimensional space [62, Chapter 4.1]. Since a k-dimensional plane is fixed by the k vectors that span it, such a plane can be represented by a $k \times n$ matrix C formed by stacking those k row vectors. The choice of spanning vectors is not unique, however: mixing the rows by a $GL(k)$ transformation leaves the plane they span invariant. The Grassmannian is therefore not the full set of $k \times n$ matrices but the space obtained by quotienting out this $GL(k)$ redundancy, of dimension $k(n-k)$.

In practice the $GL(k)$ redundancy is eliminated by fixing certain columns as an identity block; a point in $G(2,5)$, for instance, admits the gauge-fixed representation [17]:

$$C = \begin{pmatrix} 1 & 0 & c_{13} & c_{14} & c_{15} \\ 0 & 1 & c_{23} & c_{24} & c_{25} \end{pmatrix} \tag{2.7}$$

[14] The causal order runs thus: the cancellation of divergences via supersymmetry comes first, and conformal invariance follows as a consequence [62, Chapter 4.4].

[15] The spectrum of $\mathcal{N} = 4$ SYM comprises sixteen massless states—a gluon of helicity ± 1, four pairs of gluinos of helicity $\pm 1/2$, and six scalars—all in the *adjoint representation* of the gauge group.

[16] A gluon is the massless, spin-1 mediator of gauge interactions in Yang-Mills theory. Helicity is the projection of spin onto the direction of momentum; for a massless spin-1 particle it takes only two values, $+1$ or -1. Amplitudes are classified by the number of negative-helicity gluons because that count fixes the amplitude's place within the supersymmetric hierarchy.

[17] The gauge-fixed representation follows the $G(2,5)$ case of [61, Equation 4.1]; fixing the leading 2×2 block as the identity is one way to eliminate the $GL(k)$ redundancy.

The $k \times k$ minor built from k columns of C selected in order-preserving fashion is denoted $(a_1 \cdots a_k)$ and called a Plücker coordinate [18]; these minors are the coordinates characterizing the plane. It is here that the positive Grassmannian $G_+(k,n)$ is defined—as the region where every ordered minor $(a_1 \cdots a_k)$, with $a_1 < \cdots < a_k$, is strictly positive, without a single exception [72, Chapter 5.3]. This is the precise, coordinate-based rendering of the *positivity* introduced conceptually in Section 2.1.2. Because a minor is antisymmetric—flipping sign under the permutation of columns—the order of the columns must be fixed immutably before positivity can be spoken of at all [19].

The definition reveals its geometry most plainly in the simplest case. For $G_+(1,n)$, where $k = 1$, the minors are just the components themselves, and the region where all are positive is a simplex in projective space—the higher-dimensional generalization of a triangle [20]. For higher k the region is no longer a simplex—its boundaries curve —but it retains the same defining feature: a domain carved out by sign conditions on the minors. Moreover, the space is partitioned, along the boundaries where certain minors vanish, into fragments called *positroid cells*, each parameterized by its own canonical positive coordinates [72, Chapter 5.3–4]. This cellular structure is what later makes the volume computation tractable. Computing the volume of an intricate figure by partitioning it into simpler pieces and summing their contributions—one may recall the triangulation used in 3D modeling to break a surface into triangles—is the same idea, save that here the object partitioned is a high-dimensional volume rather than a surface, and each piece is not a triangle but a cell bearing its own coordinates.

The material space to harbor the amplitudes is thus prepared—the region where the ordered-column minors are all positive, together with the positroid cells into which it divides. The positive Grassmannian is not yet the amplituhedron, however; the amplituhedron is obtained by projecting this space into another via external data. How that projection is carried out, and how the amplituhedron is thereby defined, are taken up next.

2.1.3.3 Definition of the Amplituhedron

The amplituhedron is defined by a single *projection*. To carry it out, external data is required in advance. This data is a collection of n vectors in a $(k+m)$-dimensional space, one for each of the n external particles in the scattering [21]. The data is not arbitrary. It must be arranged to satisfy a positivity condition. One canonical way to guarantee this is to place the points on a moment curve: writing each vector as $(1, t, t^2, \ldots, t^{k+m-1})$ and ordering the parameter t secures the positivity of the data automatically[22].

[18] Minors are quantities whose ratios remain invariant under $GL(k)$ transformations, fully capturing the geometric information carried by the plane C [61, Equation 4.2].

[19] This antisymmetry stems from the sign change of a determinant under column exchange (distinct from the non-commutative geometry of earlier sections). A *cyclic ordering* of columns is thereby established as an essential structure of the positive Grassmannian, and this cyclic structure leads naturally to the cyclic form of the canonical form addressed later.

[20] $G_+(1,3)$, for example, is the region of the projective plane $\mathbb{RP}^2$ where all three homogeneous coordinates are positive, a single triangle (a simplex) [72, Chapter 5.3].

[21] The external data Z is defined as a set of $(k+m)$-dimensional vectors, following [61, Chapter 3]. These correspond to entities in momentum-twistor space; their concrete construction is not needed for this study and is omitted.

[22] The convex hull of n ordered points on a moment curve is a cyclic polytope, and placing points on this curve is one canonical way to ensure the positivity of Z [61, Equation 3.5]. The foundational paper notes this as one method among several.

The amplituhedron is now defined. Let C be a $k \times n$ matrix representing a point in the positive Grassmannian $G_+(k,n)$. Acting C upon the external data Z gives a projection [23]:

$$Y_\alpha^I = \sum_{a=1}^{n} C_{\alpha a} Z_a^I \tag{2.8}$$

The image of this projection is the amplituhedron. As C sweeps the entire positive Grassmannian, the region filled by the resulting Y's forms a single entity within $G(k, k+m)$, the space of k-dimensional planes in $(k+m)$ dimensions. This entity is the tree amplituhedron, denoted $\mathcal{A}_{n,k,m}(Z)$ [61, Equation 3.8–9].

Three integers (n,k,m) govern it. n is the number of external particles in the scattering. k is the helicity hierarchy of Section 2.1.3.1, with $k=0$ for MHV, $k=1$ for NMHV, and $k=2$ for N^{2}MHV. m fixes the external space as $(k+m)$-dimensional, and $m=4$ is the value corresponding to physically meaningful amplitudes [24]. The amplituhedron itself is then k·m-dimensional. Once the three are fixed, the shape of the amplituhedron is fixed, and with it its volume.

That volume, however, is not yet the amplitude. The amplituhedron so far is only a geometric region. The amplitude is extracted from a *form* corresponding to the region's volume, and to obtain that form the object must be dissected into the positroid cells of the previous section.

One caution first. The *volume* here is not the ordinary area or volume of elementary mathematics. The relevant geometries live in projective space, where a conventional volume with units carries no projective meaning. Instead, the amplituhedron is assigned a rational *form*, the *canonical form* Ω, which plays the role of volume [25]. It is fixed by a single property: it has *logarithmic singularities* on every boundary of the amplituhedron, and nowhere else[26]. The canonical form is therefore determined entirely by the boundary structure of the object. This is where the earlier claim of Section 2.1.2.2—that boundaries correspond to physical structures—becomes actual computation.

The form is computed explicitly by *cell decomposition*. Dissect the amplituhedron into non-overlapping positroid cells that cover it entirely. The amplituhedron is dissected into non-overlapping positroid cells that cover it entirely (Figure 2.6) To each cell corresponds a simple form with logarithmic singularities on that cell's boundaries.

How is the collection of cells obtained? In the physical case $m=4$, the BCFW recursion relation supplies exactly this decomposition [27]. BCFW appeared earlier, in this Section 2.1.3.3, as a way to reconstruct complex amplitudes from a minimal number of terms. Here it returns in a new role: the algorithm that designates the

[23] In (2.8), α is the row index of the Grassmannian plane, a the index of the external particle summed over, and I the index of the $(k+m)$-dimensional external space.

[24] This is specified in [61, Chapter 3].

[25] Defining the canonical form as the "volume" of the amplituhedron follows [61, Chapter 6]. The form is tied to the area of the object's dual polytope, but this dual interpretation is not needed here.

[26] A logarithmic singularity means the form diverges as $d\alpha/\alpha$ as the boundary is approached; this property fixes the canonical form uniquely [61, Equation 6.1–2].

[27] For $m=4$, the BCFW decomposition of tree amplitudes yields a collection of $4k$-dimensional cells of the positive Grassmannian that non-overlappingly cover the amplituhedron under the projection $Y = C \cdot Z$; see [61, Chapter 5]. The derivation of the BCFW recursion itself—the complex momentum shift and reconstruction—is not needed here.

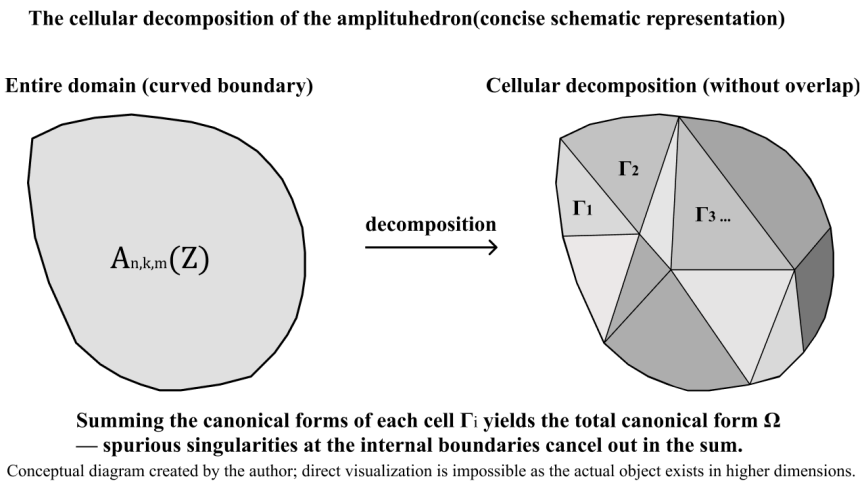


FIGURE 2.6: The cellular decomposition of the amplituhedron (concise schematic representation)

cells composing the amplitude. Each cell is parameterized by canonical positive coordinates $(\alpha_1, \ldots, \alpha_{4k})$, and its form is a logarithmic differential ($d\log$) form [28]:

$$\Omega^\Gamma = \frac{d\alpha_1}{\alpha_1} \wedge \frac{d\alpha_2}{\alpha_2} \wedge \cdots \wedge \frac{d\alpha_{4k}}{\alpha_{4k}} \tag{2.9}$$

Summing these cellular forms gives the canonical form of the whole amplituhedron. Written in minors, the coefficient of each cell's form is a rational function, which we denote ω_Γ:

$$\Omega^\Gamma \sim \frac{\langle Z_I \rangle^m}{\langle 12\cdots\rangle\langle 23\cdots\rangle\cdots} \tag{2.10}$$

The numerator holds the m-th power of minors built from the cell's points; the denominator, a product of *cyclic* minors [29]. This rational function is the exact target the simulation will compute.

The decomposition raises one problem, which then resolves itself: *spurious poles*. Each cell's form has singularities on all of its boundaries. Some of these are internal boundaries created by the decomposition, not boundaries of the amplituhedron itself. They are physically non-existent—spurious poles. Yet in the sum over all cells, these spurious poles cancel pairwise and vanish completely [61, Equation 6.6–7]. The total form retains only the singularities of the amplituhedron's true boundaries, and the result is unique regardless of how the cells were partitioned. This—artificial singularities that appear in the parts but vanish in the sum—is structurally the same situation as unphysical singularities cancelling across summed Feynman diagrams.

One final property matters most for this study. The form is defined over the positive data Z, but what it yields is a rational function with rational coefficients. Such a function is not tied to any particular number system: wherever its denominator does not vanish, it takes a value over any field. This is what makes the canonical form comparable across the number systems used to compute it—the property on

[28] In 2.6, $4k$ is the dimension mk of each cell for $m = 4$.

[29] The explicit canonical form in minors for $k = 1$, $m = 4$ is given in [61, Equation 6.6]: the numerator is the fourth power of a five-point minor, the denominator a product of cyclic minors. The notation here generalizes that structure; the exact form is fixed by (n, k, m), and the concrete substitution for this study's toy model is treated in the next section.

which the simulations of this study depend. Its expression also admits analytic continuation to arbitrary complex values [61, Section 6], though that extension is not what this study exploits. With this, a machine is complete: for any (n, k, m), it outputs the amplitude as one rational function. The next section substitutes this study's toy model, $(6, 2, 4)$, into that machine, fixing the concrete target of the simulations.

2.1.3.4 $(6, 2, 4)$ Toy Model

The toy model is the $(n, k, m) = (6, 2, 4)$ amplituhedron. $n = 6$ means six external particles take part in the scattering. $k = 2$ places the helicity hierarchy at N^2MHV, following the momentum-twistor convention fixed in Section 2.1.3.1. $m = 4$ marks a physically meaningful amplitude [30].

Why this model? Because it is the simplest case in which the amplituhedron's nontrivial structure appears. At the lowest hierarchy, MHV, the canonical form is a mere constant, and no algebraic structure can be examined within it [31]. At $k = 1$ a nontrivial rational function does emerge, but the object remains a cyclic polytope with flat facets. Only at $k = 2$ does the amplituhedron forsake flat facets for the curved structure that reflects the nonlinearity of the positive Grassmannian. And for $k = 2$ with $m = 4$, the number of external particles cannot fall below $n = k + m = 6$. $(6, 2, 4)$ is therefore the minimal case in which that structure can be examined at all. The choice rests on that minimality. Whether the structure exposed at $(6, 2, 4)$ persists under extension to higher hierarchies is not settled here; it is taken up in Chapter 4 and left open in Chapter 5.

Substituting $(6, 2, 4)$ into the formulation of the previous section, the canonical form takes the following shape. For each cell of the amplituhedron, the numerator carries the fourth power ($m = 4$) of minors formed by the cell's points; the denominator, a product of contiguous 2×2 cyclic minors ($k = 2$):

$$\Omega^{\Gamma} \sim \frac{\langle Z_{a_1} \dots Z_{a_6} \rangle^4}{\langle a_1 a_2 \rangle \langle a_2 a_3 \rangle \cdots} \tag{2.11}$$

Summing these cellular contributions over all cells designated by the BCFW decomposition yields the total canonical form Ω [32]. This Ω is the function held as the "black box" foreshadowed in Section 2.1.3, handled hereafter as a single fixed deterministic function. Its concrete numerical values, and the algebraic properties they reveal, are results of the computation itself; they are not previewed here but deferred to Chapter 4.

Beginning from the macroscopic perspectives of emergent spacetime and positive geometry, in turn: $\mathcal{N} = 4$ SYM as the computational stage, the positive Grassmannian as its material, the amplituhedron via projection, and the canonical form via cell decomposition. The whole framework has been condensed to a single concrete target: the $(6, 2, 4)$ amplituhedron. On which number system the canonical form Ω is computed, and how the idiosyncrasies of that system induce variation in

[30] In the $(6, 2, 4)$ notation, $m = 4$ is the physical case of the earlier sections. As this study is confined to tree level, the loop order L is not introduced; the model is fully specified as the tree-level $(n, k, m) = (6, 2, 4)$ amplituhedron.

[31] That the canonical form of the lowest hierarchy is trivial reflects the structural simplicity of the amplituhedron at that level; that $k = 2$ is the first case with a non-trivial canonical form is confirmed in [61, Chapter 5–6].

[32] The concrete computation of the cellular forms and their summation follows the principles of the previous section. The constitution and enumeration of the cells for $(6, 2, 4)$, and the summation of each cell's contribution, belong to the simulation's implementation and are addressed in Chapter 4.

the result, is the core inquiry of this study. Before that, the physical environment where the scattering actually occurs—the low-temperature plasma—must be examined [33].

[33] Recent work extends the amplituhedron perspective to cosmological observables, such as cosmological polytopes; as this lies outside the present scope, it is mentioned only briefly in the Future Work section of Chapter 5.

2.2 The Nature of Plasma

Chapter 1 traced this study's origin to the stochastic fluctuations—the stochastic limit—of the plasma etching process. This section asks what the plasma itself is, the stage on which those fluctuations unfold. It does not attempt the whole of plasma physics. The exposition is confined to what is needed for two things: the computational approaches to plasma treated later, and the e^--e^- Møller scattering on which this study ultimately converges.

Plasma is often called the *fourth state of matter*. Just as heat turns a solid to a liquid and a liquid to a gas, sufficient energy imparted to a gas induces *ionization*—electrons break free of their nuclei, leaving a mixture of free electrons, positive ions, and neutral particles [8, Chapter 1][73]. Most ordinary matter in the universe exists in this state.

Two properties matter for what follows. The first is *electron temperature* (T_e). Electrons in a plasma move at varying velocities, and since each cannot be tracked individually, their distribution is characterized by a single temperature—not heat in the colloquial sense, but a measure of average kinetic energy, expressed in electron-volts (eV) [8, Chapter 2.4]. In a low-temperature plasma the electron temperature far exceeds that of the ions and neutrals, and this eV-scale electron energy is what later fixes the energy regime of e^--e^- scattering (Section 2.2.4.1). The second is *collision*. The behavior of a plasma is dictated by collisions—between electrons and ions, electrons and neutrals, ions and neutrals—through which energy transfers and particles are created and annihilated [8, Chapter 3.1]. These collisions do not follow determined trajectories; they occur *probabilistically*. When and where a particle collides is described only through probability, and it is this that forms one microscopic root of the stochastic fluctuations of Chapter 1. Every collision is at bottom a scattering event— the class of process this study takes up, in idealized form, through the amplituhedron (Section 2.2.4.1).

2.2.1 Why Plasma?

Plasma has become indispensable across modern industry—in surface treatment, thin-film deposition, sterilization, display manufacturing—but the domain where it is decisive is *semiconductor fabrication* [34]. The microscopic circuits of semiconductor devices are largely fabricated through plasma-assisted processes, and in etching—which carves the wafer surface into precise morphologies—plasma is all but irreplaceable.

This is where the rationale for the study's focus on plasma lies. The stochastic limit posited in Chapter 1 as the point of departure is, fundamentally, a predicament of this very etching process. The next section examines what that process entails and how its stochastic fluctuations are generated.

2.2.2 Plasma Etching in Semiconductor Manufacturing

Plasma etching carves the wafer surface into desired morphologies using the charged particles and reactive species of a plasma. It combines two mechanisms: *physical etching*, in which accelerated cations bombard and dislodge material, and *chemical etching*, in which radicals react with the surface to remove it [9][8, Chapter 15.2]. As

[34] For plasma's industrial applications, particularly in semiconductor manufacturing, see [8, Chapter 1][74]

circuits approach the Ångström scale, atomic-precision techniques such as Atomic Layer Etching have become central to next-generation processing [75].

This process is the actual site where the *stochastic limit* of Chapter 1 manifests; the aim here is therefore not a general introduction to etching but an interrogation of the physical origins of those fluctuations. The key to the current limits of miniaturization lies in lithography and etching—notably, how plasma etching manages and transfers the roughness formed during extreme ultraviolet (EUV) lithography [3]. Three characteristics account for this limit. Building on the properties of plasma just examined, this section traces how they arise physically, following an ion to the surface: first the uncertainty in the *number* of particles arriving, then the uncertainty in their *direction*, and finally the uncertainty in the *reaction* they induce.

The first characteristic lies in the **number of particles** reaching the surface. In plasma kinematics, particles are conventionally treated as a continuous *flux*, since tracking each is prohibitive. As circuits shrink to the 2 nm scale, however, the number participating in a single etch sector falls to mere tens or hundreds, and they must be treated not as a continuous flow but as countable, discrete events [3]. When rare discrete events occur around a fixed average, their frequency follows a *Poisson distribution*[35]: for an average of λ, the probability of exactly N arrivals is:

$$P(N) = \frac{\lambda^N e^{-\lambda}}{N!} \tag{2.12}$$

Its variance equals its mean, so the fluctuation is $\sigma = \sqrt{\lambda}$. What matters is not the absolute fluctuation but its size relative to the mean—the *relative* noise. With flux Φ, area A, and time t, the mean is $\lambda = \Phi A t$, so:

$$\frac{\sigma}{\lambda} = \frac{\sqrt{\Phi A t}}{\Phi A t} = \frac{1}{\sqrt{\Phi A t}} \tag{2.13}$$

Because relative noise scales inversely with the square root of the arriving count, shrinking the area A paradoxically increases it. Miniaturization reduces A directly, and so amplifies relative noise. The escalation of stochastic fluctuation with miniaturization is thus not a symptom of immature technology, but a fundamental property of Poisson statistics.

The second characteristic is **direction**. The ions that govern etch directionality should ideally strike the surface perpendicularly, but several mechanisms deflect them. Ions accelerated across the sheath form a narrow cone about the vertical, while unaccelerated radicals arrive isotropically, and when particles strike at non-vertical angles the patterned mask casts a *shadow* that abruptly reduces the flux at certain locations [76]. Two further mechanisms compound this: *Knudsen diffusion*, in which particles entering a narrow, deep trench may rebound back out, so that a higher aspect ratio curtails the flux reaching the bottom [36]; and *local charging*, in which lightweight electrons gather on the mask's upper corners while heavier vertical ions penetrate deeper, the resulting local field bending the trajectories of trailing ions into sidewall bowing or bottom-corner notching [78]. Reflection adds a final layer: not all ions react, and at the nanoscale mask-edge roughness (LER) randomly distorts the reflection normal, concentrating reflected ions on the sidewalls

[35] For the Poisson distribution as the limit of rare independent events, and for the property $\sigma^2 = \lambda$, see [36, Section 4.7]; the shot-noise characteristic—fluctuation given by the square root of the mean—follows directly.

[36] The transmission probability through a high-aspect-ratio trench follows the Santeler approximation [77]; combining this with Poisson statistics gives a relative uncertainty scaling as $\sqrt{1 + 0.375\,(\mathrm{AR})}$.

and causing micro-trenching [8, Chapter 9]. These mechanisms differ in origin, but they share a character: none of them fixes an ion's trajectory, and each leaves it to be settled probabilistically. The direction of an ion is dominated by probability both before and after arrival.

The third characteristic is the **reaction** itself. Even an ion that reaches the surface does not necessarily react. Chemical etching requires that the ion clear a reaction *energy threshold* E_{th} and strike a site where a radical is adsorbed; below that threshold, no reaction occurs [37]. Whether a given ion clears the threshold is set by the ion energy distribution, and thus is itself probabilistic. This characteristic, too, is probabilistic in kind: whether an ion clears the threshold is set by the ion energy distribution, and whether it meets an adsorbed radical is set by the local concentration.

The three characteristics—the scarcity of arriving particles, the non-uniformity of incident direction, and the uncertainty of the reaction— arise from different physics, yet they are alike in one respect: each is described only through probability. The first of them is quantifiable directly, through Poisson statistics; the other two are not, but they are no less probabilistic for it. This leaves one question standing: how have these stochastic fluctuations been computed and predicted thus far?

2.2.3 Computational Methods for Plasma

This section surveys the representative methods —the kinetic equation that underlies them, the numerical schemes that solve it, and the recent data-driven alternatives— not for their own sake, but to locate the single point they all share and from which this study parts.

The kinetic foundation is the *Boltzmann transport equation*. Because a plasma is an assembly of vast numbers of charged particles, tracking each individually is impossible; the description proceeds instead from a *distribution function* $f(\mathbf{r}, \mathbf{v}, t)$ on phase space, whose evolution the Boltzmann equation governs [38]:

$$\frac{\partial f}{\partial t} + \mathbf{v} \cdot \nabla_{\mathbf{r}} f + \frac{\mathbf{F}}{m} \cdot \nabla_{\mathbf{v}} f = \left(\frac{\partial f}{\partial t} \right)_{\text{coll}} . \tag{2.14}$$

The left-hand terms describe deterministic flow through phase space; the collision term on the right carries the redistribution of f through collisions. It is precisely in this collision term that the probabilistic character of the plasma originates [39]. In its complete form this equation is a nonlinear integro-differential equation on a six-dimensional phase space, not analytically solvable under realistic discharge conditions. A numerical route is therefore required.

That route is the *Monte Carlo* method, and its plasma-specific form, *PIC/MCC*. Rather than solving the Boltzmann equation directly, one simulates the probabilistic

[37] In the Steinbrüchel model [79], the etch yield is $Y(E) = A\left(E^{1/2} - E_{\text{th}}^{1/2}\right)$ for $E \geq E_{\text{th}}$ and zero otherwise; read as a single-particle reaction probability, this takes the Heaviside form $P_{\text{react}}(E) = \Theta(E - E_{\text{th}})\,\eta(E)$, with $\eta(E)$ the energy-dependent efficiency.

[38] For the kinetic description built on the distribution function, and the Boltzmann equation, see [8, Chapter 2.3].

[39] The collision term is an integral over velocity space through the scattering cross-section; for long-range Coulomb collisions it is standardly approximated in Fokker–Planck form. Its explicit expansion is not needed here; see [8, Chapter 3]. The equilibrium solution of this equation is the *Maxwell–Boltzmann distribution*, whose width is set by the electron temperature of Section 2.2. Discharge plasmas, however, are driven *non-equilibrium* states, so their behavior requires the full equation including the collision term; the Boltzmann equation applied in this setting is what is meant by the *discharge equation*.

process it describes: random numbers drawn from the relevant distributions decide each particle's collisions, and the macroscopic behavior is reconstructed from the statistics of many particles. The manner of solving is thereby replaced—from deterministic integration to probabilistic sampling. This carries a limit of principle. The statistical error of a Monte Carlo estimate falls only as $1/\sqrt{N}$ in the sample size N [80, Chapter 1.1], so capturing a rare event—such as the fluctuations that leave circuit-fatal defects in etching—demands an unrealistic number of samples:

$$\epsilon \propto \frac{1}{\sqrt{N}} \tag{2.15}$$

The form is worth noting. It is the same as the relative noise of Equation 2.13, where the fluctuation of the arriving particles fell as one over the square root of their number. The two are not the same thing: one is a property of the process, the other of the estimate. But the method here reproduces the shape of the noise it was built to measure, and it does so because both are sums of independent events. Sampling a probabilistic process yields a probabilistic answer, and the price of accuracy is paid in the same currency.

PIC/MCC, which tracks the plasma self-consistently by solving the field on a grid alongside the particle motion, compounds the cost: its time step and grid must resolve the plasma frequency and the Debye length, and as miniaturization progresses this burden grows severe [40]. In the end, such a probabilistic scheme can only return a statistical statement—that the plasma lies, with a certain probability, within a certain range. So long as the manner of computation is probabilistic sampling, its result cannot escape the language of probability.

The most recent attempts seek to move beyond this computational cost through a *data-driven* approach—neural-network surrogate models and, more recently, large language models trained on process data [81, 6]. These achieve real gains in efficiency, predicting outcomes orders of magnitude faster than physics-based simulation and improving predictably with data and model scale under a *scaling law* [7]. Yet they do not move beyond the limit that concerns this study. A data-driven model captures the statistical correlations in its data but does not elucidate their physical causation, and its reliability collapses when extrapolating beyond the training range [41]. More fundamentally, the very data these models learn from—whether Monte Carlo output or process measurement—already bears the particle-level probabilistic fluctuation of Section 2.2 imprinted upon it. The model's task is to reproduce that probability distribution more accurately, not to ask where it comes from.

Here the matter widens into the question this study has pursued. The phenomenon and every method that describes it share the same probabilistic structure: Monte Carlo handles probability as samples, data-driven models learn that probabilistic data, and neither asks why the probability is there at all. It remains, untouched, as noise. If this probability is not the essence of the microscopic world but the product of the mathematical language chosen to describe it, is it acceptable to leave the approach there? This is the point at which this study diverges from every computational scheme examined thus far.

[40] In the standard explicit PIC scheme the time step must resolve the electron plasma period and the grid the electron Debye length; on this constraint and its computational burden, see [4].

[41] On the gap between correlation and causation in such models, see [82]; on shortcut learning and the memorization of noise, [83, 84].

2.2.4 Low-Temperature Plasma

Through the preceding sections a single picture has taken shape. The stochastic fluctuation of plasma etching is irreducibly probabilistic, and the computational schemes that treat it— Monte Carlo, PIC/MCC, the data-driven models— all rest on that same probabilistic structure.

This is where the study parts ways. As set out in Chapter 1, it attends to the possibility that this probabilistic character is not the essence of the microscopic world but the product of the mathematical language chosen to describe it, and it adopts the amplituhedron (Section 2.1.3), which reduces a scattering amplitude to the volume of a geometric object, as its instrument. The amplituhedron computes a scattering amplitude, and two-particle scattering of the same kind occurs ceaselessly in a plasma. The plasma is not what that instrument models. It is where the question came from, and the concrete setting in which the scattering of interest occurs is the *low-temperature plasma*.

A low-temperature plasma—also called non-equilibrium or non-thermal—is one in which the electron temperature far exceeds that of the ions and neutrals [42]. At low pressure under a high field, the light electrons are accelerated to roughly 1–10 eV while the heavier ions and neutrals remain near 0.1 eV; the glow discharge used in semiconductor processing is the representative instance [43]. This eV-scale electron energy is precisely the regime, foreshadowed in Section 2.2, in which electron–electron scattering takes place. *Møller scattering*—the elastic scattering of two electrons— is the process that supplies this study with its input regime, and the next section sets out what it is.

2.2.4.1 Electron - Electron Møller Scattering

Within a low-temperature plasma the electrons move ceaselessly at energies of roughly 1–10 eV, colliding with one another and with ions and neutrals. The collision between two electrons—the process in which they scatter through the Coulomb interaction, $e^-e^- \to e^-e^-$—is called *Møller scattering* in quantum electrodynamics, a purely pointlike exchange of a photon and one of the most fundamental QED processes [44].

This scattering is where the two trajectories of this chapter meet. It is one concrete instance of the plasma collisions of Section 2.2, whose root is the stochastic limit. It is also a process that carries a scattering amplitude— the class of quantity the amplituhedron of Section 2.1.3 computes. The correspondence stops there. Møller scattering is not an $N = 4$ process, and this study does not compute it; the amplituhedron serves as a controlled surrogate in which the arithmetic question can be posed exactly. What the two share is the kind of quantity, not the theory. This standing is examined in full in Chapter 5.

Why Møller scattering, among the plasma's many collisions? The reason is bound to the character of this study. This study does *not* seek to predict a particular process value with precision; it seeks to test a single possibility—whether the origin of the probabilistic character lies in the mathematical language. For that purpose

[42] For the non-equilibrium character of a low-temperature plasma, in which the electron energy is of the order of the ionization potential (about 10 eV), see [85]; for the general context, [8, Chapter 11.2].

[43] On the glow discharge as a representative non-equilibrium low-temperature plasma, formed when a low-pressure gas is driven by a discharge voltage, see [8, Chapter 14].

[44] On Møller scattering as a fundamental QED process, see [55, Chapter 5.4]; on its status as the dominant channel in low-energy (< 100 MeV) electron scattering, see [86].

the object of computation should be as physically simple, and its theoretical expression as clear, as possible. Møller scattering answers to this exactly. The scattering of two electrons is among the cleanest processes in quantum electrodynamics, its spin-averaged differential cross-section given in closed form at the textbook level [45], and unlike scattering that involves the internal structure of ions or neutrals, it is a purely pointlike interaction between two particles. This follows the same principle of simplification by which the study adopts $\mathcal{N} = 4$ SYM—the simplest gauge theory—and, within it, the minimal non-trivial toy model $(6, 2, 4)$.

This choice carries a limitation that must be stated candidly. The object is electron–electron scattering in the eV range of a low-temperature plasma, and in this range precise measurements of Møller scattering effectively *do not exist*. Such measurements are carried out at far higher energies: the SLAC E158 experiment in the tens of GeV [88], the MOLLER experiment proposed at Jefferson Lab likewise in the GeV range [89], and the lowest among them—a measurement at about 2.5 MeV at the MIT High Voltage Research Laboratory—still some six orders of magnitude above the eV range treated here [86]. This gap follows from the very principle of simplification: choosing the low-energy scattering of a low-temperature plasma to keep the computation simple leads, conversely, to a region where precise measurement is scarce. What significance and what constraint this carries for interpreting the results is discussed in Chapter 5, where the simulation is treated in earnest.

With this, the physical object of computation is fixed: the electron–electron Møller scattering within a low-temperature plasma. from which this study draws the regime of its computation. The discussion so far has answered what the question is about; there remains the question of the *language* in which it is posed. he following section takes up that language—non-standard analysis and the p-adic number system.

2.3 Mathematical Languages

This study identified two languages to stand in place of the real numbers, both non-Archimedean: *non-standard analysis* and the *p-adic number system*. Non-standard analysis is the language this study declared, in Chapter 1, as the place where its problem-awareness first arose. The question this study posed—what is the minute discrepancy that the real number system fails to capture in describing the microscopic world?—touches exactly on what this language treats. It is the language in which this study reasons about that discrepancy, and the core concept sustaining that reasoning is the *monad.monad*.

2.3.1 Non-standard analysis and Monad

What non-standard analysis restored is the *infinitesimal*. The infinitesimal was, for much of the history of calculus, the intuition that guided computation; but its logical status was deemed obscure, and in the nineteenth-century rigorization of analysis it was displaced by the notion of the *limit*. The achievement of non-standard analysis was to restore the infinitesimal as a rigorously defined mathematical object [46]. To this end it introduces the *hyperreal* system $\mathbb{R}^*$, an extension of $\mathbb{R}$. Within this

[45] For the spin-averaged Møller cross-section in the centre-of-mass frame, retaining the electron mass, see [87, Exercise 3.8, Eq. 18-22]; the crossing-symmetry machinery and Mandelstam variables underlying it follow [55, Chapter 5.4].

[46] Non-standard analysis arose from Robinson's attempt to resolve rigorously the logical contradictions that infinitesimal calculus had incurred; [48] is the founder's own account of the theory's foundations and applications. For a more pedagogical treatment, see [49].

extended system there exist numbers that are non-zero yet smaller than any positive real—the infinitesimals. That such a number lies below every positive real is exactly what shows the system to be non-Archimedean [47]. On the real line a finite distance separates any two points; in the hyperreal system there exist numbers infinitely close to a point yet not equal to it.

The structure these infinitesimals form around a real point is the *monad*. The monad of a real point is the collection of hyperreals lying an infinitesimal away from it—its *infinitesimal neighborhood* [48]. The real number system collapses this neighborhood into a single point, absorbing its fine interior structure through the operation of the limit. The monad, by contrast, lays that structure open. This view, in which a single point in fact harbors a whole world of infinitesimals, connects directly to the argument of Chapter 1. When this study, treating the discrepancy between reality and language, denoted its unit by m, that m was drawn from precisely this monad [49]. The region of infinitesimals that the language of the reals crushes into one point is exactly where this study located the discrepancy of the microscopic world.

Here the *monad* meets the study's problem head-on. As a language for reasoning about the discrepancy the reals cannot hold, nothing is more direct than non-standard analysis and its monad. Yet this language carries, for this study's purpose, one decisive limitation: the construction of the hyperreal system is *non-constructive*. Its existence is proved, but individual infinitesimals cannot be held as concrete values and placed into a computation [50]. Non-standard analysis therefore remains a language of diagnosis. What this study ultimately seeks is to verify a possibility *through computation*, and that requires a *computable* number system. That language is the p-adic number system, treated in the next section.

2.3.2 p-adic Number System

That system is the p-adic numbers. Like non-standard analysis, the p-adic system is non-Archimedean; unlike the hyperreals, its elements can be handled as explicit numerical values and placed into actual computation.

Introducing p-adic numbers into physics is neither new nor isolated. The lineage begins with Volovich, who in 1987 proposed a p-adic string theory defining scattering amplitudes over the p-adic numbers rather than the reals [51]. The application of p-adic numbers to quantum mechanics and field theory continued steadily thereafter, settling into a research tradition now called p-adic mathematical physics [52]. This study's attempt to treat, over a different number system, the scattering amplitude

[47] The Archimedean property states that for any positive a, b there is always a natural number n with $na > b$. An infinitesimal ε never exceeds a given positive real however large the n multiplying it, and so violates this property directly; this is precisely what it means for the hyperreal system to be non-Archimedean. See Section 1.4.2 (1.1).

[48] For the formalization of the monad, see [48, Chapter 3]: the monad of a point x is the set of all hyperreals whose difference from x is infinitesimal.

[49] See Section 1.4.1. The unit m, introduced for the discrepancy between reality and the language of the reals, corresponds to the infinitesimal region that the real number system absorbs into a single point; its notation derives from the monad.

[50] The standard construction of the hyperreals quotients the set of real sequences by an ultrafilter, and the required free ultrafilter depends on the axiom of choice and cannot be specified explicitly. This is why the construction is called non-constructive, and it is the source of the computational constraint that individual infinitesimals cannot be given as explicit numerical values. See [48, Chapters 2–3].

[51] [90]. Volovich set out the underlying viewpoint in a preprint of the same year, later published as [91].

[52] For standard reviews of the formation and development of p-adic mathematical physics, see [92]; for a more modern overview, [93].

once defined over the reals stands—broadly—in the line of the problem-awareness this tradition first raised: the idea of changing the number system that underlies our description of the world.

More recently, attempts have appeared that use p-adic numbers directly in the computation of scattering amplitudes. In 2022, De Laurentis and Page introduced p-adic numbers—closely tied to finite fields—as a tool for numerical evaluation, characterizing how the rational coefficients of a scattering amplitude diverge near a singular surface [53]. How these recent attempts use p-adic numbers, and how their purpose differs fundamentally from that of this study, is discussed in detail in the conclusion, once all results are in hand. In this study, the p-adic numbers are introduced not to replace or approximate non-standard analysis, but as a separate language—one that implements the same non-Archimedean problem-awareness in computable form. What, then, is this p-adic system concretely, and how does it hold discreteness and computability at once? The next section takes this up.

2.3.3 Discrete Metrics and Precision in p-adic System

The p-adic system parts from the reals at the very point of how distance is measured. The real system $\mathbb{R}$ is the completion of the rationals $\mathbb{Q}$ with respect to the ordinary absolute value; the p-adic system $\mathbb{Q}_p$ is the completion of the same rationals with respect to an entirely different one [54]. Fix a prime p. Writing a non-zero rational x as $x = p^v \cdot (a/b)$, with a, b not divisible by p, the exponent v is the p-adic valuation of x, written $v_p(x)$ [55] Through it the p-adic absolute value is defined as [56],

$$|x|_p = p^{-v_p(x)}, \qquad |0|_p = 0 \tag{2.16}$$

By this definition, the more divisible a number is by p—the larger $v_p(x)$—the *smaller* it becomes in p-adic distance. This absolute value satisfies a condition stronger than the ordinary triangle inequality:

$$|x + y|_p \leq \max(|x|_p, |y|_p) \tag{2.17}$$

An absolute value satisfying this *strong triangle inequality* is called non-Archimedean, and the distance it defines is an *ultrametric* [57].

From this ultrametric property follows the decisive feature of the p-adic system. The values the p-adic absolute value can take are, as $|x|_p = p^{-v_p(x)}$ shows, only integer powers of p: $\ldots, p^{-1}, 1, p, p^2, \ldots$. Unlike the real absolute value, which takes

[53] [94]. Exploiting the stability of p-adic numerical evaluation near singular surfaces, they gave a proof of concept reconstructing the rational coefficients of a two-loop pentagon function in fewer than a thousand evaluations.

[54] For a standard introduction to the p-adic system, see [95]; for the classic widely cited in mathematical physics, [51]. The account of p-adic definitions and basic properties below follows Gouvêa, Chapter 2.1–3.2.

[55] For the basic properties of the p-adic valuation v_p are the following: Lemma 2.1.3 For all $x, y \in \mathbb{Q}$, we have $i) v_p(xy) = v_p(x) + v_p(y)$ and $ii) v_p(x + y) \geq \min v_p(x), v_p(y)$, with the obvious conventions with respect to $v_p(0) = +\infty$.

[56] For the definition of the p-adic valuation, see [95, Definition 2.1.4]; for its basic properties—$v_p(xy) = v_p(x) + v_p(y)$ and $v_p(x + y) \geq \min\{v_p(x), v_p(y)\}$, with $v_p(0) = +\infty$ by convention—see Lemma 2.1.3 of the same work.

[57] For the strong triangle inequality and the definition of a non-Archimedean absolute value, see [95, Definition 2.1.1, (iv)]; for the properties of ultrametric spaces, Chpater 2.3 (Lemma 2.3.3). The real absolute value does not satisfy this condition.

every continuous value from zero upward, the p-adic distance has only this discrete hierarchy of values and never anything between them. Distance itself does not run continuously but breaks into discrete strata. Here the contact with the problem-awareness of Chapter 1 appears. Where Chapter 1 argued that the continuous distance structure of the reals may fail to hold a discrete microscopic world, this discrete structure of the p-adic distance is exactly where an alternative may lie [58].

Discreteness, however, is not the only reason the p-adic system is adopted here. The hyperreal system, being non-constructive, could not reach actual computation—and it is exactly here that the p-adic numbers part from it. A p-adic number is expressed by a p-adic expansion $x = \sum_{i \geq v} a_i p^i$ (with $0 \leq a_i < p$), and truncating this series at a finite number of digits places it into actual computation at any desired precision [59]. The representative theorem underwriting this computability is *Hensel's lemma*. For a polynomial $F(X)$ with p-adic integer coefficients, if a p-adic integer α_1 satisfies

$$F(\alpha_1) \equiv 0 \pmod{p}, \qquad F'(\alpha_1) \not\equiv 0 \pmod{p} \tag{2.18}$$

that is, if α_1 is a simple root modulo p—then there exists a unique p-adic integer α with $\alpha \equiv \alpha_1 \pmod{p}$ and $F(\alpha) = 0$ [60]. This unique root α is built by an iterative procedure that, starting from α_1, fixes one p-adic digit of the solution at each step—a procedure essentially identical to Newton's method for finding roots over the reals [61].

In short, the p-adic numbers meet the requirement the hyperreals could not—actual computability—and this is why this study chose them as its language of verification. One more bridge is needed, however, to connect this computable discrete system to the positivity conditions of the amplituhedron, the actual object of this study's computation. That bridge is the tropical geometry of the next section.

2.3.4 Tropical Geometry

To connect the p-adic system to the amplituhedron —this study's actual object of computation—one more bridge is needed: something that brings the valuation structure of the p-adic numbers into contact with the language of *positivity* on which the amplituhedron stands That bridge is *tropical geometry*.

Tropical geometry is a language that redraws algebraic objects, defined over a field equipped with a valuation, as *piecewise-linear* structures through that valuation.

[58] See Chapter 1. The argument for why the continuous distance structure of the reals is limited in describing a discrete microscopic world was made there and is not repeated here.

[59] That a p-adic number has the expansion $x = \sum_{i \geq v} a_i p^i$ ($0 \leq a_i < p$, $v = v_p(x)$) is given in [95, Chapter 4.3]. Truncating the series to finitely many terms approximates and computes it to any p-adic precision. That this study's simulation can compute the canonical form over the rationals by exact arithmetic and then analyze the result p-adically rests on **the rationals sitting jointly in both the real and the p-adic completions**.

[60] For the exact statement and proof of Hensel's lemma, see [95, Theorem 4.5.2]. Here $F'(X)$ is the formal derivative, and the simple-root condition $F'(\alpha_1) \not\equiv 0 \pmod{p}$ guarantees a unique lift; this form of the lemma holds for every prime, $p = 2$ included. Only when the simple-root condition fails does the refinement require a stronger hypothesis, and $p = 2$ is the case where that hypothesis is most restrictive.

[61] That the proof of Hensel's lemma rests on an iterative construction essentially identical to Newton's method is explicit in the proof of [95, Chapter 4.5]. As Newton's method over the reals raises the precision of the solution at each iteration, Hensel lifting fixes one p-adic digit of the solution at each step.

The valuation at its starting point is a function $\mathrm{val} : K \to \mathbb{R} \cup \{\infty\}$ on a field K satisfying $\mathrm{val}(ab) = \mathrm{val}(a) + \mathrm{val}(b)$ and $\mathrm{val}(a+b) \geq \min\{\mathrm{val}(a), \mathrm{val}(b)\}$ [62]. As the definition makes plain, the representative example of a valuation is precisely the p-adic valuation v_p of the previous section; indeed, the standard exposition of tropical geometry takes the p-adic valuation as its prototypical case [63]. This valuation moreover induces a non-Archimedean norm $|a| = \exp(-\mathrm{val}(a))$, exactly the structure of the p-adic absolute value $|x|_p = p^{-v_p(x)}$ of the previous section. The discrete distance structure of the p-adic numbers and the piecewise-linear structure of tropical geometry are thus two faces of a single root: the valuation.

It is this tropical geometry that meets the positivity condition of the amplituhedron. As seen in Section 2.1.3, the tree amplituhedron is obtained by projecting the positive Grassmannian $G_+(k, n)$—the region where the ordered minors are all positive. Rendering this Grassmannian in the language of tropical geometry gives the *tropical Grassmann-ian*, and the *positive* tropical Grassmannian —the one carrying the positivity condition as well—has recently been shown to mesh with the combinatorial structure of the amplituhedron in the $m = 2$ case [64]. In other words, the tropical geometry that follows naturally from the p-adic valuation has a point of contact with the very positivity structure on which the amplituhedron stands. This is the conceptual bridge that allows p-adic numbers to be brought into the computation of the amplituhedron.

The purpose of this section reaches only so far as establishing, conceptually, that this bridge exists. Its concrete implementation lies beyond the scope of this study: the simulation that follows takes the existence of the connection as background and computes the canonical form Ω directly. With this, both questions have been answered—what this study computes (Section 2.1) and in which language it computes (Section 2.3). The object of computation and the language of computation are both in hand; what remains is to mesh them and see the result. In the next chapter, this study computes the canonical form of the amplituhedron over two arithmetic systems—the *rational system* and a *p-adic perturbation* of it —and examines how the structure of the result diverges between them.

[62] For the axiomatic definition of a valuation, see [96, Chapter 2.1]: $\mathrm{val} : K \to \mathbb{R} \cup \{\infty\}$ satisfies (1) $\mathrm{val}(a) = \infty \Leftrightarrow a = 0$, (2) $\mathrm{val}(ab) = \mathrm{val}(a) + \mathrm{val}(b)$, and (3) $\mathrm{val}(a+b) \geq \min\{\mathrm{val}(a), \mathrm{val}(b)\}$.

[63] That the p-adic valuation is presented as the prototypical example of a valuation is seen in [96, Chapter 2.1]. The same work shows that this valuation induces a non-Archimedean norm of the form $|a| = \exp(-\mathrm{val}(a))$, the same structure as the p-adic absolute value.

[64] For the definition of the tropical Grassmannian, see [96, Chapter 4.3]. That the positive tropical Grassmannian meshes with the combinatorial structure of the amplituhedron (regular positroid subdivisions) is confirmed in [97] and [98]; on the connection between tropical geometry, cluster algebras, and scattering amplitudes, see also [99]. The concrete implementation of this synthesis lies beyond the scope of this study, whose simulation takes the existence of the connection as background and focuses on the direct computation of the canonical form.

Chapter 3

Simulation Construction

Chapter 2 answered two questions: *what* is to be computed, and *in which language*. It narrowed the object of computation from quantum field theory down to the amplituhedron, and further to its canonical form; it established the p-adic numbers—a non-Archimedean number system —as the language in which that object would be described.

This contact is aimed at the central hypothesis of Chapter 1: that the probabilistic character of the microscopic world is not intrinsic to nature, but arises from the limits of the real-number language chosen to describe it[1]. If this holds, then computing the same physical object over a number system other than the reals ought to produce a qualitative difference in the structure of the result. The simulation of this chapter attempts to observe that difference directly, on the canonical form of the amplituhedron.

To this end, two simulations suggest themselves. The first (Simulation 1[2]) computes the amplitude $\Omega(n, k)$ of the amplituhedron over two arithmetic systems—the rational system (QQ) and a system carrying a p-adic perturbation (QP)—and compares how the two *diverge*. The second treats the inverse problem: the *reconstruction of the input* from the amplitude. That problem demands a separate treatment of its own—concerning the qualitative difference in structure between the real and p-adic descriptions—so this thesis carries out only the first, Simulation 1, and defers the inverse problem to *future work*[3].

This chapter proceeds as follows. It first presents the hypothesis from which the simulation departs (Section 3.1), then defines the concrete goal and the parameters that follow from it (Section 3.2), and finally describes the computational framework that implements the amplituhedron, set against the definition of the original paper (Section 3.3).

3.1 The Precision Hypothesis

The hypothesis is the following: the p-adic-based (QP) amplituhedron will measure the amplitude more precisely than the rational-based (QQ) amplituhedron. Should it hold, a clue would follow toward reducing probabilistic uncertainty— one step closer to a more predictable, more *deterministic* picture of spacetime.

In this study, precision denotes not the usual sense of how *close* a computed amplitude lies to its true value, but the *depth* to which a number system captures, without loss, the information the amplitude carries. Numerical computation over the real

[1] The formulation *probability = the noise of the reality-language mapping* is given in Section 1.4.1

[2] Simulation 1 is referred to simply as *the simulation* throughout this study

[3] The inverse problem lies outside the scope of this thesis; the forward simulation examined here is its necessary precondition. See the concluding chapter for further discussion.

field inevitably discards minute information—truncating irrationals to finite digits, accumulating floating-point operations. If what is thus discarded amounts not merely to computational error but to the microscopic discrepancy the real-number language is in principle unable to hold—as argued in Chapter 1—then a number system able to capture that discrepancy possesses precision of a different order from the reals[4]. The QQ track stands in for that real-number computation. The rationals lie within the reals, so the rational track is taken here to inherit the limits of the real-number language, and the comparison is posed as QP against QQ rather than against the reals themselves. It is precision in this sense that the hypothesis attributes to the QP track.

This expectation is not a groundless intuition. The non-Archimedean structure of the p-adic numbers has in fact been employed as a tool for circumventing difficulties peculiar to real-based numerical computation. As seen in Section 2.3.3, the p-adic distance is an ultrametric satisfying the strong triangle inequality; errors do not accumulate in the ordinary manner but are controlled discretely, by the hierarchy of valuation. By virtue of this, p-adic computation can remain stable near the singularities where real computation breaks down. A recent study has introduced p-adic numbers directly into the computation of scattering amplitudes, stably reconstructing rational coefficients near singular surfaces [94]. This observation—that the p-adic system exhibits numerical behavior distinct from the real system—forms the background of the present hypothesis: that in the amplitude computation of the amplituhedron, the QP track will likewise capture information in a manner different from QQ[5].

This hypothesis makes direct contact with the central problem of Chapter 1. There, this study advanced the possibility that the probabilistic character of the microscopic world originates not in nature itself but in the discrepancy—the *mapping noise*—between physical reality and the real-number language chosen to describe it[6]. If this is real, then computing the same physical object over a number system other than the reals ought to make some part of that discrepancy manifest in a different form. The precision hypothesis translates this discrepancy into a concrete, measurable quantity—the difference between QP and QQ—and the present simulation is the first attempt to verify that difference on the canonical form of the amplituhedron.

The purpose of this simulation is not to assert the final conclusion that the p-adic framework supplants the real-number system, but to *explore its possibility*. The hypothesis is therefore not a settled proposition awaiting confirmation, but a starting point that sets the direction of exploration. What this chapter must determine is how the hypothesis manifests in actual computation—and, should the outcome diverge from the prediction, what that divergence reveals.

To put the hypothesis to the test, two things must be specified: what is taken as input, and by what quantity the difference between the two tracks is measured. The following section defines the concrete goal of the simulation and the parameters it requires.

[4]The argument that the real-number language is in principle unable to hold microscopic discrepancy is developed in Chapter 1.

[5]How the purpose for which this study introduces p-adic numbers differs fundamentally from such prior numerical uses is discussed in the concluding chapter, once the full results are in view.

[6]The formulation *probability = the noise of the reality–language mapping* is given in Section 1.4.1.

3.2 Goal: Characterizing the discrepancy between QQ and QP

The primary goal of this simulation is to *characterize the discrepancy* between the amplitudes computed over the two arithmetic systems, QQ and QP. Concretely, this study tracks how the discrepancy patterns with the number of particles n and the helicity grade k. Should such a pattern emerge, it is a crucial clue that the p-adic structure actively participates in the amplitude computation [7].

This exploration is conducted within an idealized theoretical framework— planar $\mathcal{N} = 4$ super Yang-Mills (SYM) theory [8]. Mapping its results directly onto the dynamics of actual spacetime may therefore leave a gap.

This idealization does not render the study meaningless, for two reasons. First, $\mathcal{N} = 4$ SYM is among the most mathematically well-controlled gauge theories in modern physics, sharing the essential perturbative structure of gauge-theory amplitudes with real-world physics [62]. Second, the amplituhedron itself is an object built upon precisely this $\mathcal{N} = 4$ SYM kinematic space [61]. For these reasons the simulation yields data that, while not strictly empirical, are theoretically meaningful. This study treats $\mathcal{N} = 4$ SYM not as a phenomenological theory requiring experimental verification, but as a single, mathematically exact, closed computational environment; the burden of verification is thereby confined to internal consistency within that environment and to the structures it reveals[9].

In accordance with this standing, the input is drawn not from complex empirical data but from a simplified conceptual model of the kind theoretical physics calls a *toy model*. Two methodological grounds justify this. First, the primary goal of Simulation is a proof of concept. Introducing empirical data at this stage would inject external physical noise—thermal fluctuations, many-body collision variables, and the like—making it impossible in principle to isolate whether an observed discrepancy stems from the physical environment or from the structure of the number system. Second, decoupling the mathematical framework from empirical complexity lets one observe the arithmetic effect of the chosen number system in pure form. An idealized model such as planar $\mathcal{N} = 4$ SYM functions as the harmonic oscillator of modern quantum field theory [100]— a structurally complete, noiseless environment in which a new mathematical tool can be rigorously validated before being scaled to macroscopic, chaotic systems.

3.2.1 Definition of Parameters

The amplituhedron is fixed by three kinematic variables—n, k, m—introduced abstractly in Section 2.1.3, together with the external data Z, here fixed by the moment-curve sequence a. Each is set to a concrete value below, with its reason. Two further parameters, absent from the definition of the amplituhedron itself, enter as well: the prime p that grounds the p-adic track, and the perturbation ε that separates QP from QQ.

Number of particles n. The number of external scattering particles n fixes the number of columns of the input matrix Z. Here n ranges over $\{5, 6, 7, \ldots, 12\}$at $k = 1$

[7] This clue connects to the number-theoretic structure actually uncovered in the results of Chapter 4.

[8] On $\mathcal{N} = 4$ SYM, see Section 2.1.3.1.

[9] The closed-simulation assumption—and the fuller defense of treating an exact theory as a self-contained computational world—is reserved for the concluding chapter.

and $\{6,7,\ldots,11\}$ at $k = 2$, bounded below by theory and above by information saturation.

The lower bound is fixed strictly by the amplituhedron definition $\Omega^{(m)}_{n,k}$ $(n \geq k+m)$ [61]; it begins at $n = 5$ for $k = 1$ and $n = 6$ for $k = 2$. The upper bound is set by information saturation. The BCFW decomposition generates a number of cells that grows polynomially in n at fixed k; the directly measured counts are:

	$k=1$	$k=2$	$k=3$	$k=4$
$n=5$	1	–	–	–
$n=6$	1	1	–	–
$n=7$	2	2	1	–
$n=8$	3	4	3	1
$n=9$	4	8	8	4
$n=10$	5	14	19	14
$n=11$	6	22	40	40
$n=12$	7	32	76	100

TABLE 3.1: Number of BCFW cells produced by this implementation. The full table for $n = 5$ to 20 is given in Appendix B.

At $n = k+m$ the amplituhedron is a single simplex and the count is 1. Beyond that threshold the count is exactly $n-5$ for $k = 1$ and $(n-6)(n-7)+2$ for $k = 2$, and grows cubically for $k = 3$. Information saturation fixes the cutoff. At $k = 1$ the recursion splits into two MHV sub-problems whose canonical forms are each unity, so for every $n > k+m$ the amplitude collapses to the cell count, $\Omega(n,1) = n-5$; the $k = 1$ track therefore serves as a degenerate control rather than as a source of arithmetic structure. At $k = 2$ the small-prime structure this study examines is present at $n = 6$ and $n = 7$ and gives way at $n = 8$ to the large-prime regime treated in Section 4.1.3. Larger n yields no new structural information. The experiments are therefore restricted to $n \leq 12$ for $k = 1$ and $n \leq 11$ for $k = 2$. This is not a limitation of the implementation but a reasoned cutoff[10].

The remaining kinematic variables are fixed directly. The helicity grade k sets the depth of the BCFW recursion; here $k \in \{1,2\}$. The bosonic dimension is fixed at $m = 4$, matching the four-dimensional spacetime of planar $\mathcal{N} = 4$ SYM. The moment-curve parameter sequence is the simplest strictly increasing integers, $a = (1,2,\ldots,n)$. This choice does double duty. The strict ordering $0 < a_1 < \cdots < a_n$, combined with the moment-curve construction $Z^j_a = a^j$, with $j = 0,1,\ldots,k+m-1$, guarantees the positivity of all ordered minors through the Vandermonde determinant—the foundational requirement of the amplituhedron[11]. The integer entries then permit exact rational arithmetic, which preserves the prime-factorization structure the p-adic analysis requires. The mechanism of this positivity is demonstrated concretely in Steps 1 and 2 of Section 3.3.3.

[10] The cell count reported is that of this implementation's specific BCFW recursion. The canonical form is independent of the cell decomposition [61, Section 6], and other schemes may yield different counts for the same Ω; the correctness of the amplitude values is verified independently through the five theoretical invariants of Appendix D. The counts also satisfy the parity relation $k \leftrightarrow n-m-k$ exactly, verified over all thirty-five conjugate pairs in the range $5 \leq n \leq 14$ with no exception; this serves as an independent structural check on the implementation.

[11] The moment-curve matrix $Z^j_a = a^j$ is a Vandermonde matrix, whose determinant is the difference product $\det(Z_I) = \prod_{i<j,\, i,j\in I}(a_j - a_i)$; [101]. For a strictly increasing sequence $0 < a_1 < \cdots < a_n$, every factor $a_j - a_i$ $(j > i)$ is positive, so the determinant is strictly positive, and all ordered minors of Z are positive.

Prime base p. The prime p for the QP track is fixed at $p = 2$ by two requirements. First, non-triviality: p must appear as a prime factor of the amplitude $\Omega(n,k)$ on the moment curve, for otherwise $v_p(\Omega) = 0$ and the p-adic valuation of $\Omega_{QP} - \Omega_{QQ}$ becomes structurally trivial. Second, maximum dynamic range: among the primes that appear, the one of highest multiplicity in the factorization of Ω affords the widest range for valuation measurement. Direct preliminary computation on the moment curve $a = (1, \ldots, n)$ shows the factorization of $\Omega(n,k)$ to be dominated by 2, 3, and 5 in the simplex configurations $n = k + m$, with $p = 2$ carrying the highest multiplicity in every tested case:

$$\Omega(5,1) = 2^{20} \cdot 3^8 \tag{3.1}$$

$$\Omega(6,2) = -2^{32} \cdot 3^{12} \cdot 5^3 \tag{3.2}$$

$$\Omega(7,3) = 2^{41} \cdot 3^{18} \cdot 5^6 \tag{3.3}$$

Choosing $p = 2$ thus maximizes the resolution of the p-adic depth measurement. By contrast, larger primes such as $p = 23$—which for $n < 23$ do not divide Ω on the moment curve—were verified to yield trivial valuations, and were excluded[12].

Perturbation and topological equivalence. The baseline input matrix is shared by both tracks, and the two diverge only in what is added to it: QQ takes the baseline as is, while QP carries a perturbation scaled to a prime power $p^{\text{prec}/2}$ rather than to an arbitrary real increment. The reason lies in the metric each track works within—the rational system rests on the ordinary Euclidean distance, while the p-adic system operates on a non-Archimedean metric, where distance is set by prime divisibility.

What is decisive here is that this choice is *not* arbitrary. By Ostrowski's theorem, the p-adic absolute values and the ordinary one exhaust the non-trivial absolute values on $\mathbb{Q}$[13]. Scaling the perturbation to its own absolute value is therefore not one option among many; it exhausts the ways a rational input can be locally displaced at all. No third topology is being passed over.

With the baseline shared, the perturbation parameters for QP are set as follows. The control parameters are $(p, \text{prec}, n_{\text{step}}) = (2, 64, 1)$: the precision $\text{prec} = 64$ sets a numerical ceiling for the depth measurement, and the step $n_{\text{step}} = 1$ is the minimal non-trivial perturbation. The component-wise perturbation function is

$$\varepsilon_{a,j} = \frac{n_{\text{step}}}{p^{\text{prec}/2}} \cdot (j+1) \cdot a \tag{3.4}$$

The factor $(j+1) \cdot a$ ensures non-uniformity across entries, preventing the cancellation of p-adic depth during fraction reduction in the BCFW computation[14]. The perturbed input matrix

$$Z^{j}_{\mathbb{Q}_p, a} = a^{j} + \varepsilon_{a,j} \tag{3.5}$$

[12] This dominance is sharpest at $n = k + m$, where a single simplex cell carries the entire amplitude, and it degrades as the recursion deepens: $\Omega(7,2)$ already carries the primes 17 and 1489, and for $n \geq 8$ at $k = 2$ the valuation $v_2(\Omega)$ falls to 1 or 2. Section 4.1.3 examines this boundary and the regime beyond it.

[13] Every non-trivial absolute value on $\mathbb{Q}$ is equivalent to $|\cdot|_p$ for some prime p, or to the ordinary absolute value ($p = \infty$); [95, Section 3.1].

[14] A uniform perturbation—a constant ε added to all entries—leads, via fraction reduction, to cancellation of the $p^{\text{prec}/2}$ factors in $\Omega_{QP} - \Omega_{QQ}$, collapsing the measurable depth to zero. The non-uniform form breaks this degeneracy and preserves the p-adic structure of the perturbation.

thus retains the baseline kinematic structure while being displaced by an increment whose p-adic valuation is fixed and known, $v_p(\varepsilon) = -\text{prec}/2$. It is this fixed valuation, rather than the size of the increment, that makes the depth of the resulting difference readable.

The parameters of Simulation are summarized in Table 3.2.

Symbol	Name	Role	Value
n	Number of particles	External scattering particles	$5 \leq n \leq 12$
k	Helicity grade	BCFW recursion depth	$k \in \{1, 2\}$
m	Bosonic dimension	Spacetime kinematics ($\mathcal{N} = 4$ SYM)	$m = 4$
a	Moment-curve params	Sequence ensuring strict positivity	$(1, 2, \ldots, n)$
p	Prime base	Base prime for p-adic valuation	$p = 2$
prec	Precision	Numerical ceiling for depth	64
n_{step}	Step	Minimal perturbation scale	1

TABLE 3.2: Summary of kinematic and p-adic perturbation parameters used in Simulation

3.3 Simulation Framework

What remains is to specify *how* the amplituhedron is actually evaluated on these inputs. This section describes the computational framework that realizes the amplituhedron, and sets out at each stage where it follows the definition given in [61] and where it departs from it. The framework is presented in four parts: the choice of implementation language (Section 3.3.1), the algorithmic pipeline and its correspondence with the original paper (Section 3.3.2), a step-by-step validation of each stage (Section 3.3.3), and the implementation itself (Section 3.3.4).

3.3.1 Rationale for using Python over SageMath

In this simulation, Python was chosen as the primary tool for implementing the amplituhedron and conducting the numerical experiments. Three methodological advantages ground this decision.

First, Python is the de facto standard for scientific computing and numerical simulation. Major scientific libraries—NumPy, SciPy, SymPy—are Python-based, and Jupyter notebooks are widely used for sharing results. This is more than convenience; it is a matter of reproducibility. Working in Python makes it practical for the entire series of simulations can be independently reproduced, verified, and extended by future researchers.

Second, Python offers a broad expansion path within a single language, from simple standard libraries to high-level computer algebra systems. The present stage relies on exact rational arithmetic, for which the standard `fractions` library suffices[15]. Should the research advance toward direct computation of the tropical geometry of high-dimensional Grassmannians, the transition to a specialized computer algebra system such as SageMath can be made within the same Python environment.

[15]The *fractions* module is part of the Python standard library. It represents each rational number as a pair of integers—numerator and denominator—in lowest terms, and performs the four arithmetic operations exactly, without floating-point rounding. This exactness is what preserves the prime-factorization structure of Ω required for the p-adic analysis of this study.

Expanding on one codebase, without replacing foundational tools at each stage, is a significant advantage for the continuity of the work.

Third, readability matters in an interdisciplinary domain where mathematical physics and number theory intersect. Python reveals algorithmic intent explicitly, through indentation-based syntax and explicit keywords. This serves not only coding convenience but external verifiability. A reader or reviewer of this study may be an expert in only one of the two domains—the amplituhedron (mathematical physics) or p-adic valuation (number theory)—or in neither. Because the code mirrors the algorithm's logical flow, researchers from either background can track and verify the computational procedure. Because the code is legible on its own terms, the algorithm flowchart and core functions are placed in the appendix rather than expanded in the body.

A word on SageMath itself is in order before those constraints are set out. SageMath is a Python-based open-source computer algebra system (CAS) that integrates a range of advanced algebraic structures within one environment—among them the p-adic numbers $\mathbb{Q}_p$, tropical semirings, and Grassmannians, all directly relevant to this study. For these strengths, a SageMath-based design was seriously considered during the initial conceptualization of this simulation.

Two practical constraints told against it. First, the development environment: SageMath ships as its own distribution and did not integrate cleanly with the standard Python toolchain used here—VS Code with WSL (Ubuntu). Second, and more decisive: the present simulation uses simple integer inputs, such as the moment-curve sequence $a = (1, 2, \ldots, n)$ of Section 3.2.1, a regime in which all arithmetic can be carried out exactly by Python's `fractions.Fraction` alone. The objectives of this simulation are thus fully met without SageMath's advanced algebraic machinery. This implementation therefore bypasses SageMath deliberately, performing p-adic valuation and exact determinant operations through custom-built `PadicValuation` and `FractionMatrix` classes[16].

This choice does not negate the value of SageMath. On the contrary, adopting a Python environment is precisely what keeps the path to SageMath open. Should the Should the research advance to direct computation on tropical-geometric structures or higher-dimensional Grassmannians[17], the simple built-in classes will no longer suffice; at that point the current codebase can transition to SageMath within the same overarching Python environment. Bypassing SageMath now is therefore not in tension with the choice of Python —it anticipates the scaling of this research.

3.3.2 Algorithm Flowchart and its Comparison with *The Amplituhedron*

Figure 3.1 presents the full algorithm of the simulation alongside its structural correspondence with [61]. The left column shows the five computational stages of this implementation, run in parallel for the QQ and QP tracks; the right column shows the corresponding constructions in the original paper.

[16]The `PadicValuation` and `FractionMatrix` classes were designed and implemented specifically for this simulation, not imported from an external module. `FractionMatrix` stores all entries as `fractions.Fraction` and evaluates determinants exactly by Gaussian elimination; `PadicValuation` computes the p-adic valuation $v_p(\cdot)$ of a rational number. Together they secure exact rational arithmetic without the overhead of a full CAS.

[17]Here higher-dimensional Grassmannians refers both to computations at higher n ($k \geq 3$) and to generalized kinematic spaces $G_+(k+m, n)$.

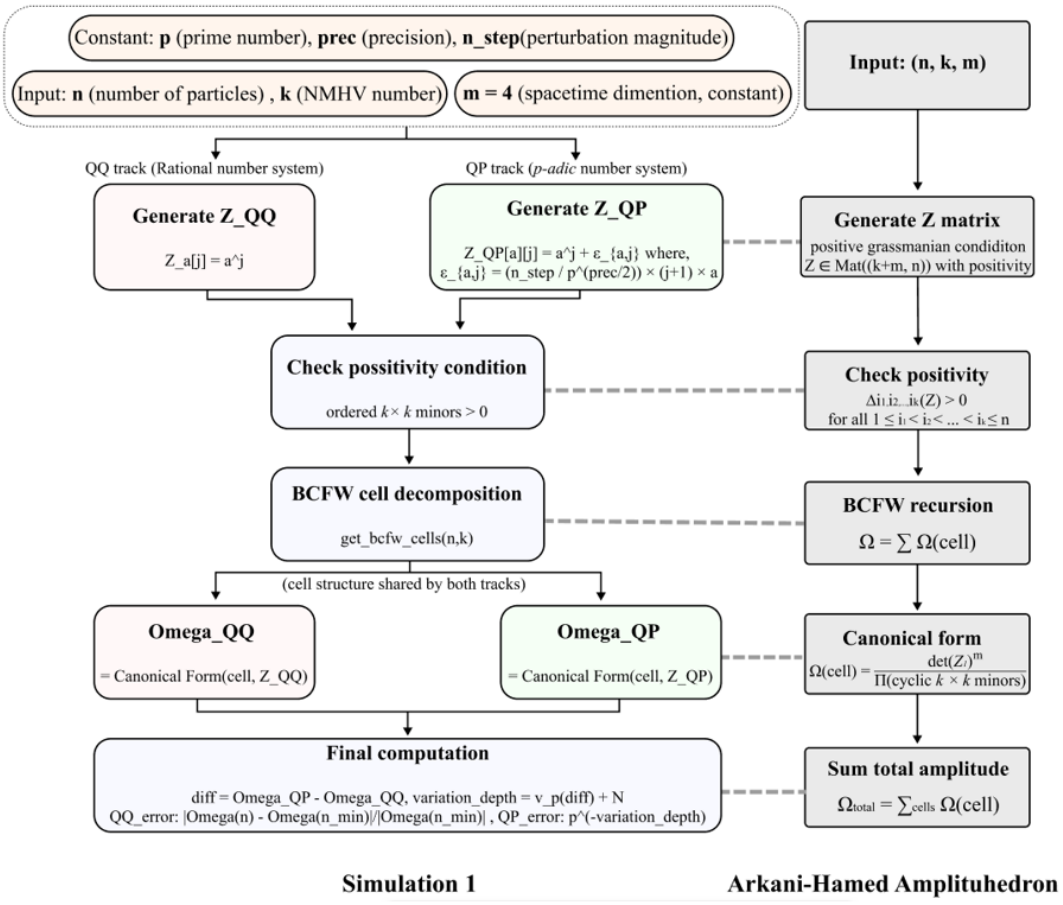


FIGURE 3.1: Flowchart of the Simulation algorithm and its correspondence with [61].

As the flowchart shows, the QQ and QP tracks share an identical pipeline. What allows both to run through the same architecture is the algebraic nature of the amplituhedron. The canonical form Ω is defined strictly as a rational function of the components of the input matrix Z. For a simplex cell, in the canonical-form stage of Figure 3.1, the form of a simplex cell is [18]:

$$\Omega(\text{cell}) = \frac{\det(Z_I)^m}{\prod(\text{cyclic } k \times k \text{ minors})} \tag{3.6}$$

Numerator and denominator are both polynomials in the components of Z, so evaluating this rational function requires nothing but the four field operations $(+, -, \times, \div)$. These formal operations behave identically regardless of the field the entries are drawn from:: the construction of polynomial rings, and of rational functions over an integral domain, does not depend on the particular field[19]. Whether Z carries the baseline moment-curve entries or the p-adically scaled perturbation of Section 3.2.1, both matrices have rational entries and pass through the same rational function; the canonical form is computed by the exact same arithmetic.

This is the core design principle of the simulation. At the code level, both tracks share the same functional modules—`get_bcfw_cells`, `compute_total_amplitude`, and the rest. Their only difference lies in the input matrix Z. The tracks branch at the initial stage alone, to generate their respective inputs: QQ uses the baseline moment-curve Z as is, while QP uses the component-wise p-adically perturbed Z. After that branching both follow the identical pipeline—positivity check, BCFW cell decomposition, canonical-form evaluation—before merging at the final output.

Because the two pathways are perfectly parallel, it suffices to demonstrate the theoretical consistency of the algorithm once. The step-by-step validation of the following section, carried out against the original amplituhedron framework, applies simultaneously to both tracks.

3.3.3 Step-by-step Algorithm and its legitimacy

This section describes the five-step algorithm on the rational (QQ) track; by the equivalence of Section 3.3.2, the account applies verbatim to QP.

Each step is explained so as to make clear how the algorithm corresponds to the definition of the amplituhedron in [61]. To keep the calculations concrete, the base case of the recursion, $(n, k, m) = (5, 1, 4)$ serves as the main walkthrough throughout. Where a step turns trivial at $k = 1$, an auxiliary case with non-trivial behavior is introduced alongside it; the full walkthrough of that larger case is organized in Appendix **??**[20].

[18] Recursive cells are not evaluated by this formula directly; each is assembled from the canonical forms of its two sub-problems, and the lowest hierarchy (MHV) contributes unity. The construction is set out in Section 3.3.3.

[19] The field-independence of polynomial and rational-function rings over a commutative ring is a standard result in commutative ring theory [102].

[20] The case $(n, k) = (5, 1)$ is the base case of the BCFW recursion—a single simplex, since $n = k + m$. The recursion itself is walked through separately in Appendix **??** on the $(n, k) = (7, 2)$ case. The $(7, 2)$ configuration is one of the core cases of this study and serves as key data in the GCD-hierarchy verification.

3.3.3.1 Step 1–Generating the moment curve Z

Given the baseline input (n, k, m, a), the QQ track generates the input matrix Z_{QQ} as a moment-curve embedding:

$$Z_{QQ}[a][j] = a^j, \quad a \in \{1, 2, \ldots, n\}, \quad j = 0, 1, \ldots, k+m-1 \tag{3.7}$$

In the matrix convention of this implementation, Z_{QQ} is a $(k+m) \times n$ matrix: the row index j is the order along the moment curve, and the column index a is the particle number. For the main walkthrough $(n, k, m) = (5, 1, 4)$, Z_{QQ} is the 5×5 matrix

$$Z_{QQ} = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 \\ 1 & 4 & 9 & 16 & 25 \\ 1 & 8 & 27 & 64 & 125 \\ 1 & 16 & 81 & 256 & 625 \end{pmatrix}. \tag{3.8}$$

Each column a is a point $Z_a = (1, a, a^2, a^3, a^4)$ on the moment curve. All entries are integers, stored as exact rationals through Python's `fractions.Fraction`.

The justification for the integer sequence $a = (1, 2, \ldots, n)$ was given in Section 3.2.1; in short, it (i) provides a canonical positive configuration in which positivity holds automatically, and (ii) renders all arithmetic exactly trackable as rationals, enabling the factorization analysis of Ω in the sections that follow.

[61, Section 3, Eq.(3.6)] introduces the cyclic polytope and its generalization as the standard configuration of positive external data, specifying the moment-curve embedding:

> n ordered points on the moment curve in $\mathbb{P}^m$, $Z_a = (1, t_a, t_a^2, \ldots, t_a^m)$, with $t_1 < t_2 < \cdots < t_n$.

This implementation applies the simplest integer choice $t_a = a$ to that definition. For $k = 1$ the two coincide exactly. For $k \geq 2$ the same construction is carried to the $(k+m)$-component moment curve, $Z_a = (1, a, \ldots, a^{k+m-1})$, which is the positive external data of the general amplituhedron. The code is handled by `generate_Z_qq(n, k, m)`, and the result is represented by the self-implemented `FractionMatrix` class.

3.3.3.2 Step 2–Testing the positivity condition

This step checks whether the Z_{QQ} of Step 1 (3.3.3.1) satisfies the positivity condition—that all ordered $k \times k$ minors are strictly positive:

$$\Delta_{i_1, i_2, \ldots, i_k}(Z_{QQ}) > 0, \quad \text{for all } 1 \leq i_1 < i_2 < \cdots < i_k \leq n \tag{3.9}$$

In the main walkthrough $(5, 1, 4)$, $k = 1$, so an ordered minor is simply a first-row entry of Z_{QQ}. The first row is $(1, 1, 1, 1, 1)$; all positive, so the condition holds.

This triviality is a feature of $k = 1$; cases with $k \geq 2$ require a non-trivial minor check. To exhibit it, consider the auxiliary case $(n, k, m) = (7, 2, 4)$[21]. Here Z_{QQ} is a 6×7 matrix (the full matrix is in Appendix **??**), and positivity requires all ordered

[21] The smallest $k = 2$ case $(6, 2, 4)$ is not used as the auxiliary case because $(6, 2)$ is itself a base case ($n = k + m$), which returns a single simplex and leaves the recursion unexercised. In $(7, 2)$ the recursion runs and the canonical form is assembled from more than one cell, so it serves as the auxiliary walkthrough throughout the main text. The toy model of Section 2.1.3.4 is chosen on a different ground— the curved boundary structure that appears at $k = 2$, which is already present in the base case.

2×2 minors to be positive. There are $\binom{7}{2} = 21$ ordered index pairs (i_1, i_2); and the general form of the minor settles all of them at once. A few examples:

$$\Delta_{1,2} = \begin{vmatrix} 1 & 1 \\ 1 & 2 \end{vmatrix} = 1 > 0, \quad \Delta_{1,5} = \begin{vmatrix} 1 & 1 \\ 1 & 5 \end{vmatrix} = 4 > 0, \tag{3.10}$$

$$\Delta_{3,7} = \begin{vmatrix} 1 & 1 \\ 3 & 7 \end{vmatrix} = 4 > 0, \quad \Delta_{4,6} = \begin{vmatrix} 1 & 1 \\ 4 & 6 \end{vmatrix} = 2 > 0 \tag{3.11}$$

In general $\Delta_{i_1,i_2}(Z_{QQ}) = i_2 - i_1$, positive for every pair with $i_1 < i_2$, so all 21 ordered 2×2 minors are thus positive. The implementation does not enumerate all of them: it evaluates the $n - k + 1$ minors on contiguous column blocks, which suffices as a guard on the monotone integer baseline.

Because the baseline uses a monotonically increasing integer sequence, all ordered minors are positive automatically by the Vandermonde structure, and the check is in principle unnecessary here. It is performed explicitly for two reasons. First, it is a defensive design: it guarantees at the algorithmic level that the positivity premise—on which BCFW decomposition and canonical-form evaluation depend—is not silently violated. Second, if the framework is later extended to non-arithmetic or non-integer baselines, positivity is no longer automatic, and the explicit check becomes an essential safeguard.

[61, Section 3, Eq. 3.5] defines the positivity of external data for general m:

> we can consider n points Z_a^I in $G(1, 1+m)$, with $I = 1, \ldots, 1+m$, which are positive: $\langle Z_{a_1} \ldots Z_{a_{1+m}} \rangle > 0$.

The bracket $\langle Z_{a_1} \ldots Z_{a_{1+m}} \rangle$ is the determinant of a $(1+m) \times (1+m)$ minor, expressing that the positive external data forms a cyclic-polytope structure in m-dimensional space. The ordered $k \times k$ minor check of this implementation is the equivalent formulation of that condition over the k-Grassmannian; for $k = 1$ it matches directly the positivity of the $(1+m)$-index bracket on the moment curve[22]. The code is handled by `is_classical_positive(Z_QQ, k, m)`.

3.3.3.3 Step 3–BCFW cell decomposition

Once the positivity of Z_{QQ} is secured in Step 2 (3.3.3.2), the amplituhedron is decomposed into a set of cells through the BCFW recursion. Each cell corresponds to a $4k$-dimensional subspace of the positive Grassmannian $G_+(k, n)$, and the collection of cells is intended to cover the tree amplituhedron in the sense of [61].

The recursion rests on two base cases: *i*) the *MHV cell* ($k = 0$), a trivial cell whose canonical form is defined to be 1; and *ii*) the *simplex cell* ($n = k + m$), decomposed into a single simplex whose canonical form is evaluated directly. If the input matches neither ($k \geq 1$ and $n > k + m$), the algorithm recurses into left and right sub-problems about a splitting point i.

In the main walkthrough $(5, 1, 4)$, $n = k + m = 5$, so the recursion does not operate and a single simplex cell is returned as a base case:

$$\text{cell} = \{\text{type} : \text{simplex}, \ \text{indices} : (1, 2, 3, 4, 5), \ k = 1\} \tag{3.12}$$

[22] The equivalence for general k remains consistent within the definition of positive external data in [61, Section 3]. For $k = 1$ and $m = 4$, the ordered 1×1 minor check validates the positivity of the first row, equivalent to the positivity of the $(1+m) = 5$-index bracket on the moment curve, by the Vandermonde structure.

Because this is a base-case property, it does not exhibit the recursion in action. The auxiliary case $(7,2,4)$ does. Since $n > k+m$ here, the recursion operates, and this implementation produces two cells:

Cell	Split i	(k_L, k_R)	Left sub-problem	Right sub-problem
0	3	$(0,1)$	MHV, particles $(1,2,3,4)$	Simplex, particles $(1,4,5,6,7)$
1	4	$(1,0)$	Simplex, particles $(1,2,3,4,5)$	MHV, particles $(1,5,6,7)$

TABLE 3.3: The two BCFW cells produced for $(n,k,m) = (7,2,4)$.

Each cell is a pair of a left and a right base case. In both, one side is MHV ($k = 0$) and the other a simplex ($n = k+m$); the decomposition therefore terminates in a single step, without further recursion[23]. That the right sub-problem's particle indices are non-contiguous— the right set $(1,4,5,6,7)$ of cell0 omits particles 2 and 3— is a natural reflection of the BCFW bridge structure: the first particle participates in both sub-problems, and so does the particle at the split point, while the particles beyond it pass to the right. The recursion tree for $(7,2)$ is shown in Figure 3.2.

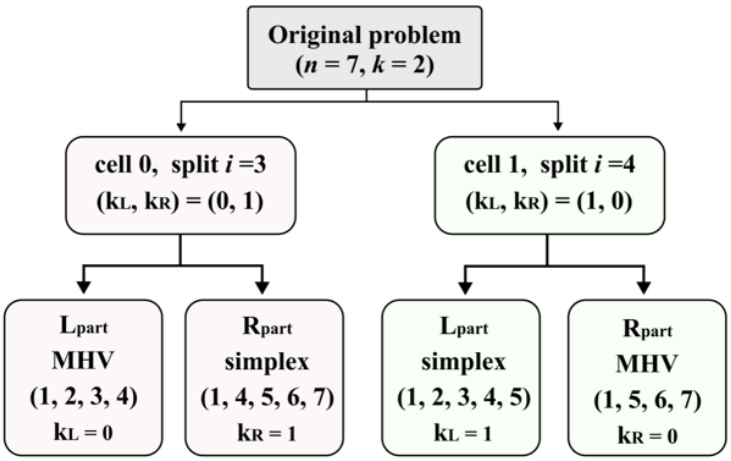


FIGURE 3.2: BCFW recursion tree for $(7,2)$. Each leaf corresponds to a base case, and no further decomposition occurs. For larger (n,k), the recursion proceeds through multiple steps, increasing the depth of the tree.

Splitting rules. The splitting rules of this implementation are as follows. Given the current input (n,k) and particle-index labels $L = (a_0, a_1, \ldots, a_{n-1})$, a combination

[23]One base case of this implementation is $k = 0$ (MHV) and the other is $n = k+m$ (simplex). In the $(7,2)$ walkthrough both cells show the pattern where one sub-problem is MHV and the other a simplex—e.g.The left of cell 0 is $(n = 4, k = 0)$, hence MHV, and the right is $(n = 5, k = 1)$, satisfying $n = k+m = 5$, hence a simplex. The symmetric appearance of the two base cases is a feature of the small size of $(7,2)$; for larger n at $k \geq 2$ the recursion runs through several steps, deepening the tree until the base cases are reached. At $k = 1$ the split always yields two MHV sub-problems, so the tree remains one level deep regardless of n.

(i, k_L, k_R) defines a valid cell if and only if it satisfies all three conditions:

$$\text{(C1) Splitting range:} \quad i \in \{2, 3, \ldots, n-2\} \tag{3.13}$$

(C2) Admissible helicity distribution:

$$k_L \in \big[\max(0,\, k-(n-i+1)),\ \min(k-1,\, i-2)\big], \quad k_R = k-1-k_L \tag{3.14}$$

(C3) Particle-number validity:

$$i+1 \geq k_L + m \ \text{(Left)}, \qquad n-i+1 \geq k_R + m \ \text{(Right)} \tag{3.15}$$

The left sub-problem has labels $L_L = (a_0, a_1, \ldots, a_i)$ of length $i+1$ and helicity k_L; the right has labels $L_R = (a_0, a_i, a_{i+1}, \ldots, a_{n-1})$ of length $n-i+1$ and helicity k_R. Conditions (C1) and (C2) delimit the combinatorially possible range of split point and helicity distribution; (C3) is an additional check guaranteeing that each resulting sub-problem forms a valid amplituhedron. Since (i, k_L) combinations can satisfy (C2) yet violate (C3), the two must be checked independently[24].

Applying the three conditions to the walkthrough $(7, 2)$ gives the following exhaustive analysis:

i	(k_L, k_R)	Left: $i{+}1$ vs. $k_L{+}m$	Right: $n{-}i{+}1$ vs. $k_R{+}m$	Result
2	$(0,1)$	$3 < 4$	$6 \geq 5$	Invalid (left insufficient)
3	$(0,1)$	$4 \geq 4$	$5 \geq 5$	**Valid → cell 0**
3	$(1,0)$	$4 < 5$	$5 \geq 4$	Invalid (left insufficient)
4	$(0,1)$	$5 \geq 4$	$4 < 5$	Invalid (right insufficient)
4	$(1,0)$	$5 \geq 5$	$4 \geq 4$	**Valid → cell 1**
5	$(0,1)$	$6 \geq 4$	$3 < 5$	Invalid (right insufficient)
5	$(1,0)$	$6 \geq 5$	$3 < 4$	Invalid (right insufficient)

TABLE 3.4: Exhaustive analysis of the splitting rules (C1)–(C3) for $(n, k) = (7, 2)$.

At $i = 2$, (C2) admits only $k_L = 0$, which then violates the left part of (C3). At $i = 3$ and $i = 4$, (C2) admits two k_L values, but only one is valid under (C3) in each case. At $i = 5$, both k_L values violate the right part of (C3). Only the two combinations $(i = 3, k_L = 0)$ and $(i = 4, k_L = 1)$ produce valid cells—exactly the two cells presented earlier.

The BCFW recursion of this implementation follows three design choices. First, the left sub-problem takes the first $i+1$ particles, and the right takes the first particle together with the particles from index i onward. This originates in the BCFW bridge structure [61, Section 3], in which the first particle participates in both sub-problems; the particle at the split point i is likewise included in both—a natural implementation of the BCFW shift. Second, at each split point i, all admissible (k_L, k_R) distributions are generated as separate cells and summed, reflecting the general structure whereby the BCFW recursion is the sum over all valid decomposition paths [61, Section 3]. Third, memoization prevents redundant evaluation of identical sub-problems: for large (n, k) the recursion tree reaches the same small sub-problems along many paths, and without caching the computation grows exponentially. This implementation realizes one particular decomposition of the BCFW recursion, and

[24] The inequalities of (C2) arise on two grounds. First, $k_L \geq 0$ and $k_R = k-1-k_L \geq 0$ must hold together, the latter giving $k_L \leq k-1$. Second, the BCFW bridge structure requires a minimum number of particles on each side of i. At $m = 4$ this second pair of bounds is strictly weaker than (C3) and never binds on its own; the implication runs one way only, and combinations satisfying (C2) may still violate (C3). Both are retained so that the constraint remains correct if m is changed.

it yields a different number of cells from the standard BCFW expansion [25]. The difference is not a choice among equivalent triangulations but a departure in how a recursive cell is assembled from its sub-problems. It therefore fixes the object on which the rest of this study operates: Ω is taken hereafter as the deterministic rational function that this implementation defines, and the arithmetic questions of Chapter 4 are posed about that function. Its internal consistency is verified against the five invariants of Appendix D.

[61, Section 5] defines the BCFW recursion as follows:

> For any k, the BCFW decomposition of tree amplitudes gives us a collection of $4k$-dimensional cells of the positive Grassmannian ... these cells are beautifully mapped into non-overlapping regions of $G(k, k+4)$ that together cover the entire tree amplituhedron.

The recursion of this implementation follows the splitting structure of that definition; where it departs is in the assembly of the cell forms, as noted above. The cell dimension $4k$ is guaranteed automatically by the splitting rules: the left dimension $4k_L$ and the right dimension $4k_R$ sum to $4(k_L + k_R) = 4(k-1)$, and the four dimensions of the BCFW bridge itself bring the total to $4k$. The original paper also specifies the non-uniqueness of the decomposition:

> These collections of cells are not unique—there are many different triangulations.

The cell structure of this implementation depends only on (n, k, m): `get_bcfw_cells` does not take Z as an argument. The two tracks of Section 3.3.2 therefore share an identical cell structure by construction. Different triangulations of the same amplituhedron yield the same canonical form; the difference registered above is of another kind, as footnote 26 sets out. The code is handled by`get_bcfw_cells(n, k, m, current_labels, cache)`.

3.3.3.4 Step 4–Evaluation of the canonical form

For each cell obtained in Step 3 (3.3.3.3), the canonical form $\Omega(\text{cell})$ is evaluated. The method depends on the cell type.

(i) *MHV cell* ($k = 0$): the canonical form is defined trivially as 1.

(ii) *Simplex cell* ($n = k + m$): the canonical form is a fraction whose numerator is the m-th power of $\det(Z_I)$ and whose denominator is the product of cyclic $k \times k$ minors:

$$\Omega(\text{simplex}) = \frac{\det(Z_I)^{n_{local}-1}}{\prod_{s=0}^{n-1} \Delta_{c_s}(Z_I)} \tag{3.16}$$

Here $I = (a_0, a_1, \ldots, a_{n_{local}-1})$ is the particle-index set of the cell, Z_I the corresponding $(k+m) \times (k+m)$ square submatrix, and c_s a k-index set defined by cyclic permutation, $c_s = (s, s+1, \ldots, s+k-1) \bmod n_{local}$[26].

[25] The $(5,1)$ simplex recurs identically inside $(7,2)$ as the left sub-problem of cell 1—a property of the splitting rules, not of the cache. Memoization ensures that this shared sub-problem is evaluated once; the GCD relation made explicit in Step 5 (3.3.3.5) rests on the structural recurrence itself.

[26] Cyclic permutation refers to forming the denominator's $k_{\text{sub}} \times k_{\text{sub}}$ minors while cycling the particle indices. For $n_{\text{local}} = 5$ and $k_{\text{sub}} = 1$, the five single-index minors are generated cyclically as $c_0 = (0), c_1 = (1), \ldots, c_4 = (4)$. For $k_{\text{sub}} \geq 2$, c_s is a modular window of k_{sub} consecutive indices, and at $c_{n_{\text{local}}-1}$ the index wraps from the end of the cell back to the beginning.

(iii) *BCFW composite cell* (recursive, non-base): the canonical form is the product of the canonical forms of the left and right sub-cells, summed over all valid cells from the splitting rules of Step 3 (3.3.3.3).

Apply the simplex formula to the main walkthrough $(5,1,4)$. In the Step 1 (3.3.3.1) matrix, Z is already a 5×5 square matrix, so $Z_I = Z_{QQ}$, and its determinant is the Vandermonde determinant:

$$\det(Z_I) = \prod_{1 \le i < j \le 5} (a_j - a_i) = 288 \tag{3.17}$$

For $a = (1,2,3,4,5)$ the product of all differences is $(1\cdot2\cdot3\cdot4)(1\cdot2\cdot3)(1\cdot2)(1) = 288$. For the denominator, since $k = 1$, the cyclic minors reduce to a product of 1×1 minors—the first-row entries of Z. As the first row is $(1,1,1,1,1)$,

$$\prod_{s=0}^{4} \Delta_{c_s}(Z_{QQ}) = 1\cdot1\cdot1\cdot1\cdot1 = 1 \tag{3.18}$$

The canonical form of the main case is therefore

$$\Omega_{(5,1)}(\text{cell}) = \frac{288^4}{1} = 6{,}879{,}707{,}136 = 2^{20}\cdot 3^8 \tag{3.19}$$

Since $(5,1)$ is a single cell, this value is exactly $\Omega^{\text{total}}_{(5,1)}$ of Step 5 (3.3.3.5).

Next, the two BCFW cells of the auxiliary walkthrough $(7,2,4)$.

(i) *Cell 0* (split $i = 3$, $(k_L, k_R) = (0,1)$): Left is MHV ($k_L = 0$), canonical form 1. Right is a simplex on particles $(1,4,5,6,7)$ with $k_R = 1$; the five non-contiguous columns of the Step 1 matrix, restricted to the top five rows, form a 5×5 square matrix to which the simplex formula applies, giving $\Omega(R) = 348{,}285{,}173{,}760{,}000$. The product is

$$\Omega(\text{cell } 0) = 1 \times 348{,}285{,}173{,}760{,}000 = 348{,}285{,}173{,}760{,}000 = 2^{20}\cdot 3^{12}\cdot 5^4 \tag{3.20}$$

(ii) *Cell 1* (split $i = 4$, $(k_L, k_R) = (1,0)$): Left is a simplex on particles $(1,2,3,4,5)$ with $k_L = 1$—structurally identical to the simplex of the main walkthrough—with canonical form 6,879,707,136. Right is MHV ($k_R = 0$), canonical form 1. The product is

$$\Omega(\text{cell } 1) = 6{,}879{,}707{,}136 \times 1 = 6{,}879{,}707{,}136 = 2^{20}\cdot 3^8 \tag{3.21}$$

Note that each cell's evaluation decomposes into a product of sub-cell canonical forms, and that the $(5,1)$ simplex of the main walkthrough recurs as the left sub-problem of cell 1—a consequence of the splitting rules of Step 3 (3.3.3.3), not of any caching. The summation of the two cells is carried out in Step 5 (3.3.3.5).

Exact specification of the canonical form. For a simplex cell with $n_{\text{local}} = k_{\text{sub}} + m$, the simplex canonical form of this implementation is

$$\Omega_{\text{simplex}}(\text{cell}) = \frac{[\det(Z_I)]^m}{\prod_{s=0}^{n_{\text{local}}-1} \det\left[Z|_{c_s}^{(k_{\text{sub}}\text{ rows})}\right]} \tag{3.22}$$

where Z_I is the $(k_{\text{sub}}+m) \times (k_{\text{sub}}+m)$ square submatrix extracted by the particle-index set $I = \{a_0, \ldots, a_{n_{\text{local}}-1}\}$; $c_s = ((s+0) \bmod n_{\text{local}}, \ldots, (s+k_{\text{sub}}-1) \bmod n_{\text{local}})$

is a k_{sub}-index sequence defined by cyclic permutation; and $Z|_{c_s}^{(k_{\text{sub}}\text{ rows})}$ is the $(k_{\text{sub}}) \times (k_{\text{sub}})$ submatrix formed from the top k_{sub} rows of Z and the columns indexed by c_s. For the BCFW composite cell, the recursive canonical form is a sum of products:

$$\Omega_{\text{bcfw}}(\text{cell}) = \sum_{(l,r)\in L\times R} \Omega(l) \cdot \Omega(r) \tag{3.23}$$

where L and R are the sets of left and right sub-cells; if both are single cells, the sum collapses to a single product—the case for both cells of the $(7, 2)$ walkthrough.

The canonical-form evaluation follows three design choices. First, simplex cells and BCFW composite cells are handled by separate functions: the simplex cell is a closed-form evaluation applying the fraction formula of [61, Section 6, Eq. 6.6] directly, whereas the composite cell recursively multiplies and sums the canonical forms of its sub-cells. The difference between the two evaluation structures is the basis for the separation. Second, in computing the numerator determinant, only the top $k + m$ rows of Z are used: the full Z is $(k_{\text{original}} + m) \times n$, but when a sub-cell's k_{sub} is smaller, only the top rows are taken to form a $(k_{\text{sub}} + m) \times (k_{\text{sub}} + m)$ square matrix—the natural dimensional reduction of the sub-cell in the BCFW decomposition. Third, all arithmetic is exact rational: since both the numerator determinant and the denominator minor product are integers or rationals, the result $\Omega(\text{cell})$ is an exact rational. This exactness is a prerequisite for the factorization analysis of this study—an exact factorization such as $288^4 = 2^{20} \cdot 3^8$ cannot be tracked in floating-point arithmetic.

[61, Section 6] defines the canonical form in two equivalent ways. The logarithmic differential form for a general cell is

$$\Omega^{\Gamma}_{n,k}(Y; Z) = \prod_{i=1}^{4k} \frac{d\alpha_i^{\Gamma}}{\alpha_i^{\Gamma}} \tag{3.24}$$

and the explicit fractional form for the $k = 1$ case is Eq.(6.6) [61, Chapter 6, Eq. 6.6]. The fractional form $(Z_I)^m / \prod \Delta_{k\times k}(Z)$ of this implementation stands in for the $\langle Y \dots \rangle$ expression on the right-hand side of Eq.(6.6) [61], evaluated on the vertex data of the cell. The correspondence is exact in the numerator: the bracket $\langle Z_{a_1} Z_{a_2} Z_{a_3} Z_{a_4} Z_{a_5} \rangle$ is the determinant of the 5×5 minor Z_I. In the denominator it is not: each $\langle YZ \dots \rangle$ factor of Eq.(6.6) [61] carries Y, whereas this implementation substitutes a cyclically permuted k-minor of Z, which at $k = 1$ reduces to unity as Equation (3.18) shows. What is computed here is therefore a numerical value extracted from the cell's vertex data, not the Y-dependent form of [61] [27]. The code is handled by `compute_simplex_canonical_form` for the simplex cell and `compute_cell_canonical_form` for the BCFW composite cell.

3.3.3.5 Step 5–Summation of the total amplitude

The total amplitude is the sum of the canonical forms of all cells evaluated in Step 4 (3.3.3.4):

$$\Omega^{\text{total}}_{QQ} = \sum_{\text{cells}} \Omega_{QQ}(\text{cell}) \tag{3.25}$$

[27] This implementation uses $\det(\cdot)$ notation rather than the paper's $\langle Y \dots \rangle$ because it evaluates the amplitude at a fixed point of the simplex cell, computing directly from the cell's vertex data without explicitly retaining Y-dependence. The present study is concerned with the numerical value of the amplitude, for factorization tracking; the differential structure in Y lies outside its scope.

For the main walkthrough $(5,1,4)$ the sum is trivial, a single cell:

$$\Omega_{QQ}^{\text{total}}(5,1) = \Omega(\text{cell}) = 6{,}879{,}707{,}136 = 2^{20} \cdot 3^8 \tag{3.26}$$

For the auxiliary walkthrough $(7,2,4)$ it is the sum of the two cells:

$$\begin{aligned}\Omega_{QQ}^{\text{total}}(7,2) &= \Omega(\text{cell } 0) + \Omega(\text{cell } 1) \\ &= 348{,}285{,}173{,}760{,}000 + 6{,}879{,}707{,}136 = 348{,}292{,}053{,}467{,}136\end{aligned} \tag{3.27}$$

The point to note is the exact coincidence $\Omega(\text{cell } 1) = 6{,}879{,}707{,}136 = \Omega_{QQ}^{\text{total}}(5,1)$. As observed in Step 4 (3.3.3.4), this follows from the left sub-problem of cell 1 sharing the structure of the main-walkthrough simplex, and it connects directly to the core observation of the GCD-hierarchy verification: the greatest common divisor of the two cell values coincides with $\Omega(5,1)$[28].

This summation is a simple rational addition, but because every cell's canonical form is an exact rational, the sum is likewise exact. The values of the walkthroughs are small enough to be held exactly in double precision, but the amplitudes of the larger configurations are not: $\Omega(10,2)$ and $\Omega(11,2)$ both exceed the range in which a double-precision float represents an integer without loss, and a single rounding there would destroy the prime factorization on which Chapter 4 depends. The exact rational arithmetic of this implementation forecloses that risk at the root.

[61, Section 6] defines the tree amplitude as the sum over the cell decomposition,

$$\Omega_{n,k}(Y;Z) = \sum_{\Gamma \subset T} \Omega_{n,k}^{\Gamma}(Y;Z) \tag{3.28}$$

and specifies that this sum is independent of the chosen decomposition [61, Section 6]:

> the form is independent of the particular cell decomposition.

The original paper thus fixes the sum over cells as the amplitude. Two points of this implementation stand in relation to it. First, the cell structure here depends only on (n,k,m)— `get_bcfw_cells` does not take Z as an argument— so the two tracks of Section 3.3.2 share an identical decomposition by construction, and the comparison of Chapter 4 is between two evaluations of one and the same cell structure. Second, the sum-invariance of Equation (3.28) holds among decompositions that tile the amplituhedron; as Section 3.3.3.3 sets out, this implementation departs from that condition, and the validity of $\Omega_{QQ^{total}}$ is therefore established not by invariance but by the invariants of Appendix D. The code is handled by `compute_total_amplitude (bcfw_cells, Z_matrix, m)`, which branches on whether the input is a single dict (base case) or a list (recursive decomposition) and sums the canonical forms over all cells.

This summed $\Omega_{QQ}^{\text{total}}$ is the final output of the QQ track, and becomes the direct input for the factorization analysis of the following chapter.

[28] The two cell values of $(7,2)$ have prime factorizations $\Omega(\text{cell } 0) = 2^{20} \cdot 3^{12} \cdot 5^4 = \Omega(5,1) \cdot (3^4 \cdot 5^4)$ and $\Omega(\text{cell } 1) = 2^{20} \cdot 3^8 = \Omega(5,1)$. Their ratio is exactly $50{,}625 = 3^4 \cdot 5^4$, and their greatest common divisor is $\gcd(\Omega(\text{cell } 0), \Omega(\text{cell } 1)) = 2^{20} \cdot 3^8 = \Omega(5,1) = 6{,}879{,}707{,}136$. This is not a coincidence but a structural consequence: as observed in Step 3 (3.3.3.3), the left sub-problem of cell 1 shares the identical structure with the $(5,1)$ simplex and its right sub-problem is MHV ($= 1$), so $\Omega(\text{cell } 1) = \Omega(5,1) \cdot 1 = \Omega(5,1)$ holds strictly, and carries no independent information. What the relation does test is the divisibility $\Omega(5,1) \mid \Omega(\text{cell } 0)$: cell 0 is a simplex on the non-contiguous particle set $(1,4,5,6,7)$, and nothing in its construction requires its canonical form to be divisible by that of the contiguous simplex. The quotient is exactly $3^4 \cdot 5^4$. This is the substance of invariant I-5 in Appendix D.

3.3.4 Python Implementation

The complete Python source implementing the five-step algorithm described above is provided in full in Appendix E. This includes the functions `generate_Z_qq`, `is_classical_positive`, `get_bcfw_cells`, `compute_simplex_canonical_form`, `compute_cell_canonical_form`, `compute_total_amplitude`, together with the `FractionMatrix` class for exact rational matrix arithmetic and the `PadicValuation` class for p-adic valuation. As noted in Section 3.3.1, the code is placed in the appendix (Appendix E) so that the main text preserves the logical flow of the algorithm, while the implementation remains fully available for verification.

Two properties of the implementation bear on the reproducibility claimed in Section 3.3.1. First, the arithmetic core depends on the Python standard library alone—`fractions`, `itertools` and `math`. No third-party package enters the computation of Ω; NumPy and Matplotlib appear only in the reporting and plotting of results. The argument of Section 3.3.1 that this simulation needs no specialized algebra system is therefore not a matter of preference but of dependency: the code has none to satisfy. Second, because every intermediate quantity is an exact rational, the computation is deterministic in the strict sense. The same (n, k, m, a) yields the same value, digit for digit, on any platform, since no floating-point rounding enters whose behaviour could differ between machines or library versions.

This is what allows the verification of Appendix D to take the form it does. Each of the five invariants is an exact identity between integers rather than an agreement within tolerance, and re-running the same functions on the configurations listed there reproduces them exactly. Within the parameter range of Table 3.2 the cost of doing so is negligible. What governs that cost is the helicity grade k rather than the number of particles n: the cell count grows polynomially in n with a degree set by k, so it is k that determines how far the recursion can be carried in practice.

Chapter 4

Results and Analysis

Chapter 3 constructed the simulation: it fixed *what* was to be computed—the canonical form Ω of the amplituhedron—and *how*, through a five-step algorithm run in parallel over the rational (QQ) and p-adically perturbed (QP) tracks. This chapter brings the two tracks into contact and examines what the comparison returns.

What it returns is not what the hypothesis of Section 3.1 anticipated. The prediction was cast in the language of computational precision—that the QP track would measure the amplitude more precisely than QQ. The simulation does not bear this out; but neither does it leave the exploration empty-handed. In place of a difference in precision, it returns a difference in *structure*—a coherent p-adic arithmetic that the rational and p-adic descriptions do not carry symmetrically. The task of this chapter is to follow that redirection: to show how the initial prediction fails, where the exploration turns, and what structure the turn brings into view.

4.1 Simulation Results and Unexpected Findings

4.1.1 The first observation: the precision hypothesis, unrealized as error

This section takes up the outcome bearing directly on the hypothesis of Section 3.1—that the QP amplituhedron would measure the amplitude more precisely than QQ. That section took the QQ track to inherit the limits of real-number computation, on the ground that the rationals lie within the reals. That step presupposes a measurable computational error in QQ against which QP could show an improvement. The results overturn that presupposition at the root.

Because both tracks evaluate the canonical form entirely through exact rational arithmetic—Python's `fractions.Fraction`—the QQ track carries *no error whatsoever*: $\Omega_{QQ}^{\text{total}}$ is an exact rational number, computed without floating-point rounding at any stage. There is, in short, no error to reduce. The comparison the hypothesis envisaged—QP more precise than QQ—cannot even be posed in terms of computational error, for the quantity it would improve upon is identically zero[1].

This outcome does *not* amount to a refutation of the study's guiding hypothesis. As stated in Section 3.1, the purpose of this simulation is to *explore the possibility* of the p-adic framework, not to establish a decisive quantitative superiority. The initial prediction was cast in the language of computational precision; that this particular

[1] This is not a defect of the implementation but a direct consequence of its exactness, which Section 3.3.3.5 established as a methodological prerequisite: exact rational arithmetic is what preserves the prime-factorization structure of Ω on which the entire p-adic analysis rests. The very property that makes the precision comparison vacuous is the property that makes the structural analysis to follow possible.

prediction is not realized redirects the exploration rather than closing it. The pertinent question therefore shifts: if the two tracks cannot be distinguished by error, are they distinguishable at all—and if so, in what?

4.1.2 The pivot: from error to p-adic structure

The redirection is furnished by the simulation itself. The absence of error in the QQ track is a statement about the *computational process*: no rounding occurs, so the amplitude is evaluated exactly. It is not a statement about the two tracks producing the same *value*. The QP track operates on a different input—the moment-curve matrix displaced by the p-adic perturbation of Section 3.2.1— and evaluating the canonical form on that displaced input yields, exactly and without rounding, a different rational number. The two facts belong to different levels: the calculation carries no error, while the inputs differ by construction.

$$\text{diff} = \Omega_{QP} - \Omega_{QQ} \neq 0 \tag{4.1}$$

and this difference is itself an exact rational quantity. The statement is confined to the non-degenerate configurations. On the $k = 1$ track above $n = k + m$ the amplitude reduces to the cell count, which does not depend on Z at all, so the two tracks return the identical value and the difference vanishes identically. The comparison that follows is therefore carried on $k = 2$, together with the simplex case (5, 1).

The two tracks are thus not identical; they simply do not differ in the dimension—computational error—that the original hypothesis anticipated. What Section 3.1 reframed as *precision in the sense of depth of information* now becomes the operative notion: rather than asking how large the difference is, the question becomes how it is *structured* in the p-adic sense. The natural instrument is the p-adic valuation v_p of Section 2.3.3, which measures not the magnitude of a quantity but the depth to which the prime p divides it. The analysis therefore turns from the magnitude of diff to its valuation $v_2(\text{diff})$. The full range over which the two were computed is set out first.

4.1.3 The ground: the two tracks across the full parameter range

The two tracks were evaluated over the full range fixed in Table 4.1: $n = 5, \ldots, 12$ at $k = 1$ and $n = 6, \ldots, 11$ at $k = 2$, with $m = 4$ throughout, the baseline sequence $a = (1, 2, \ldots, n)$, and the perturbation of Section 3.2.1 scaled to $p = 2$. Table 3.2 collects the results. For each configuration it records the number of BCFW cells, the prime factorization of the rational amplitude, and the two valuations whose comparison Section 4.1.2 identified as the operative measure.

(n,k)	cells	Ω_Q (factorization)	$v_2(\Omega_Q)$	v_2(diff)
$(5,1)$	1	$2^{20} \cdot 3^8$	20	20
$(6,1)$	1	1	0	—
$(7,1)$	2	2	1	—
$(8,1)$	3	3	0	—
$(9,1)$	4	2^2	2	—
$(10,1)$	5	5	0	—
$(11,1)$	6	$2 \cdot 3$	1	—
$(12,1)$	7	7	0	—
$(6,2)$	1	$-2^{32} \cdot 3^{12} \cdot 5^3$	32	32
$(7,2)$	2	$2^{21} \cdot 3^8 \cdot 17 \cdot 1489$	21	21
$(8,2)$	4	$2 \cdot 23 \cdot 37 \cdot 2{,}218{,}831 \cdot 2{,}733{,}749$	1	21
$(9,2)$	8	$2 \cdot 3 \cdot 7 \cdot 53 \cdot 259{,}841 \cdot 285{,}580{,}987$	1	20
$(10,2)$	14	$2^2 \cdot 3 \cdot 17 \cdot 1549 \cdot 5{,}487{,}449{,}645{,}429$	2	20
$(11,2)$	22	$2^2 \cdot 7 \cdot 392{,}003{,}819 \cdot 1{,}218{,}986{,}233$	2	20

TABLE 4.1: The full parameter sweep of Table 3.2 at $p = 2$. A dash in the last column marks the configurations for which the difference vanishes identically, so that no valuation is defined. The two valuations coincide at $(5,1)$, $(6,2)$ and $(7,2)$, and part company from $n = 8$ at $k = 2$.

The $k = 1$ track returns nothing to compare. Beyond the simplex at $n = 5$ the amplitude reduces to the cell count, $\Omega(n,1) = n - 5$, and the cell count is fixed by (n,k,m) alone. The external data never enters. The perturbed input therefore yields the identical value, and the difference vanishes identically rather than approximately. This is the degeneracy anticipated in Section 3.2.1, now visible as an empty column: the $k = 1$ track serves as a control, and every configuration carrying information sits at $k = 2$ or at the base simplex $(5,1)$.

At $k = 2$ the amplitude does not vary smoothly with n. The cell count grows steadily—1, 2, 4, 8, 14, 22— but the valuation does not follow it. The valuation stands at 32 and 21 for $n = 6$ and $n = 7$, falls abruptly to 1 at $n = 8$, and remains at 1 or 2 thereafter. The prime factorizations show what happens at that step. Through $n = 7$ the amplitude is assembled from 2, 3, 5 and the two moderate primes 17 and 1489; from $n = 8$ onward factors running to seven digits and beyond appear, and the small primes survive only to the first or second power. The pattern that Section 3.2 set out to track is therefore not a trend but a threshold, and the threshold falls between $n = 7$ and $n = 8$.

The valuation of the difference does not fall with it. It remains at 20 or above across the whole $k = 2$ range, and from $n = 7$ onward it varies only between 20 and 21. The two columns of Table 4.1 therefore behave quite differently as n grows, and what that difference in behaviour means is the subject of the following section.

4.1.4 The finding: conservation of the arithmetic skeleton

Computing the p-adic valuation of the difference against that of the amplitude itself reveals an exact coincidence. For the configurations in which every cell retains the small-prime content deposited by the moment curve, the valuation of the difference equals the valuation of the amplitude:

$$v_p\left(\Omega_{QP} - \Omega_{QQ}\right) = v_p\left(\Omega_{QQ}\right) \tag{4.2}$$

Table 4.1 shows this in three of its rows. At $(5,1)$ both valuations are 20; at $(6,2)$ both are 32; at $(7,2)$ both are 21.

This coincidence carries a definite meaning. The p-adic perturbation, though it displaces the amplitude to a different value, does so *without disturbing the depth to which p divides it*: the difference it induces sits at exactly the same p-adic depth as the amplitude's own arithmetic core. In this precise sense the perturbation preserves the *arithmetic skeleton* of Ω—the p-adic structure of the amplitude proves robust enough to be reproduced, at the level of valuation, even in the perturbed track. This is the first concrete sense in which the p-adic structure is actively imprinted on the amplitude calculation, rather than merely tolerated by it.

The boundary of this observation must be noted candidly. The coincidence $v_p(\mathrm{diff}) = v_p(\Omega_{QQ})$ is manifest precisely while no cell of the decomposition reduces to unity. From $n = 8$ onward at $k = 2$ the valuation $v_2(\Omega)$ falls to 1 or 2 while $v_2(\mathrm{diff})$ stays at 20 or 21, as the last two columns of Table 4.1 show, and the clean coincidence no longer holds [2]. This is consistent with the rationale for the choice $p = 2$ in Section 3.2.1—a choice justified precisely by the a choic small-prime dominance of Ω—and it marks the regime in which the present observation is sharpest; the large-prime regime is left to future work.

4.1.5 The check: what the coincidence depends on

An exact agreement observed at three configurations invites an obvious objection: that it has been tuned. The perturbation of Section 3.2.1 carries two free parameters —the step n_{step} and the precision prec that fixes its depth— and it is scaled to a prime that need not be the prime at which the valuation is read. Whether the coincidence survives variation in these is a question the simulation can answer directly, since each run is exact and costs almost nothing.

Varying the two parameters leaves the coincidence untouched. The computation was repeated over the grid $n_{\mathrm{step}} \in \{1,2,3,5,7\}$ and $\mathrm{prec} \in \{32,64,128,256,512,1024\}$, thirty runs for each configuration. In every configuration $v_2(\mathrm{diff})$ took a single value across all thirty: 20 at $(5,1)$, 32 at $(6,2)$, 21 at $(7,2)$. The depth of the perturbation ranges from 2^{-16} to 2^{-512}, and the valuation of the difference does not move. Whatever fixes it, the size of the displacement is not it [3].

[2] What breaks the coincidence is thus the collapse of $v_2(\Omega)$, not any change in the difference. That collapse arises within this implementation—from n = 8 onward some cells reduce to unity through the degenerate sub-problems of Section 3.3.3.3, so the amplitude becomes a sum of one highly 2-divisible term and several small ones, and the common 2-content is lost. A structural account of this transition lies beyond the scope of the present study.

[3] This also disposes of a reading suggested by the perturbation itself. The increment carries $v_2(\varepsilon) = -\mathrm{prec}/2$, so a difference inheriting its depth from the perturbation would track prec directly. It does not.

$v_2(\text{diff})$ with the perturbation scaled to q						
(n,k)	$v_2(\Omega_Q)$	$q=2$	$q=3$	$q=5$	$q=7$	$q=23$
$(5,1)$	20	**20**	16	16	16	16
$(6,2)$	32	**32**	35	35	35	35
$(7,2)$	21	**21**	15	15	15	15

TABLE 4.2: Dependence of the coincidence on the prime to which the perturbation is scaled. The valuation is read at $p = 2$ throughout. It agrees with $v_2(\Omega_Q)$ only when the perturbation is scaled to the same prime; for every other choice it settles on a single value that misses in either direction.

The prime is another matter. Table 4.2 repeats the computation with the perturbation scaled to $q = 3, 5, 7$ and 23, reading the valuation at $p = 2$ throughout. The agreement disappears. For every $q \neq 2$ the difference settles on one value per configuration, and that value misses $v_2(\Omega_Q)$ in either direction: below it at $(5,1)$ and $(7,2)$, above it at $(6,2)$. Only when the perturbation is scaled to the prime at which the valuation is read do the two agree. Section 3.2.1 argued, on Ostrowski's theorem, that scaling the perturbation to an absolute value exhausts the ways a rational input can be locally displaced; the present check shows that the choice among those ways is not indifferent.

Taken together the two checks give the coincidence a definite shape. Its magnitude is irrelevant and its prime is decisive— which is the non-Archimedean character of Section 2.3.3 appearing in the result rather than in the construction. A displacement of arbitrary size in the 2-adic metric reproduces the arithmetic depth of the amplitude exactly; a displacement scaled to any other prime does not. This is the sense in which the coincidence is a property of the amplitude and the metric together, and not an artifact of how far the input was moved. That the prime is decisive does not make the result a construction of the perturbation. A displacement built to carry the coincidence would have to carry its depth as well, and the first check shows that it does not: the agreement holds at a fixed value while the depth of the displacement varies by a factor of thirty-two. What the second check adds is that the agreement is not generic either.

4.1.6 The source of the structure: prime factorization and the Vandermonde origin

The conservation observed above raises an immediate question: if the p-adic depth of Ω is robust enough to survive perturbation, where does that depth originate? The answer lies in the arithmetic of Ω itself, prior to any perturbation, and traces back directly to the moment-curve construction of Step 1 (3.3.3.1).

Recall from Section 3.3.3.4 that for a simplex cell the numerator of the canonical form is $\det(Z_I)^m$, and that on the moment curve $Z_a^j = a^j$ this determinant is a Vandermonde determinant,

$$\det(Z_I) = \prod_{i<j} (a_j - a_i) \tag{4.3}$$

For the baseline $a = (1, 2, \ldots, n)$, every factor $a_j - a_i$ equals the gap $j - i$, a small integer no larger than $n - 1$. The difference product is therefore assembled entirely

from small integers, and its prime content is confined to the small primes they carry. Table 4.3 makes this assembly explicit for the two baseline cases $n = 5$ and $n = 6$.

gap $j - i$	factor	multiplicity	contribution
$n = 5$: $a = (1,2,3,4,5)$			
1	1	4	1
2	2	3	2^3
3	3	2	3^2
4	4	1	2^2
product $= 2^5 \cdot 3^2 = 288$			
$n = 6$: $a = (1,2,3,4,5,6)$			
1	1	5	1
2	2	4	2^4
3	3	3	3^3
4	4	2	2^4
5	5	1	5
product $= 2^8 \cdot 3^3 \cdot 5$			

TABLE 4.3: Assembly of the Vandermonde difference product $\prod_{i<j}(a_j - a_i)$ from integer gaps, for $n = 5$ and $n = 6$. Each gap value g occurs with multiplicity $n - g$; the prime content of the product is fixed by the small integers so assembled. At $n = 6$ the gap 5 appears for the first time, introducing the prime 5.

The table exposes the mechanism directly. Each gap value g contributes $n - g$ copies of the integer g, so the smallest gaps dominate the multiplicity and the product stays confined to small primes. At $n = 5$ the largest gap is 4, and the product $2^5 \cdot 3^2 = 288$ carries only 2 and 3; at $n = 6$ the gap 5 enters for the first time, and the prime 5 appears in the product[4].

At $(6,2)$ the amplitude is again a single simplex cell, but the denominator now leaves a trace. The Vandermonde product at $n = 6$ is $34{,}560 = 2^8 \cdot 3^3 \cdot 5$, so the numerator contributes $34{,}560^4 = 2^{32} \cdot 3^{12} \cdot 5^4$. The denominator is the product of the six cyclic 2×2 minors of Z, five of which equal unity; only the wrap-around minor on columns $(6,1)$ differs, contributing -5. Hence, $\Omega(6,2) = 34{,}560^4/(-5) = -2^{32} \cdot 3^{12} \cdot 5^3$. The prime 5, absent at $n = 5$, enters with the gap 5 in the difference product and is then reduced by one power through the cyclic denominator, which also fixes the sign of the amplitude. The origin is the same in both configurations: the prime content of Ω is inherited from the small-integer difference products of the moment-curve parameters, modified only by the cyclic minors of the denominator. The small-prime dominance exploited throughout this chapter, and the ground on which $p = 2$ was chosen in Section 3.2.1, is thus a structural consequence of the Vandermonde origin, not a numerical accident. The p-adic depth found to be conserved is, at root, the depth deposited into Ω by the differences of consecutive integers.

4.1.7 The GCD hierarchy: number-theoretic structure across the recursion

A second, independent manifestation of the same structure appears when the BCFW recursion produces more than one cell. In the auxiliary case $(7,2)$, the two

[4] This is the same mechanism registered in Section 3.2.1, where the positivity of the Vandermonde determinant justified the baseline choice $a = (1,2,\ldots,n)$. There the difference product fixed the sign of the determinant; here the same product is seen to fix the prime content of the amplitude.

cells evaluated in Section 3.3.3.5 carry the factorizations,

$$\Omega(\text{cell } 0) = 2^{20} \cdot 3^{12} \cdot 5^4, \qquad \Omega(\text{cell } 1) = 2^{20} \cdot 3^8 \tag{4.4}$$

whose greatest common divisor is

$$\gcd\left(\Omega(\text{cell } 0),\, \Omega(\text{cell } 1)\right) = 2^{20} \cdot 3^8 = 6{,}879{,}707{,}136 = \Omega(5,1) \tag{4.5}$$

The greatest common divisor of the two cells of $\Omega(7,2)$ is *exactly* the amplitude of the lower configuration $\Omega(5,1)$. As Section 3.3.3.5 showed, this is no coincidence: the left sub-problem of cell 1 is structurally identical to the $(5,1)$ simplex, and its right sub-problem is MHV (contributing a factor of 1), so $\Omega(\text{cell } 1) = \Omega(5,1)$ holds by construction and carries no independent information. The substance of the relation lies in the other cell. Cell 0 is a simplex on the non-contiguous particle set $(1,4,5,6,7)$, and nothing in its construction requires its canonical form to be divisible by that of the contiguous simplex; it nonetheless carries the same $2^{20} \cdot 3^8$ core beneath its additional factor $3^4 \cdot 5^4$. Figure 4.1 makes the shared core visible.

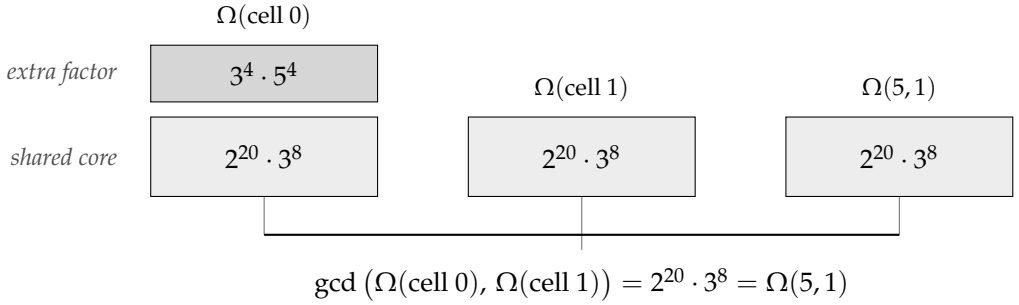


FIGURE 4.1: The shared arithmetic core of the two BCFW cells of $\Omega(7,2)$. Both cells carry the common core $2^{20} \cdot 3^8$; cell 0 differs only by the extra factor $3^4 \cdot 5^4$. The greatest common divisor is therefore the core itself, which equals the lower amplitude $\Omega(5,1)$—the arithmetic content of $\Omega(5,1)$ surviving as a divisor of both cells of the higher amplitude.

The relation is not confined to $(7,2)$. At $(8,2)$, $(9,2)$ and $(10,2)$ the same divisor recurs: in each case the cells that do not reduce to unity share the greatest common divisor $2^{20} \cdot 3^8 = \Omega(5,1)$, the amplitude of the base simplex. The qualification is necessary. From $n = 8$ onward the decomposition contains cells whose canonical form is unity— the degenerate sub-problems noted in Section 3.3.3.3— and the greatest common divisor taken over all cells therefore collapses to 1. Taken over the remaining cells it does not move: the same divisor appears at $(7,2)$ with two cells, at $(8,2)$ with four, at $(9,2)$ with eight and at $(10,2)$ with fourteen. The recursion, therefore, does not merely reproduce a number; it carries the entire arithmetic content of the smaller amplitude forward as a common divisor of the cells from which the larger one is assembled.

This GCD hierarchy matters because it exhibits the p-adic structure as a property of the *recursive architecture* of the amplitude, not only of individual values. The number-theoretic core of a lower amplitude survives as a divisor threading through the cells of a higher one—a hierarchical nesting that mirrors, in the language of divisibility, the recursive nesting of the BCFW decomposition itself[5]. Together with

[5]This relation is invariant I-5 of Appendix D, one of the five used to verify the implementation independently of the standard BCFW cell count. That the same equality serves both as a correctness check

the skeleton conservation and the Vandermonde origin above, it forms a coherent picture: the amplitude carries a p-adic arithmetic structure that is deposited by its construction, propagated by its recursion, and robust under perturbation.

4.1.8 Synthesis: the possibility restated at the level of structure

The three findings of this chapter converge on a single picture. The amplitude Ω carries a p-adic arithmetic structure that is *deposited* by its moment-curve construction (Section 4.1.6), *propagated* through its recursive decomposition as a divisibility hierarchy (Section 4.1.7), and *conserved* under p-adic perturbation of the input (Section 4.1.4). None of these was the outcome the initial hypothesis anticipated. The hypothesis looked for a difference in computational precision; what the simulation returned instead was a coherent number-theoretic structure that two tracks do not carry symmetrically.

This does *not* establish that the p-adic framework yields a more accurate amplitude, nor that probabilistic uncertainty has been reduced in any measurable sense; the exactness of the rational arithmetic made that comparison vacuous from the outset. What it establishes is narrower, but not negligible: that the p-adic structure is not inert with respect to the amplitude calculation. The valuation of the amplitude is a robust invariant of its arithmetic—reproduced under perturbation, threaded through the recursion—which is to say that the choice of number system is not a passive notational matter but leaves a definite, measurable imprint on the object computed. This is the minimal sense in which the possibility set out in Section 3.1 is sustained: not as precision, but as *structural imprint*.

This minimal result connects back to the guiding hypothesis of Chapter 1. There, the study advanced the possibility that the probabilistic character of the microscopic world may originate not in nature itself but in the gap—the *mapping noise*—between physical reality and the real-number language chosen to describe it. The present chapter does not confirm that hypothesis, and does not claim to. But it supplies a concrete instance of the premise on which the hypothesis rests: that the same physical object, described over two different number systems, need not carry the same structure across them. If a difference in number system can imprint a measurable arithmetic structure even on an idealized amplitude, then the conjecture that the real-number description omits something present in the phenomenon—something a non-Archimedean description might retain—is shown to be, at the least, a well-posed and computationally tractable question, rather than a purely philosophical supposition[6].

The natural continuation is therefore twofold: to characterize the transition to the large-prime regime where skeleton conservation weakens, and to move from the forward computation examined here toward the inverse problem—the reconstruction of the input from the amplitude—where the divergence between the real and p-adic descriptions is expected to be most pronounced. Both are deferred to future work[7]. What this chapter has secured is the foundational step on which such continuations depend: the demonstration, within an exact and fully controlled setting,

and as evidence of number-theoretic structure is not accidental: both rest on the recursion preserving the arithmetic content of its sub-problems exactly.

[6] The distinction between the purpose of the present p-adic analysis and that of prior uses of p-adic numbers in amplitude computation—notably the numerical reconstruction discussed in Section 3.1—is taken up in the concluding chapter, once the full results are in view.

[7] The inverse problem lies outside the scope of this thesis; the forward simulation examined here is its necessary precondition.

that the p-adic structure of the amplitude as this implementation defines it is present, reproducible, and worth pursuing.

Chapter 5

Conclusion

5.1 Summary of the study

This thesis began with a question raised by a concrete physical difficulty. In the plasma etching process at nanometer scales, stochastic fluctuations impose a limit that does not vanish however finely the process is controlled. Chapter 1 asked whether this probabilistic character is truly intrinsic to nature, or whether it might instead be an artifact of the mathematical language through which the microscopic world is described. From this question the guiding hypothesis of the study was formulated. The probabilistic uncertainty of quantum mechanics may not be a property of nature itself. It may instead be noise introduced in the act of mapping nature onto the real number system.

To examine this hypothesis, the study had to answer two prior questions: *what* to compute, and in *which* language. Chapter 2 addressed both. It narrowed the object of computation from quantum field theory, through the emergence of the amplituhedron, down to its canonical form— a quantity defined purely by the positive geometry of scattering, independent of spacetime. And it established the p-adic numbers, together with the diagnostic vocabulary of non-standard analysis, as an alternative mathematical language: a non-Archimedean number system whose notion of distance is governed not by magnitude but by divisibility. With the object and the language both in place, Chapter 3 brought them into contact, constructing the simulation that computes the canonical form of the amplituhedron over the rational (QQ) and p-adically perturbed (QP) tracks. Chapter 4 compared the two.

The first outcome was negative. The perturbed track did not compute the amplitude more precisely than the rational one, and the reason was structural: both tracks carry rational entries, and neither approximates anything. The difference between them had to be read in another way, and the p-adic valuation supplied it.

Read that way, the two tracks were compared across the full range fixed in Table 3.2. The comparison did not yield a smooth trend in n and k. At $k = 1$ the amplitude reduces to the number of cells and no longer depends on the input at all, so the two tracks return the identical value and there is nothing to compare. At $k = 2$ the behaviour is sharply divided, and the division falls between $n = 7$ and $n = 8$: below it the valuations of the amplitude and of the difference agree exactly, above it they do not. What Section 3.2 sought as a pattern in n and k turned out to be a threshold.

Below the threshold the agreement is exact, and it is not fragile. Two checks bore this out. The agreement holds at a single fixed value while the depth of the perturbation varies by a factor of thirty-two, and it disappears the moment the perturbation is scaled to a prime other than the one at which the valuation is read. The size of the displacement is irrelevant; the prime is decisive.

The structure has a definite origin. The prime content of the amplitude is inherited from the difference products of the moment-curve parameters, which are small integers, and is modified only by the cyclic minors of the denominator. It also propagates. The amplitude of the smallest configuration reappears as a common divisor of the cells from which every larger one at $k = 2$ is assembled.

What this establishes is narrower than the original hypothesis proposed, but not without weight. It does not show that the p-adic framework computes the amplitude more accurately, nor that probabilistic uncertainty has been reduced. What it shows is that the p-adic structure is not inert with respect to the amplitude: the choice of number system leaves a definite, reproducible, and measurable imprint on the object computed. The possibility set out in Chapter 1 is thus sustained— not, as first imagined, at the level of precision, but at the level of *structural imprint*. The study does not confirm that the probabilistic character of the microscopic world arises from the limits of the real-number language; it was never positioned to do so within a single idealized simulation. But it furnishes a concrete instance of the premise on which that conjecture rests: that one and the same mathematical object, described over two different number systems, does not carry the same structure across them. In demonstrating this within an exact and fully controlled setting, the study moves the conjecture from a philosophical supposition toward a well-posed and computationally tractable question—which is precisely what a first step is meant to do.

Three questions remain, and the sections that follow take them up in turn. The first is *what kind of world* produced this result. The second is *where the study stands* in relation to the work that surrounds it. The third is *what the result does not reach*, and *where it might be carried next*.

5.2 The closed simulation and the standing of the result

It remains to ask what *kind* of result the simulation returned. Chapter 4 did not test the amplitude against a laboratory measurement of electron scattering, nor could it have. The $\mathcal{N} = 4$ super Yang–Mills amplitude computed here is not a prediction awaiting experimental confirmation. It is an idealized theoretical object, chosen precisely because its canonical form is exact, closed, and free of the empirical noise that would otherwise bury the question the study set out to pose. To read the result correctly, then, one must first be clear about the kind of world in which it was obtained.

That world is best described as a *closed simulation*. The analogy is to a physics engine of the sort that drives a modern game environment such as Unity or Omniverse: within the engine, bodies fall, collide, and rebound according to rules that are computed self-consistently and to full internal precision, answerable to nothing outside the engine. The engine does not measure the world; it *is* a world, complete in its own terms. The present simulation was built in the same spirit. The amplituhedron, its moment curve, and its canonical form together constitute a self-contained arithmetic universe, and the computation asks not whether that universe matches the plasma on the bench, but what structure the universe carries when it is described over one number system rather than another. The closure is not a limitation to be apologized for but the very condition of the experiment. Only in a setting sealed against empirical noise can the effect of the number system be isolated from every other source of variation and observed on its own. Had the object been an actual measured cross-section, the arithmetic imprint of the p-adic description would have

been buried beneath experimental uncertainty long before it could be seen. This closure fixes what kind of result Chapter 4 returned. It also fixes where the study stands in relation to the work around it.

5.3 The study in relation to existing work

The p-adic numbers have already entered the computation of scattering amplitudes. They entered it as a tool. The reconstruction discussed in Section 3.1 uses them to evaluate an amplitude near a singular surface, where ordinary floating-point arithmetic loses its accuracy [94]. A later method chooses its evaluation points by their p-adic valuation, and recovers the rational function in a far more compact form[1] [104]. In both the p-adic field is a means, and the end is a rational answer, obtained more reliably or written more briefly. The valuation is a parameter the method sets.

This study reverses that relation. It works inside a closed simulation, so there is no answer to recover more reliably. The rational track already gives the exact value. The question is a different one. It asks whether the number system leaves a trace on the object itself, and the p-adic numbers enter not as a means of computing a value but as the *subject* of the inquiry. The valuation is not set but read. What separates this study from the work before it is not technique but purpose.

It is this reorientation—from tool to question—that gives the modest result of Chapter 4 its weight. The finding that the choice of *number system* leaves a measurable imprint on the amplitude would be a minor curiosity if the goal were merely accurate computation. It becomes significant only against the question the closed simulation was built to pose: whether the probabilistic description of the microscopic world reflects the world or the language chosen for it. By showing, in an exact and controlled setting, that a change of number system does alter the structure carried by one and the same object, the study lifts that question out of philosophy and sets it on computational ground. It does not answer the question. What it establishes is that the question can be asked in a form that admits calculation.

Section 2.2 left a gap open. The object of computation is electron–electron scattering in the eV range of a low-temperature plasma, and in that range precise measurements of Møller scattering *do not exist*. The measurements that do exist sit far above it. The high-energy experiments at SLAC and Jefferson Lab measure a parity-violating asymmetry rather than a cross-section, at tens of GeV and at GeV respectively [88, 89]. The lowest measurement of the spectrum itself was made at about $2.5 MeV$, six orders of magnitude above the regime treated here [86]. That section deferred the meaning of this gap to the present chapter.

The meaning is this. Had the study sought agreement with measurement, the gap would be fatal, since there is nothing to agree with. But the study sought a structure inside an exactly computed object, and for that purpose the absence of a measurement is not an obstacle. The closed simulation is in part a response to this absence. Where no experiment can arbitrate, an object whose value is exact and whose arithmetic is fully controlled is what remains, and it is enough to ask whether the number system leaves a trace.

A second gap is harder to close. What this study computed is a scattering amplitude, and a scattering amplitude is defined *without spacetime* at all. The S-matrix

[1] Both stand on the finite-field reconstruction now standard in multi-loop computation [103]; the p-adic numbers are closely tied to finite fields, as noted in Section 2.3.2.

gives the amplitude for a transition from a past state of free particles to a future one, and says nothing of the interval between them— a point Appendix A states plainly. The amplituhedron carries this further and dispenses with spacetime from the outset. The etching of a plasma is not such a process. It *unfolds in time*, and no single amplitude describes it. The distance here is not one of accuracy but of dimension.

The field itself is pressing against this limit. The wavefunction of the universe, unlike a scattering amplitude, is a time-dependent quantity, and it too has been given a positive geometry: a polytope whose canonical form is the wavefunction, and one of whose faces returns the scattering amplitude [105]. The construction has since been extended to further polytopes for the wavefunction and for correlation functions [106, 107]. These remain toy models of scalar theory, and the extension to $\mathcal{N} = 4$ has not been made[2]. Their significance for this study lies elsewhere. The object computed here is a face of a larger and time-dependent one, and the arithmetic question raised in Chapter 4 does not depend on which of them is taken. The canonical form of a time-dependent geometry is a rational function of its own, and one may ask of it exactly what was asked here: whether the number system in which it is written leaves a trace. That is the direction in which the closed simulation opens onto the phenomenon that raised the question—and it is on the limits of the present formulation, and the directions in which they might be pressed, that the final section turns.

5.4 Limitations and future work

The result of this study is a first step, and a first step is defined as much by what it leaves undone as by what it secures. Three limitations bound the claim, and each opens a direction in which the work can be carried forward.

The first limitation is one of idealization. The object computed here is the canonical form of an $\mathcal{N} = 4$ super Yang–Mills amplitude—an exact, closed quantity deliberately chosen for the reasons set out in Section 5.2. That same closure, however, places a distance between the object and the plasma scattering that originally motivated the study. The Møller scattering of electrons in a low-temperature plasma is not an $\mathcal{N} = 4$ process, and the amplituhedron does not describe it directly; it serves here as a controlled surrogate in which the arithmetic question can be posed exactly, not as a model of the physical interaction itself. The imprint demonstrated in Chapter 4 is therefore a fact about the idealized object, and its bearing on the physical case remains a conjecture rather than a result. Narrowing that distance—identifying a physical amplitude, or a discretized process model, in which the same p-adic structure can be sought without the shelter of the closed simulation—is the natural direction in which the physical motivation of the study would be reengaged.

The second limitation is one of arithmetic regime. The conservation of the arithmetic skeleton, on which the central finding rests, was observed clearly only where no cell of the decomposition reduces to unity. As noted in Section 4.1.4, from $n = 8$ at $k = 2$ such cells appear, the 2-content common to the cells is lost, and the clean coincidence of valuations weakens. The skeleton conservation is thus not a universal law but a feature of a particular regime, and the study does not yet explain why the cells should degenerate where they do. A structural account of that transition is the first question left open at the level of the arithmetic itself. Whether the weakening

[2]The models capture the poles of the wavefunction but not the numerators, which is where the structure of Yang-Mills and gravity resides [105].

reflects a limit of the object or the departure in construction described next is not decided here.

The third limitation lies in the construction of the amplitude itself. The canonical form of a recursive cell is assembled in this implementation as the product of its subproblem forms, without a factor corresponding to the BCFW bridge. The cell counts and the amplitude values therefore differ from the standard BCFW expansion, and at $k = 1$ the amplitude degenerates to the cell count, so that the perturbed and unperturbed tracks return the identical value and the difference vanishes identically. The same departure accounts for the cells that reduce to unity from $n = 8$ onward, and so for the arithmetic boundary just described. The findings of Chapter 4 are properties of the function this implementation defines. Whether they persist once the bridge factor is restored is not settled here, and settling it is the most immediate technical task the present construction leaves behind.

Beyond these three limitations lies the direction toward which the whole of the present construction points. Section 3.2 set out two simulations and carried out only the first. The second is the **inverse problem**, and it is left to future work.

The simulation examined here is a forward computation. It begins with the moment-curve data and produces the amplitude; it does not attempt the *reverse passage*, from a given amplitude back to the descriptions of the input consistent with it. It is in this reverse passage that the divergence between the real and p-adic descriptions is expected to be most pronounced. Over the reals the reconstruction is governed by continuity, and the descriptions consistent with a given amplitude form a continuum. Over the p-adic numbers the ultrametric structure and the discreteness of the valuation are expected to constrain them far more rigidly, resolving that *continuum* into a *discrete set*. The two number systems would part company most visibly precisely where the input is recovered from the output.

The forward computation established in this thesis is the necessary precondition for that inquiry: one cannot study the inversion of a map before the map itself is constructed and shown to carry structure. Several technical extensions come with it—a full search over $\mathbb{Q}_p$ rather than the perturbative displacement used here, the lifting of multiple roots modulo p, and the extension of the measurement slice beyond the configurations examined in this study.

Taken together, these limitations and directions mark the boundary of what a single idealized simulation can establish, and no more than that. The thesis does not claim to have located the origin of quantum discreteness, nor to have shown that the probabilistic character of the microscopic world is an artifact of the real-number language. It claims something narrower and, within its bounds, secure: that the question can be posed exactly, that a change of number system leaves a definite and reproducible mark on an exactly computed amplitude, and that this mark has a definite arithmetic origin and a definite recursive structure. If the probabilistic description of nature does in part reflect the language in which nature is written rather than nature itself, then the structure exhibited here is the kind of trace that possibility would leave—and the task of the work to come is to follow that trace out of the closed simulation and toward the phenomenon that first raised the question.

Appendix A

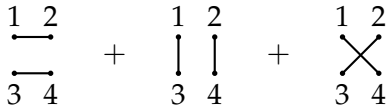
AppendixA

A.1 *S*-matrix (scattering matrix)

To handle the collisions and interactions of particles, the *S-matrix (scattering matrix)* was introduced. This provides an algebraic method to describe solely the probability amplitude of transitioning from a past state of free particles to a future state of free particles, circumventing the need to explicitly solve the labyrinthine internal processes occurring at the instant of collision [55, Chapter 4.5-6]. The matrix element itself is given in Equation (2.3).

To compute each element of this *S*-matrix, *Feynman diagrams* were introduced, visually representing particle interactions as topological combinations of lines and vertices[1].

$$\langle 0|T\{\phi_1\phi_2\phi_3\phi_4\}|0\rangle =$$

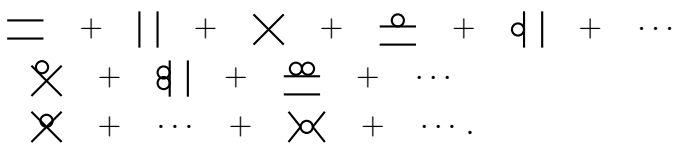


(A.1)

The generalization to higher correlation functions is [2]:

$$\langle \Omega|T\{\phi(x_1)\cdots\phi(x_n)\}|\Omega\rangle = \begin{pmatrix} \text{sum of all connected diagrams} \\ \text{with } n \text{ external points} \end{pmatrix} \tag{A.2}$$

$$\langle \Omega|T\phi_1\phi_2\phi_3\phi_4|\Omega\rangle =$$



(A.3)

[1] Redrawn from the four-point expansion in [55, Equation 4.42]

[2] Redrawn from the higher-point expansion in [55, Equation 4.57-8]

Appendix B

AppendixB

B.1 Full table of BCFW cell counts

Table B.1 lists the number of cells produced by this implementation's BCFW recursion (Section 3.3.3.3) for $5 \leq n \leq 20$ and $1 \leq k \leq 4$, at fixed $m = 4$. The counts reflect this implementation's specific decomposition. As noted in Section 3.2.1, the canonical form is independent of the cell decomposition [61, Section 6], so alternative schemes may yield different counts for the same Ω.

$n \backslash k$	$k = 1$	$k = 2$	$k = 3$	$k = 4$
$n = 5$	1	–	–	–
$n = 6$	1	1	–	–
$n = 7$	2	2	1	–
$n = 8$	3	4	3	1
$n = 9$	4	8	8	4
$n = 10$	5	14	19	14
$n = 11$	6	22	40	40
$n = 12$	7	32	76	100
$n = 13$	8	44	132	222
$n = 14$	9	58	213	448
$n = 15$	10	74	324	834
$n = 16$	11	92	470	1450
$n = 17$	12	112	656	2380
$n = 18$	13	134	887	3722
$n = 19$	14	158	1168	5588
$n = 20$	15	184	1504	8104

TABLE B.1: Number of BCFW cells produced by this implementation, for $5 \leq n \leq 20$ and $1 \leq k \leq 4$ at $m = 4$. Dashes mark configurations excluded by $n \geq k + m$.

Appendix C

Complete walkthrough of the (7,2,4) case

This appendix provides the detailed computation for the auxiliary case $(n, k, m) = (7, 2, 4)$, whose intermediate steps were referenced but not fully expanded in Section 3.3.3. It supplies the full input matrix, the exhaustive positivity check, and the explicit canonical-form evaluation of both BCFW cells.

C.1 The input matrix Z_{QQ}

For $(n, k, m) = (7, 2, 4)$ the moment-curve embedding $Z_{QQ}[a][j] = a^j$ yields a $(k + m) \times n = 6 \times 7$ matrix:

$$Z_{QQ} = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 1 & 4 & 9 & 16 & 25 & 36 & 49 \\ 1 & 8 & 27 & 64 & 125 & 216 & 343 \\ 1 & 16 & 81 & 256 & 625 & 1296 & 2401 \\ 1 & 32 & 243 & 1024 & 3125 & 7776 & 16807 \end{pmatrix} \tag{C.1}$$

Each column a is the point $Z_a = (1, a, a^2, a^3, a^4, a^5)$ on the moment curve.

C.2 Positivity check

The positivity condition requires every ordered 2×2 minor to be positive. With the first two rows equal to $(1, \ldots, 1)$ and $(1, 2, \ldots, 7)$, the minor reduces to $\Delta_{i_1,i_2} = i_2 - i_1$. The $\binom{7}{2} = 21$ values are therefore strictly positive:

$$\begin{array}{llllll} \Delta_{1,2} = 1 & \Delta_{1,3} = 2 & \Delta_{1,4} = 3 & \Delta_{1,5} = 4 & \Delta_{1,6} = 5 & \Delta_{1,7} = 6 \\ \Delta_{2,3} = 1 & \Delta_{2,4} = 2 & \Delta_{2,5} = 3 & \Delta_{2,6} = 4 & \Delta_{2,7} = 5 & \\ \Delta_{3,4} = 1 & \Delta_{3,5} = 2 & \Delta_{3,6} = 3 & \Delta_{3,7} = 4 & & \\ \Delta_{4,5} = 1 & \Delta_{4,6} = 2 & \Delta_{4,7} = 3 & & & \\ \Delta_{5,6} = 1 & \Delta_{5,7} = 2 & & & & \\ \Delta_{6,7} = 1 & & & & & \end{array}$$

The implementation does not evaluate all of them. As Section 3.3.3.2 notes, it checks the $n - k + 1 = 6$ minors on contiguous column blocks, $\Delta_{1,2}$ through $\Delta_{6,7}$, each equal to 1. The remaining values follow from these by additivity: $\Delta_{i_1,i_3} = \Delta_{i_1,i_2} + \Delta_{i_2,i_3}$.

C.3 BCFW decomposition

As derived in Section 3.3.3.2, the recursion produces two cells:

Cell	i	(k_L, k_R)	Left sub-problem	Right sub-problem
0	3	$(0,1)$	MHV, $(1,2,3,4)$	Simplex, $(1,4,5,6,7)$
1	4	$(1,0)$	Simplex, $(1,2,3,4,5)$	MHV, $(1,5,6,7)$

C.4 Canonical-form evaluation of the two cells

C.4.1 Cell 0 ($i = 3$, $(k_L, k_R) = (0,1)$)

The left sub-problem is MHV, so $\Omega(L) = 1$. The right sub-problem is a simplex on particles $(1,4,5,6,7)$ with $k_R = 1$. Extracting columns $\{1,4,5,6,7\}$ and the top five rows of Z_{QQ} gives a 5×5 submatrix Z_I with

$$\det(Z_I) = 4320, \qquad \det(Z_I)^m = 4320^4 = 348{,}285{,}173{,}760{,}000 \tag{C.2}$$

Since $k = 1$, the denominator (product of first-row entries) equals 1, so,

$$\Omega(\text{cell } 0) = 1 \times 348{,}285{,}173{,}760{,}000 = 348{,}285{,}173{,}760{,}000 = 2^{20} \cdot 3^{12} \cdot 5^4 \tag{C.3}$$

C.4.2 Cell 1 ($i = 4$, $(k_L, k_R) = (1,0)$)

The left sub-problem is a simplex on particles $(1,2,3,4,5)$ with $k_L = 1$, structurally identical to the main walkthrough of Section 3.3.3.4; its canonical form is $\det(Z_I)^4 = 288^4 = 6{,}879{,}707{,}136$. The right sub-problem is MHV, contributing $\Omega(R) = 1$. Hence,

$$\Omega(\text{cell } 1) = 6{,}879{,}707{,}136 \times 1 = 6{,}879{,}707{,}136 = 2^{20} \cdot 3^8 = \Omega(5,1) \tag{C.4}$$

C.5 Total amplitude and GCD hierarchy

Summing the two cells (Section 3.3.3.5) gives

$$\begin{aligned} \Omega^{\text{total}}_{QQ}(7,2) &= 348{,}285{,}173{,}760{,}000 + 6{,}879{,}707{,}136 \\ &= 348{,}292{,}053{,}467{,}136 = 2^{21} \cdot 3^8 \cdot 17 \cdot 1489 \end{aligned} \tag{C.5}$$

The greatest common divisor of the two cell values is

$$\gcd\big(\Omega(\text{cell } 0),\, \Omega(\text{cell } 1)\big) = 2^{20} \cdot 3^8 = 6{,}879{,}707{,}136 = \Omega(5,1) \tag{C.6}$$

and their ratio is exactly $\Omega(\text{cell } 0)/\Omega(\text{cell } 1) = 50{,}625 = 3^4 \cdot 5^4$. This is the GCD-hierarchy relation discussed in Section 4.1.7.

Appendix D

Appendix D

D.1 Verification invariants of the implementation

The implementation described in Chapter 3 is checked against five invariants. Each follows from the definition of the amplituhedron or from the moment-curve construction of the input, and each can be evaluated independently of the amplitude values themselves. All five are reproduced by the source code of Appendix D.

	Invariant	Configuration	Result
I-1	$\Omega = \det(Z_I)^m$ with $\det(Z_I) = \prod_{i<j}(a_j - a_i)$	$(5,1,4)$	$288^4 = 6{,}879{,}707{,}136$
I-2	Ordered maximal minors of Z positive, and matching the closed form of Equation (D.1)	$(7,2,4)$	7 of 7 exact
I-3	$n = k+m$ returns a single simplex cell	$k = 1,\ldots,5$	1 cell each
I-4	Cell counts satisfy the parity relation $k \leftrightarrow n-m-k$	$5 \le n \le 14$	35 pairs, 0 exceptions
I-5	$\Omega(5,1)$ divides $\Omega(cell0)$ of $(7,2)$; hence the gcd of the two cells equals $\Omega(5,1)$	$(7,2,4)$	quotient $3^4 5^4 = 50,625$

TABLE D.1: The five invariants used to verify the implementation.

I-1 and I-2 test the moment-curve construction of Step 1 3.3.3.1 and the positivity condition of Step 2 3.3.3.2. I-3 and I-4 test the splitting rules of Step 3 against structural properties of the amplituhedron that hold independently of this implementation; the cell counts on which I-4 is evaluated are those of Appendix B. I-5 tests that the recursion carries the arithmetic content of a lower configuration into a higher one, and is discussed in Section 4.1.7.

On I-2, at $(7,2,4)$ the matrix Z is 6×7, so every ordered maximal minor is obtained by omitting a single column c. The Vandermonde product then gives a closed form:

$$\det(Z_{\hat{c}}) = \frac{S}{(c-1)!\,(n-c)!} = 34{,}560\binom{6}{c-1}, S = \prod_{1\le i<j\le n}(a_j - a_i) = 24{,}883{,}200 \tag{D.1}$$

The seven minors are therefore $34{,}560 \cdot (1,6,15,20,15,6,1)$, that is 34,560, 207,360, 518,400, 691,200, 518,400, 207,360, 34,560, and the computed values reproduce all seven exactly. The sequence is a palindrome because the reflection $i \mapsto n+1-i$ leaves every pairwise difference $|a_j - a_i|$ unchanged and only reverses their order, so an ordered maximal minor equals that of the reflected index set. This reflection

invariance was verified over 673 maximal minors across the configurations $(7,2)$, $(8,2)$, $(9,2)$, $(10,2)$, $(8,1)$, $(10,1)$ and $(9,3)$, with no exception. I-2 therefore tests not only the sign of the minors but their exact values.

Appendix E

Appendix E

E.1 The complete Python source code

The listing below is the complete source of `experiment1_standalone.py`, the implementation of the five-step algorithm described in Section 3.3.3. All figures reported in Chapter 4 and all invariants of Appendix D are reproduced by this code.

```python
"""
experiment1_standalone.py
=========================
Amplituhedron Experiment 1: QQ / QP error comparison as particle number n increases

Runs in a standard Python environment (VSCode + WSL Ubuntu) without SageMath.

Required packages:
    pip install numpy matplotlib

Run:
    python experiment1_standalone.py
"""

import gc
import json
import math
import os
import time
from fractions import Fraction
from itertools import combinations

import numpy as np
import matplotlib.pyplot as plt
import matplotlib
matplotlib.rcParams['font.family'] = 'DejaVu Sans'  # use English labels


# ================
# 1. Number-system implementation
# ================

class QQField:
    """
    Rational field (exact arithmetic via Fraction).
    Replacement for SageMath's QQ.
    """
    @staticmethod
    def zero():
        return Fraction(0)

    @staticmethod
    def one():
        return Fraction(1)

    @staticmethod
    def from_int(n):
        return Fraction(n)

    @staticmethod
    def to_float(x):
        return float(x)


class PadicValuation:
    """
    p-adic valuation calculator.
    v_p(x) = the number of times p divides x.

    Replacement for SageMath Qp's .valuation().
    For a rational a/b, v_p(a/b) = v_p(a) - v_p(b).
    """
    def __init__(self, p):
        self.p = p

    def valuation(self, x):
        """
        x: Fraction or int
        returns: v_p(x) (integer or inf)
        """
        if x == 0:
            return float('inf')
        if isinstance(x, Fraction):
            return self._val_int(x.numerator) - self._val_int(x.denominator)
        return self._val_int(int(x))

    def _val_int(self, n):
        if n == 0:
            return float('inf')
        n = abs(n)
        count = 0
        while n % self.p == 0:
            n //= self.p
            count += 1
        return count


# ================
# 2. Matrix operations (exact rational arithmetic)
# ================

class FractionMatrix:
    """
    Exact matrix operations based on Fraction.
    Replacement for SageMath matrix(QQ, ...).
    Computes the determinant as an exact rational (no floating-point error).
    """
    def __init__(self, rows):
        """
        rows: list of list of Fraction (int/float are converted to Fraction)
        """
        self.data = [[Fraction(x) for x in row] for row in rows]
        self.nrows = len(rows)
        self.ncols = len(rows[0]) if rows else 0

    def matrix_from_rows_and_columns(self, row_indices, col_indices):
        """Extract a submatrix from the given rows/columns."""
        rows = []
        for r in row_indices:
            rows.append([self.data[r][c] for c in col_indices])
        return FractionMatrix(rows)

    def matrix_from_columns(self, col_indices):
        """Extract a submatrix from the given columns."""
        return self.matrix_from_rows_and_columns(range(self.nrows), col_indices)

    def det(self):
        """
        Fraction determinant via LU decomposition.
        Guarantees an exact rational result.
        """
        n = self.nrows
        if n != self.ncols:
            raise ValueError(
                f"det() applies only to square matrices. size: {self.nrows}x{self.ncols}"
            )
        if n == 0:
            return Fraction(1)
        if n == 1:
            return self.data[0][0]
        if n == 2:
            return (self.data[0][0] * self.data[1][1]
                    - self.data[0][1] * self.data[1][0])

        # Gaussian elimination with Fraction (exact)
        M = [row[:] for row in self.data]
        det = Fraction(1)
        for col in range(n):
```

```
            # pivot selection
            pivot_row = None
            for row in range(col, n):
                if M[row][col] != 0:
                    pivot_row = row
                    break
            if pivot_row is None:
                return Fraction(0)
            if pivot_row != col:
                M[col], M[pivot_row] = M[pivot_row], M[col]
                det = -det

            det *= M[col][col]
            pivot = M[col][col]
            for row in range(col + 1, n):
                if M[row][col] == 0:
                    continue
                factor = M[row][col] / pivot
                for c in range(col, n):
                    M[row][c] -= factor * M[col][c]

        return det


# ========================
# 3. Momentum-twistor Z generation (moment curve)
# ========================
def generate_Z_qq(n, k, m=4):
    """
    Rational track: Z_a = (1, a, a^2, ..., a^(k+m-1))
    returns: FractionMatrix (k+m) x n
    """
    rows = []
    for a in range(1, n + 1):
        row = [Fraction(a ** j) for j in range(k + m)]
        rows.append(row)
    # transpose to (k+m) x n
    transposed = [[rows[a][j] for a in range(n)] for j in range(k + m)]
    return FractionMatrix(transposed)


# ========================
# 4. Positivity condition check
# ========================
def is_classical_positive(Z_matrix, k, m=4):
    """
    Check ordered k x k minor > 0 -- definition eq.(2.4) of the paper.

    [Theoretical basis]
    The positivity condition of the amplituhedron requires that only
    'ordered' minors with a1 < a2 < ... < ak be positive.
    Postnikov (2006)'s sufficient condition is likewise based on ordered minors.

    [Cause of a previous bug]
    Using cyclic-consecutive indices (including wrap-around)
    includes reversed combinations such as cols=(n-1, 0), causing det < 0.
    The moment curve guarantees positivity only for ordered minors;
    the sign of a wrap-around minor is arbitrary depending on k, n.

    [Fix]
    iterate a1 over range(n - k + 1) -> always guarantees a1 < a2 < ... < ak.
    no wrap-around -> passes correctly for all k=1,2,3,...

    [Squareness]
    use the top k rows x k columns for the k x k minor.
    """
    n_cols = Z_matrix.ncols

    if k == 0:
        return True  # G(0,n): always positive

    # ordered: a1 < a2 < ... < ak, no wrap
    for a1 in range(n_cols - k + 1):
        cols = list(range(a1, a1 + k))
        # top k rows x k cols -> guarantees k x k square
        sub = Z_matrix.matrix_from_rows_and_columns(list(range(k)), cols)
        val = sub.det()
        if val <= 0:
            return False
    return True


# ========================
# 5. BCFW cell decomposition (with memoization)
# ========================

def get_bcfw_cells(n, k, m=4, current_labels=None, cache=None):
    """
    BCFW recursive cell decomposition.
    Memoization: cached with (labels_tuple, k) as key.
    """
    if cache is None:
        cache = {}
    if current_labels is None:
        current_labels = list(range(n))

    cache_key = (tuple(current_labels), k)
    if cache_key in cache:
        return cache[cache_key]

    # physical validity
    if n < k + m:
        raise ValueError(f"physically invalid: n={n} < k+m={k+m}")

    # base case 1: k == 0 -> MHV
    if k == 0:
        result = {"type": "MHV", "indices": tuple(current_labels), "k": 0}
        cache[cache_key] = result
        return result

    # base case 2: n == k + m -> simplex
    if n == k + m:
        result = {
            "type":    "simplex",
            "indices": tuple(sorted(current_labels)),
            "k":       k
        }
        cache[cache_key] = result
        return result

    all_cells = []
    for i in range(2, n - 1):
        k_L_min = max(0, k - (n - i + 1))
        k_L_max = min(k - 1, i - 2)

        for k_L in range(k_L_min, k_L_max + 1):
            k_R = k - 1 - k_L
            l_labels = current_labels[:i + 1]
            r_labels = [current_labels[0]] + current_labels[i:]

            if len(l_labels) < k_L + m:
                continue
            if len(r_labels) < k_R + m:
                continue

            l_res = get_bcfw_cells(len(l_labels), k_L, m, l_labels, cache)
            r_res = get_bcfw_cells(len(r_labels), k_R, m, r_labels, cache)

            l_list = l_res if isinstance(l_res, list) else [l_res]
            r_list = r_res if isinstance(r_res, list) else [r_res]

            for l_cell in l_list:
                for r_cell in r_list:
                    all_cells.append({
                        "type":    "bcfw",
                        "L_part":  l_cell,
                        "R_part":  r_cell,
                        "split_at": i,
                        "k_map":   (k_L, k_R),
                        "k":       k
                    })

    cache[cache_key] = all_cells
    return all_cells


# ========================
# 6. Canonical form computation
# ========================
def _slice_Z(Z_matrix, k_sub, indices, m=4):
    """
    slice Z rows to k_sub -> guarantees (k_sub+m) x (k_sub+m) square.
    """
    k_plus_m = k_sub + m
    n_rows = Z_matrix.nrows

    if n_rows == k_plus_m:
        return Z_matrix.matrix_from_columns(indices)
    elif n_rows > k_plus_m:
        return Z_matrix.matrix_from_rows_and_columns(
            list(range(k_plus_m)), indices
        )
    else:
        raise ValueError(
            f"Z row count({n_rows}) < k_sub+m({k_plus_m}). "
            f"k_sub={k_sub}, indices={indices}"
        )


def compute_simplex_canonical_form(cell, Z_matrix, m=4):
    """
    Compute the canonical form of a simplex cell.
    Omega = det(Z_I)^m / prod(cyclic consecutive k_sub x k_sub minors)

    [Squareness note]
    - numerator det: (k_sub+m)x(k_sub+m) -> use _slice_Z
    - denominator minor: k_sub x k_sub -> use only the top k_sub rows
    """
    indices = list(cell["indices"])
    k_sub   = cell["k"]
    n_local = len(indices)  # == k_sub + m

    # numerator: det(Z_sq)^m -- (k_sub+m)x(k_sub+m) square
    Z_sq    = _slice_Z(Z_matrix, k_sub, indices, m)
    det_val = Z_sq.det()
    numerator = det_val ** m

    # denominator: product of cyclic-consecutive k_sub x k_sub minors
    # must be k_sub rows x k_sub cols -> k_sub x k_sub square
```

```
    denominator = Fraction(1)
    for start in range(n_local):
        local_cols  = [(start + offset) % n_local for offset in range(k_sub)]
        global_cols = [indices[c] for c in local_cols]

        # top k_sub rows x k_sub cols -> guarantees k_sub x k_sub square
        sub       = Z_matrix.matrix_from_rows_and_columns(
                        list(range(k_sub)), global_cols
                    )
        minor_val = sub.det()
        if minor_val == 0:
            return Fraction(0)
        denominator *= minor_val

    if denominator == 0:
        return Fraction(0)
    return numerator / denominator


def compute_cell_canonical_form(cell, Z_matrix, m=4):
    """
    Recursively compute the canonical form by cell type.
    returns: Fraction
    """
    cell_type = cell.get("type")

    if cell_type == "MHV":
        return Fraction(1)

    if cell_type == "simplex":
        return compute_simplex_canonical_form(cell, Z_matrix, m)

    if cell_type == "bcfw":
        total = Fraction(0)
        l_cells = cell["L_part"] if isinstance(cell["L_part"], list) else [cell["L_part"]]
        r_cells = cell["R_part"] if isinstance(cell["R_part"], list) else [cell["R_part"]]
        for l_cell in l_cells:
            for r_cell in r_cells:
                l_val = compute_cell_canonical_form(l_cell, Z_matrix, m)
                r_val = compute_cell_canonical_form(r_cell, Z_matrix, m)
                total += l_val * r_val
        return total

    return Fraction(0)


def compute_total_amplitude(bcfw_cells, Z_matrix, m=4):
    """
    Sum the canonical forms of all BCFW cells -> total amplitude Omega.
    returns: Fraction
    """
    cells = bcfw_cells if isinstance(bcfw_cells, list) else [bcfw_cells]
    total = Fraction(0)
    for cell in cells:
        total += compute_cell_canonical_form(cell, Z_matrix, m)
    return total


# ======================
# 7. p-adic error computation
# ======================

def compute_padic_error(omega_qq, omega_qp, p, prec):
    """
    Compute the p-adic error via the relative-depth method.

    [Key idea]
    with eps = Fraction(n_step, p^N), v_p(eps) = -N.
    diff = omega_qp - omega_qq ~ (d omega/d eps) * eps

    raw depth = v_p(diff) = v_p(d omega/d eps) + v_p(eps)
                          = v_p(partial omega/partial eps) - N
    -> depends on N, so not comparable across n.

    relative depth = v_p(diff) - v_p(eps)
                   = v_p(diff) + N   (since v_p(eps) = -N)
                   = v_p(partial omega/partial eps)
    -> p-adic structure of the amplitude itself, independent of N (the eps setting)
    -> a meaningful quantity that varies with n

    QP_error = p^(-relative_depth)
    the larger relative_depth is, the smaller QP_error is
    (stronger p-adic structure of the amplitude -> perturbation is p-adically aligned)

    Parameters
    ----------
    omega_qq : Fraction  rational amplitude
    omega_qp : Fraction  perturbed amplitude (includes eps = 1/p^N)
    p        : int       prime
    prec     : int       precision (N = prec//2)

    Returns
    -------
    (relative_depth, qp_error, raw_depth)
    """
    pval = PadicValuation(p)
    N    = prec // 2           # v_p(eps) = -N

    diff = omega_qp - omega_qq

    if diff == 0:
        # exact match: use the theoretical ceiling prec as the relative depth
        return prec, 0.0, prec

    raw_depth = pval.valuation(diff)

    if raw_depth == float('inf'):
        return prec, 0.0, prec

    raw_int = int(raw_depth)

    # relative depth = v_p(diff) - v_p(eps) = raw_depth - (-N) = raw_depth + N
    relative_depth = raw_int + N

    if relative_depth <= 0:
        # perturbation has no p-adic structure -> QP_error = 1 (maximum error)
        qp_error = 1.0
    else:
        qp_error = float(p) ** (-relative_depth)

    return relative_depth, qp_error, raw_int


def generate_Z_with_monad(n, k, m=4, n_step=1, p=23, prec=1024):
    """
    Z matrix with monad perturbation applied (QP track).

    [Bug history]
    v1: eps = p^(prec//2) (large integer) -> Z information is buried in eps
    v2: eps = 1/p^N (small rational, uniform) -> fraction reduction fixes v_p(diff) = 0

    [Root cause of v2]
    adding the same eps to all Z entries means that, when computing omega_qp - omega_qq,
    fraction reduction cancels 23^N from both numerator and denominator.
    result: v_p(diff) = 0 -> relative_depth = 128 (a constant independent of n)

    [v3 fix: per-entry distinct eps]
    eps_{row, col} = eps_base \times (row+1) \times (col+1)
    = (n_step / p^N) \times (row+1) \times (col+1)

    reason:
    - (row+1), (col+1) are integers in 1..(k+m), 1..n
    - since n < p=23, (col+1) <= n < 23 -> no multiple of 23
    - (row+1) <= k+m <= 7 -> no multiple of 23
    - hence v_p(eps_{row,col}) = v_p(eps_base) = -N
    - but since each entry's eps differs, in the determinant expansion
      the eps factors of each term differ, so cancellation is incomplete
    - -> v_p of diff = omega_qp - omega_qq acquires n-dependence

    physical meaning:
    (row+1): weight along the moment-curve order (higher orders perturbed more)
    (col+1): weight along the particle number (later particles perturbed more)
    this models a non-uniform off-shell deviation as particles are added.
    """
    eps_base = Fraction(n_step, p ** (prec // 2))

    rows = []
    for a in range(1, n + 1):          # a = col+1 (1-indexed)
        row = []
        for j in range(k + m):         # j = row index (0-indexed)
            eps_ij = eps_base * (j + 1) * a   # per-entry distinct eps
            row.append(Fraction(a ** j) + eps_ij)
        rows.append(row)
    transposed = [[rows[a][j] for a in range(n)] for j in range(k + m)]
    return FractionMatrix(transposed)


# ======================
# 8. Experiment 1 main function
# ======================
def run_experiment_1(
    k=1,
    n_list=None,
    n_step_monad=1,
    p_val=23,
    precision=1024,
    log_path="experiment1_log.json",
    resume=True,
    m=4
):
    """
    Experiment 1: compare QQ / QP error as particle number n increases.

    Parameters
    ----------
    k            : NMHV degree (default 1)
```

```
    n_list       : list of n to test. If None, defaults to [k+m, ..., k+m+5].
    n_step_monad : monad perturbation strength
    p_val        : p-adic prime (default 23, fixed throughout)
    precision    : upper bound for variation_depth (default 1024)
    log_path     : path to the JSON log file
    resume       : if True, resume from an existing log
    m            : spacetime dimension (fixed 4)
    """
    n_min_valid = k + m
    if n_list is None:
        n_list = list(range(n_min_valid, n_min_valid + 6))

    invalid = [n for n in n_list if n < n_min_valid]
    if invalid:
        raise ValueError(f"invalid n included: {invalid}. minimum={n_min_valid}")

    # -- load existing log --
    completed = {}
    omega_ref = None

    if resume and os.path.exists(log_path):
        try:
            with open(log_path, "r", encoding="utf-8") as f:
                log_data = json.load(f)
            for row in log_data.get("rows", []):
                completed[row["n"]] = row
            omega_ref = log_data.get("omega_ref")
            print(f"[Resume] '{log_path}' -> {len(completed)} results loaded.")
            remaining = [n for n in n_list if n not in completed]
            print(f"  Done: {sorted(completed.keys())}")
            print(f"  Remaining: {remaining}")
            if not remaining:
                print("  All done. Returning cached results.")
                rows = [completed[n] for n in sorted(completed) if n in n_list]
                return _build_result(k, p_val, precision, n_step_monad,
                                     n_list, rows, omega_ref, log_path)
            print()
        except Exception as e:
            print(f"[Warning] Log read failed ({e}). Starting fresh.")
            completed = {}
            omega_ref = None

    # -- print experiment header --
    print("=" * 70)
    print("  Experiment 1: QQ vs QP Error Rate by Particle Number n")
    print(f"  k={k}, m={m}, p={p_val} (fixed), precision={precision} (fixed)")
    print(f"  n_list={n_list}, n_step_monad={n_step_monad}")
    print(f"  Log: {log_path}")
    print("=" * 70)
    print(f"  {'n':>3} | {'cells':>5} | {'Omega_QQ':>18} | "
          f"{'QQ_err':>12} | {'depth':>6} | {'QP_err':>14} | {'time':>6} | status")
    print("-" * 100)

    rows_all = list(completed.values())
    n_min_in_list = min(n_list)

    for n in n_list:
        if n in completed:
            r = completed[n]
            print(f"  {n:>3} | [skipped - already in log]")
            continue

        row = {
            "n":          n,
            "p_used":     p_val,
            "prec_used": precision,
            "status":     "ok",
            "timestamp": time.strftime("%Y-%m-%d %H:%M:%S"),
        }
        t0 = time.time()
        bcfw_cells = None
        Z_qq       = None
        Z_qp       = None

        try:
            # -- 1. generate positive Z matrix --
            Z_qq = generate_Z_qq(n, k, m)

            # verify positivity condition
            if not is_classical_positive(Z_qq, k):
                raise ValueError("Z_qq does not satisfy the positivity condition.")

            # -- 2. monad-perturbed Z (QP track) -- pass precision to fix eps direction
            Z_qp = generate_Z_with_monad(n, k, m, n_step_monad, p_val, precision)

            # -- 3. BCFW cell decomposition --
            cache = {}
            bcfw_cells = get_bcfw_cells(n, k, m, cache=cache)
            n_cells = len(bcfw_cells) if isinstance(bcfw_cells, list) else 1
            row["n_cells"] = n_cells
            del cache  # release cache immediately

            # -- 4. amplitude computation (QQ track) --
            omega_qq_frac = compute_total_amplitude(bcfw_cells, Z_qq, m)
            omega_qq_f    = float(omega_qq_frac)
            row["omega_qq"] = omega_qq_f

            # -- 5. amplitude computation (QP track) --
            omega_qp_frac = compute_total_amplitude(bcfw_cells, Z_qp, m)
            omega_qp_f    = float(omega_qp_frac)
            row["omega_qp_real"] = omega_qp_f

            # -- 6. p-adic variation depth (relative-depth method) --
            rel_depth, qp_err, raw_depth = compute_padic_error(
                    omega_qq_frac, omega_qp_frac, p_val, precision
                    )
            row["variation_depth"] = rel_depth   # relative depth (captures n-dependence)
            row["raw_depth"]       = raw_depth   # raw depth (for reference)
            row["qp_error"]        = qp_err

            # -- 7. QQ error computation --
            # reference: Omega_QQ of the first successful n (proceed to next n if n_min fails)
            if omega_ref is None:
                omega_ref       = omega_qq_f if omega_qq_f != 0 else 1.0
                row["qq_error"] = 0.0
            elif omega_ref != 0:
                row["qq_error"] = abs(omega_qq_f - omega_ref) / abs(omega_ref)
            else:
                row["qq_error"] = float('inf')

        except Exception as e:
            row["status"] = f"ERROR: {e}"
            for key in ["n_cells", "omega_qq", "omega_qp_real",
                        "variation_depth", "qq_error", "qp_error"]:
                row.setdefault(key, None)

        # -- 8. release memory immediately --
        del bcfw_cells
        del Z_qq
        del Z_qp
        gc.collect()

        row["elapsed_sec"] = round(time.time() - t0, 2)

        # -- 9. one-line output --
        omega_s  = f"{row.get('omega_qq'):.6e}"    if row.get('omega_qq')       is not None else "N/A"
        qq_s     = f"{row.get('qq_error'):.4e}"    if row.get('qq_error')       is not None else "N/A"
        qp_s     = f"{row.get('qp_error'):.4e}"    if row.get('qp_error')       is not None else "N/A"
        dp_s     = str(row.get('variation_depth',  'N/A'))
        cells_s  = str(row.get('n_cells', '-'))

        print(f"  {n:>3} | {cells_s:>5} | {omega_s:>18} | "
              f"{qq_s:>12} | {dp_s:>6} | {qp_s:>14} | "
              f"{row['elapsed_sec']:>5}s | {row['status']}")

        # -- 10. save log immediately --
        rows_all.append(row)
        completed[n] = row
        _save_log(log_path, k, p_val, precision, n_step_monad,
                  n_list, rows_all, omega_ref)

    rows_sorted = [completed[n] for n in sorted(completed) if n in n_list]
    return _build_result(k, p_val, precision, n_step_monad,
                         n_list, rows_sorted, omega_ref, log_path)


# ==============================
# 9. log-saving / result-assembly helpers
# ==============================
def _save_log(log_path, k, p, precision, n_step, n_list, rows, omega_ref):
    """Save JSON immediately after each n completes."""
    safe_rows = []
    for row in rows:
        safe_row = {}
        for key, val in row.items():
            if val == float('inf'):
                safe_row[key] = "Inf"
            elif val != val:  # NaN
                safe_row[key] = "NaN"
            else:
                try:
                    json.dumps(val)
                    safe_row[key] = val
                except Exception:
                    safe_row[key] = str(val)
        safe_rows.append(safe_row)
```

```
    data = {
        "experiment": "experiment_1",
        "k":          k,
        "p":          p,
        "precision":  precision,
        "n_step":     n_step,
        "n_list":     n_list,
        "omega_ref":  omega_ref,
        "saved_at":   time.strftime("%Y-%m-%d %H:%M:%S"),
        "rows":       safe_rows,
    }
    try:
        with open(log_path, "w", encoding="utf-8") as f:
            json.dump(data, f, ensure_ascii=False, indent
=2)
    except Exception as e:
        print(f"  [Log save failed] {e}")


def _build_result(k, p, precision, n_step, n_list, rows,
omega_ref, log_path):
    return {
        "k":          k,
        "p":          p,
        "precision": precision,
        "n_step":    n_step,
        "n_list":    n_list,
        "omega_ref": omega_ref,
        "log_path":  log_path,
        "rows":      rows,
    }


# ============================
# 10. result output
# ============================

def print_experiment_1(results):
    """Print a summary of Experiment 1 results."""
    k    = results["k"]
    p    = results["p"]
    prec = results["precision"]

    print()
    print("=" * 80)
    print("  Experiment 1 Results: QQ vs QP Error Rate")
    print(f"  k={k}, p={p} (fixed), precision={prec} (
fixed), m=4")
    print(f"  Reference Omega_QQ (n_min) = "
          f"{results['omega_ref']:.6g}" if results.get('
omega_ref') is not None
          else "  Reference Omega_QQ (n_min) = N/A")
    print("=" * 80)
    print(f"  {'n':>3} | {'cells':>5} | {'Omega_QQ':>16} |
 "
          f"{'QQ_error':>12} | {'depth':>6} | {'QP_error
':>14} | interpretation")
    print("-" * 90)

    for row in results["rows"]:
        n = row["n"]
        if row.get("status", "ok") != "ok":
            print(f"  {n:>3} | ERROR: {row['status']}")
            continue

        omega_s = f"{row.get('omega_qq', 0):.6g}"
        qq_s    = f"{row.get('qq_error', 0):.4e}"   if row
.get('qq_error')  is not None else "N/A"
        qp_s    = f"{row.get('qp_error', 0):.4e}"   if row
.get('qp_error')  is not None else "N/A"
        dp_s    = str(row.get('variation_depth', 'N/A'))

        # interpretation
        qe = row.get('qq_error')
        pe = row.get('qp_error')
        if pe is not None and qe is not None:
            if pe == 0.0:
                interp = "QP == QQ (exact)"
            elif pe < 1e-10:
                interp = "QP near-exact"
            elif pe < qe:
                interp = "QP < QQ error"
            elif pe > qe:
                interp = "QP > QQ error"
            else:
                interp = "QP = QQ error"
        else:
            interp = ""

        print(f"  {n:>3} | {row.get('n_cells','-'):>5} | {
omega_s:>16} | "
              f"{qq_s:>12} | {dp_s:>6} | {qp_s:>14} | {
interp}")

    print("=" * 80)

    # trend summary
    valid = [r for r in results["rows"]
             if r.get("status", "ok") == "ok"
             and r.get("qq_error") is not None
             and r.get("qp_error") is not None]
    if len(valid) < 2:
        return

    qq_errs = [r["qq_error"] for r in valid]
    qp_errs = [r["qp_error"] for r in valid]
    depths  = [r["variation_depth"] for r in valid if r.
get("variation_depth") is not None]

    def trend(vals):
        if all(vals[i] <= vals[i+1] for i in range(len(
vals)-1)):
            return "monotone increasing"
        elif all(vals[i] >= vals[i+1] for i in range(len(
vals)-1)):
            return "monotone decreasing"
        return "irregular"

    print()
    print("  [Summary]")
    print(f"  QQ_error trend : {trend(qq_errs)}")
    print(f"  QP_error trend : {trend(qp_errs)}")
    qp_lt = sum(1 for r in valid if r["qp_error"] < r["
qq_error"])
    print(f"  QP < QQ error  : {qp_lt}/{len(valid)} cases"
)
    if depths:
        print(f"  depth range    : {min(depths)} ~ {max(
depths)}")
        if len(set(depths)) == 1:
            print(f"  depth stable ({depths[0]}) ->
structural perturbation (Hypothesis B)")


# ============================
# 11. plot output
# ============================

def plot_experiment_1(results, save_path=None):
    """
    Grouped bar chart + log y-axis.
    - blue bars: QQ_error
    - orange bars: QP_error
    - green line (right axis): variation_depth
    """
    valid = [r for r in results["rows"]
             if r.get("status", "ok") == "ok"
             and r.get("qq_error") is not None
             and r.get("qp_error") is not None]

    if not valid:
        print("No valid data for plot.")
        return

    ns        = [r["n"]               for r in valid]
    qq_errors = [max(r["qq_error"],  1e-300) for r in
valid]
    qp_errors = [max(r["qp_error"],  1e-300) for r in
valid]
    depths    = [r.get("variation_depth") for r in valid]

    fig, ax1 = plt.subplots(figsize=(max(8, len(ns) * 1.4)
, 5.5))

    x = np.arange(len(ns))
    w = 0.30

    bars_qq = ax1.bar(x - w/2, qq_errors, w,
                      label="QQ_error (rational)", color="
#3B82F6", alpha=0.85)
    bars_qp = ax1.bar(x + w/2, qp_errors, w,
                      label=f"QP_error (p={results['p']}-
adic)", color="#F97316", alpha=0.85)

    ax1.set_yscale("log")
    ax1.set_xlabel("Particle number n", fontsize=11)
    ax1.set_ylabel("Error rate (log scale)", fontsize=11)
    ax1.set_xticks(x)
    ax1.set_xticklabels([str(n) for n in ns])
    ax1.grid(axis="y", alpha=0.3, linestyle="--")
    ax1.set_ylim(bottom=1e-300 * 0.1)

    # value labels
    for bar, val in zip(bars_qq, [r["qq_error"] for r in
valid]):
        if val is not None and val > 1e-100:
            ax1.text(bar.get_x() + bar.get_width()/2,
                     bar.get_height() * 2,
                     f"{val:.1e}", ha="center", va="bottom
",
                     fontsize=7, color="#1E40AF")
    for bar, val in zip(bars_qp, [r["qp_error"] for r in
valid]):
        if val is not None and val > 1e-100:
            ax1.text(bar.get_x() + bar.get_width()/2,
                     bar.get_height() * 2,
                     f"{val:.1e}", ha="center", va="bottom
",
                     fontsize=7, color="#C2410C")

    # secondary axis: variation_depth
    ax2 = ax1.twinx()
    valid_d = [(i, d) for i, d in enumerate(depths) if d
is not None]
    if valid_d:
        xi, di = zip(*valid_d)
        ax2.plot(list(xi), list(di), "g-o", linewidth=1.8,
                 markersize=5, label="variation_depth",
alpha=0.85)
        ax2.set_ylabel("variation_depth  v_p(Omega_QP -
Omega_QQ)",
                       fontsize=10, color="green")
    ax2.tick_params(axis="y", labelcolor="green")

    lines1, labels1 = ax1.get_legend_handles_labels()
    lines2, labels2 = ax2.get_legend_handles_labels()
    ax1.legend(lines1 + lines2, labels1 + labels2, loc="
upper left", fontsize=9)

    title = (f"Experiment 1: QQ vs QP Error Rate by n\n"
             f"k={results['k']}, m=4, p={results['p']} (
fixed), "
             f"precision={results['precision']}, "
             f"n_step={results['n_step']}")
```

```
    ax1.set_title(title, fontsize=10, pad=10)

    plt.tight_layout()
    if save_path:
        plt.savefig(save_path, dpi=150, bbox_inches="tight
↪ ")
        print(f"Graph saved: {save_path}")
    else:
        plt.savefig("experiment1_result.png", dpi=150,
↪ bbox_inches="tight")
        print("Graph saved: experiment1_result.png")
    plt.show()
    plt.close()


# ========================
# 12. estimate n limit
# ========================

def estimate_n_limit(k=1, n_max=15, m=4):
    """Estimate the feasible n limit by counting BCFW
↪ cells only."""
    cache = {}

    def count(n, k):
        key = (n, k)
        if key in cache: return cache[key]
        if n < k + m: return 0
        if k == 0: return 1
        if n == k + m: return 1
        total = 0
        for i in range(2, n - 1):
            k_L_min = max(0, k - (n - i + 1))
            k_L_max = min(k - 1, i - 2)
            for k_L in range(k_L_min, k_L_max + 1):
                k_R = k - 1 - k_L
                if i+1 >= k_L+m and n-i+1 >= k_R+m:
                    total += count(i+1, k_L) * count(n-i
↪ +1, k_R)
        cache[key] = total
        return total

    print(f"\n  n limit estimate (k={k}, m={m})")
    print(f"  {'n':>4} | {'cells':>8} | {'verdict':>20}")
    print("  " + "-" * 40)
    prev = 1
    for n in range(k+m, n_max+1):
        c = count(n, k)
        g = c / prev if prev > 0 else 0
        prev = max(c, 1)
        v = ("recommended" if c <= 50 else
             "feasible"    if c <= 500 else
             "slow"        if c <= 5000 else "too slow")
        print(f"  {n:>4} | {c:>8,} | {v:>20}  (x{g:.1f})")


# ====================
# 13. entry point
# ====================

if __name__ == "__main__":
    print()
    print("Amplituhedron Experiment 1 - Standalone Python"
↪ )
    print("p=23 fixed, comparing QQ (rational) vs QP (p-
↪ adic) error rates")
    print()

    # -- preliminary check: n limit --
    estimate_n_limit(k=1, n_max=12)
    print()

    # -- run experiment --
    # specify n_list directly; run only the desired n
↪ values.
    # precision: 1024 (default) or 2048 (more precise,
↪ slower)
    results = run_experiment_1(
        k           = 1,
        n_list      = [5, 6, 7, 8],   # <- specify the
↪ desired n list directly
        n_step_monad = 1,
        p_val       = 23,
        precision   = 1024,
        log_path    = "experiment1_log.json",
        resume      = True,
    )

    # -- print results --
    print_experiment_1(results)

    # -- save plot --
    plot_experiment_1(results, save_path="
↪ experiment1_result.png")


"""
experiment2_matrix.py
====================
Experiment 2: precision x n_step matrix experiment

purpose:
  (1) effect of precision on depth -> is k=1 depth=64 the
↪ theoretical ceiling?
  (2) effect of n_step on depth -> discriminate hypothesis
↪  A / B

design:
    precision in {64, 128, 256}
    n_step    in {1, 2, 4}
    k         in {1, 2}
  p=2 fixed, m=4 fixed

discrimination logic:
  - if k=1 depth rises with precision -> k=1's QP=QQ is a
↪ precision artifact
  - if k=1 depth stays fixed as precision rises ->
↪ structural p-adic completeness of MHV (confirms
↪ hypothesis B)
  - if k=2 depth is linear in n_step -> hypothesis A (off-
↪ shell deviation)
  - if k=2 depth is independent of n_step -> hypothesis B
↪ (structural perturbation)

Run:
  python experiment2_matrix.py
"""

import os
import sys
import time
import json

# reuse the functions of experiment1_standalone.py
sys.path.insert(0, os.path.dirname(os.path.abspath(
↪ __file__)))
from experiment1_standalone import (
    run_experiment_1, generate_Z_qq, generate_Z_with_monad
↪ ,
    is_classical_positive, get_bcfw_cells,
↪ compute_total_amplitude,
    compute_padic_error, PadicValuation, FractionMatrix
)


def run_experiment_2(
    k_list=(1, 2),
    precision_list=(64, 128, 256),
    n_step_list=(1, 2, 4),
    n_range_k1=(5, 10),
    n_range_k2=(6, 10),
    p_val=2,
    log_dir="exp2_logs",
):
    """
    Run the precision x n_step matrix.

    Run run_experiment_1 for each (k, precision, n_step)
↪ combination.
    Save results in exp2_logs/.
    """
    os.makedirs(log_dir, exist_ok=True)

    all_results = {}
    total_configs = len(k_list) * len(precision_list) *
↪ len(n_step_list)
    done = 0
    t_total_start = time.time()

    print("=" * 75)
    print(f"  Experiment 2: precision x n_step matrix")
    print(f"  k={list(k_list)},  precision={list(
↪ precision_list)},  n_step={list(n_step_list)}")
    print(f"  {total_configs} combinations total")
    print("=" * 75)

    for k in k_list:
        if k == 1:
            n_min, n_max = n_range_k1
        else:
            n_min, n_max = n_range_k2
        n_list = list(range(n_min, n_max + 1))

        for prec in precision_list:
            for nstep in n_step_list:
                done += 1
                config_key = f"k{k}_prec{prec}_nstep{nstep
↪ }"
                log_path = os.path.join(log_dir, f"{
↪ config_key}.json")

                print(f"\n[{done}/{total_configs}] {
↪ config_key} - n={n_list}")
                print("-" * 70)
                t0 = time.time()

                try:
                    result = run_experiment_1(
                        k=k,
                        n_list=n_list,
                        n_step_monad=nstep,
                        p_val=p_val,
                        precision=prec,
                        log_path=log_path,
                        resume=False,
                    )
                    all_results[config_key] = result
                    elapsed = time.time() - t0
                    print(f"  elapsed: {elapsed:.1f}s")
                except Exception as e:
                    print(f"  failed: {e}")
                    all_results[config_key] = {"error":
↪ str(e)}

    # save the full matrix results
    summary = {
        "meta": {
            "k_list": list(k_list),
            "precision_list": list(precision_list),
            "n_step_list": list(n_step_list),
            "p": p_val,
            "total_configs": total_configs,
            "total_elapsed_sec": round(time.time() -
↪ t_total_start, 1),
```

```
        },
        "results": {}
    }

    for key, result in all_results.items():
        if "error" in result:
            summary["results"][key] = {"error": result["error"]}
            continue
        # per-config summary of (depth, qp_error, omega_qq) by n
        rows_summary = []
        for row in result.get("rows", []):
            rows_summary.append({
                "n":                row.get("n"),
                "depth":            row.get("variation_depth"),
                "qp_error":         row.get("qp_error"),
                "qq_error":         row.get("qq_error"),
                "omega_qq":         row.get("omega_qq"),
                "status":           row.get("status"),
            })
        summary["results"][key] = {
            "k":         result.get("k"),
            "precision": result.get("precision"),
            "n_step":    result.get("n_step"),
            "rows":      rows_summary,
        }

    summary_path = os.path.join(log_dir, "SUMMARY.json")
    with open(summary_path, "w", encoding="utf-8") as f:
        json.dump(summary, f, ensure_ascii=False, indent=2, default=str)

    print()
    print("=" * 75)
    print(f"  all done. total elapsed {summary['meta']['total_elapsed_sec']}s")
    print(f"  summary saved: {summary_path}")
    print("=" * 75)

    return summary


def analyze_experiment_2(summary):
    """
    Analyze matrix results -- apply discrimination logic.

    output:
      1. k=1 precision responsiveness (precision artifact vs structural)
      2. k=2 n_step responsiveness (hypothesis A vs B)
      3. conclusion
    """
    print()
    print("=" * 75)
    print("  Analysis 1: precision responsiveness of k=1 depth")
    print("=" * 75)
    print("  n_step=1 fixed; trend of depth(n) as precision varies")
    print()
    print(f"  {'n':>3} | " +
          " | ".join(f"prec={p:>3}" for p in [64, 128, 256]))
    print("  " + "-" * 50)

    # k=1 precision matrix (n_step=1 fixed)
    for n in range(5, 11):
        row_vals = [str(n)]
        for prec in [64, 128, 256]:
            key = f"k1_prec{prec}_nstep1"
            rows = summary["results"].get(key, {}).get("rows", [])
            match = next((r for r in rows if r.get("n") == n), None)
            row_vals.append(f"{match['depth']:>3}" if match and match.get("depth") is not None else "  -")
        print("  " + " | ".join(f"{v:>7}" for v in row_vals))

    # k=1 discrimination
    print()
    print("  discrimination:")
    k1_reactive = False
    for n in [6, 7, 8, 9]:
        depths = []
        for prec in [64, 128, 256]:
            key = f"k1_prec{prec}_nstep1"
            rows = summary["results"].get(key, {}).get("rows", [])
            match = next((r for r in rows if r.get("n") == n), None)
            if match and match.get("depth") is not None:
                depths.append(match["depth"])
        if len(depths) == 3 and (depths[1] > depths[0] or depths[2] > depths[1]):
            k1_reactive = True
            print(f"    n={n}: depth increase confirmed {depths} -> responds to precision")
    if not k1_reactive:
        print("    depth rises in proportion to precision for all n -> precision artifact")
        print("    OR depth stays fixed regardless of precision -> structural completeness (confirms hypothesis B)")

    print()
    print("=" * 75)
    print("  Analysis 2: n_step responsiveness of k=2 depth (discriminate hypothesis A vs B)")
    print("=" * 75)
    print("  precision=128 fixed; trend of depth(n) as n_step varies")
    print()
    print(f"  {'n':>3} | " +
          " | ".join(f"nstep={s:>1}" for s in [1, 2, 4]))
    print("  " + "-" * 45)

    for n in range(6, 11):
        row_vals = [str(n)]
        for nstep in [1, 2, 4]:
            key = f"k2_prec128_nstep{nstep}"
            rows = summary["results"].get(key, {}).get("rows", [])
            match = next((r for r in rows if r.get("n") == n), None)
            row_vals.append(f"{match['depth']:>4}" if match and match.get("depth") is not None else "   -")
        print("  " + " | ".join(f"{v:>7}" for v in row_vals))

    # k=2 n_step response discrimination
    print()
    print("  discrimination:")
    linear_count = 0
    stable_count = 0
    for n in [7, 8, 9, 10]:
        depths = []
        for nstep in [1, 2, 4]:
            key = f"k2_prec128_nstep{nstep}"
            rows = summary["results"].get(key, {}).get("rows", [])
            match = next((r for r in rows if r.get("n") == n), None)
            if match and match.get("depth") is not None:
                depths.append(match["depth"])
        if len(depths) == 3:
            d_stable = len(set(depths)) == 1
            # if depth drops by 1, 2 as n_step doubles 1->2->4, then hypothesis A
            d_linear = (depths[0] - depths[1]) >= 1 and (depths[1] - depths[2]) >= 1
            if d_stable:
                stable_count += 1
                print(f"    n={n}: depth {depths} -- independent of n_step (supports hypothesis B)")
            elif d_linear:
                linear_count += 1
                print(f"    n={n}: depth {depths} -- depth drops as n_step rises (supports hypothesis A)")
            else:
                print(f"    n={n}: depth {depths} -- irregular")

    print()
    print("=" * 75)
    print("  overall conclusion")
    print("=" * 75)

    if stable_count > linear_count:
        print("  * k=2 depth is largely independent of n_step -> supports hypothesis B (structural perturbation)")
    elif linear_count > stable_count:
        print("  * k=2 depth drops as n_step rises -> supports hypothesis A (off-shell deviation)")
    else:
        print("  * k=2 shows both A and B across n ranges -- finer analysis needed")

    print(f"  * hypothesis A evidence: {linear_count} n values")
    print(f"  * hypothesis B evidence: {stable_count} n values")

    return {
        "k1_precision_reactive":    k1_reactive,
        "k2_linear_count":          linear_count,
        "k2_stable_count":          stable_count,
    }


if __name__ == "__main__":
    print("Experiment 2: precision x n_step matrix start\n")

    summary = run_experiment_2(
        k_list         = (1, 2),
        precision_list = (64, 128, 256),
        n_step_list    = (1, 2, 4),
        n_range_k1     = (5, 10),
        n_range_k2     = (6, 10),
        p_val          = 2,
        log_dir        = "exp2_logs",
    )

    # analysis
    verdict = analyze_experiment_2(summary)

    print()
    print("next steps:")
    if verdict["k1_precision_reactive"]:
        print("  * if k=1 also responds to precision, hypothesis B needs review")
    else:
        print("  * k=1 p-adic completeness confirmed -> structural property of the MHV amplitude")

    if verdict["k2_linear_count"] > verdict["k2_stable_count"]:
        print("  * hypothesis A dominant at k=2 -- off-shell deviation is the primary interpretation")
```

```
    elif verdict["k2_stable_count"] > verdict["k2_linear_count"]:
        print("  * hypothesis B dominant at k=2 -- structural perturbation is the primary interpretation")
```

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